



Spectra Vista Corporation

SVC XHR-1024i / SVC HR-1024i
SVC HR-768i / SVC HR-768si
SVC HR-640i
SVC HR-512i

USER MANUAL

Revision 1.17

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Section 1 - Spectroradiometer

1. INTRODUCTION

This User Manual applies to the following SVC field-portable spectroradiometer models:

Instrument Model	Wavelength Range (nm)	Total Bands	Spectrometers	Si Bands	InGaAs #1 Bands	InGaAs #2 Bands
XHR-1024i	350 - 2500	1024	3	512	256	256
HR-1024i	350 - 2500	1024	3	512	256	256
HR-768i	350 - 2500	768	3	512	128	128
HR-640i	350 - 2500	640	3	512	64	64
HR-768si	350 - 1900	768	2	512	256	n/a
HR-512i	350 - 1050	512	1	512	n/a	n/a

Each instrument includes internal memory and a display for stand-alone operation as well as capability for personal computer (PC) and personal data assistant (PDA) assisted operation. PC communications to the instrument are available through its USB client port, a Bluetooth wireless connection, or via RS-232. A PDA may control the instrument via a Bluetooth connection or via RS-232.

The instruments include a second Bluetooth channel that may optionally communicate with an external sensor suite.

When operated in stand-alone mode, up to 1000 spectral readings can be stored within the instrument for subsequent down loading and analysis using a PC. All spectroradiometer identification and calibration information is stored on board, so there are no external files to manage.

The instrument contains an internal GPS device to record the latitude, longitude, and time of each spectral reading; it also optionally includes an inclinometer to record the instrument's orientation during scanning.

This instrument is self contained and incorporates an easily replaceable battery pack. The battery chargers supplied with the instrument have the capability to charge the LI-Ion batteries from line voltage with the AC adapter.

The computer based operation allows for real time data display and data analysis. Radiance and percent reflectance are provided within the acquisition software. Stand-alone operation makes this same information available for display on the instrument's LCD (liquid crystal display), which is located on the rear panel on the instrument.

A full range of options are available. These options include alternate foreoptics, fiber optic light guides, irradiance spheres, cosine diffusers, Reflectance Probe, back packs, white reference panels, tripods and other custom engineered field devices.

2. HARDWARE DESCRIPTION

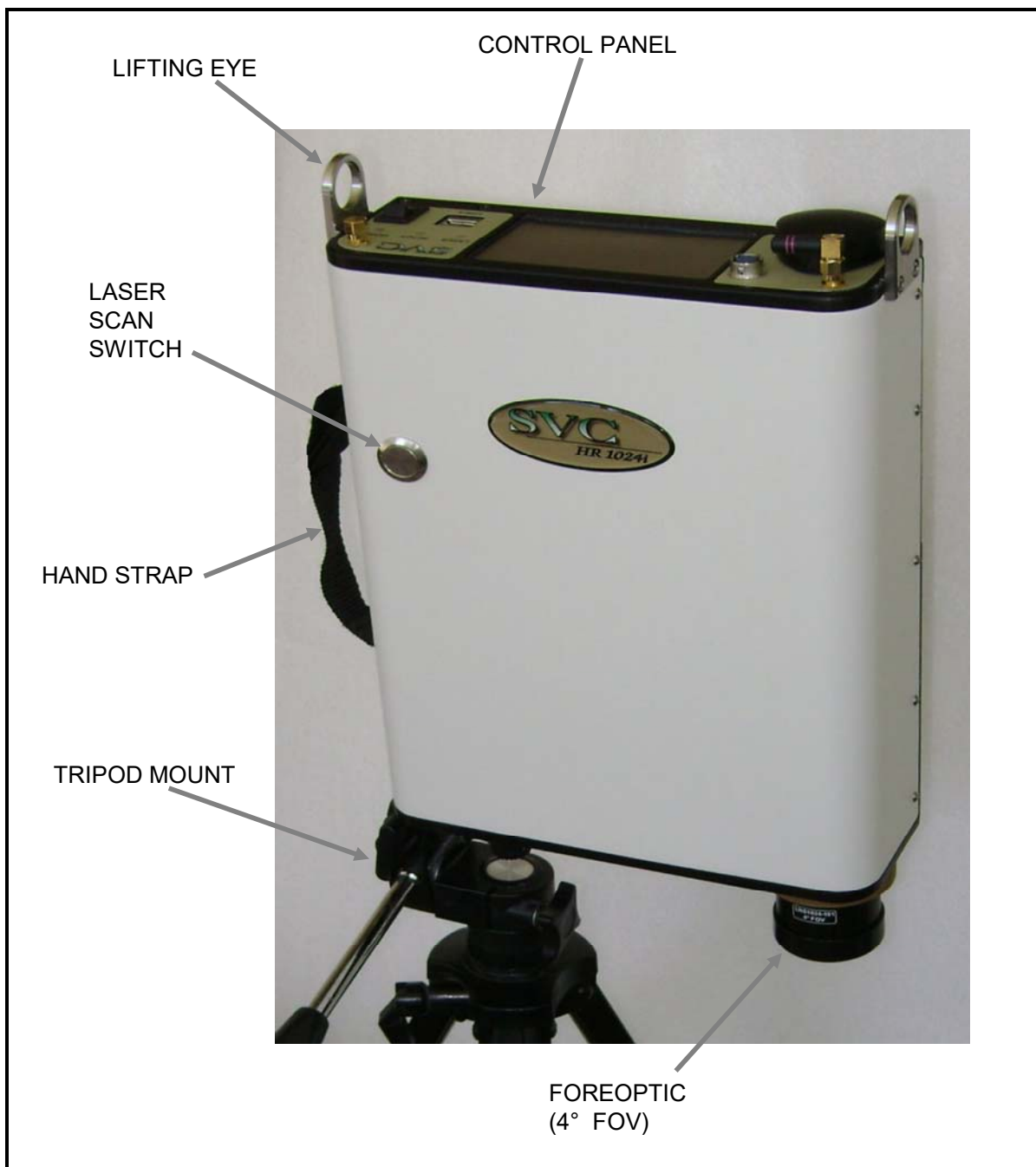


Figure 1 Instrument Attached To Tripod Via Front Panel

Tripod Mounting Options

The instrument can be mounted to a tripod on the front panel (as shown in Figure 1 above) using one or more of five 1/4-20 UNC threaded holes that are compatible with most conventional photographic or video tripods. Alternatively, a tripod can be mounted at the bottom of the optical plate (as shown in

Figure 2 below) where the holes' locations are near the center of gravity of the instrument with battery pack installed.

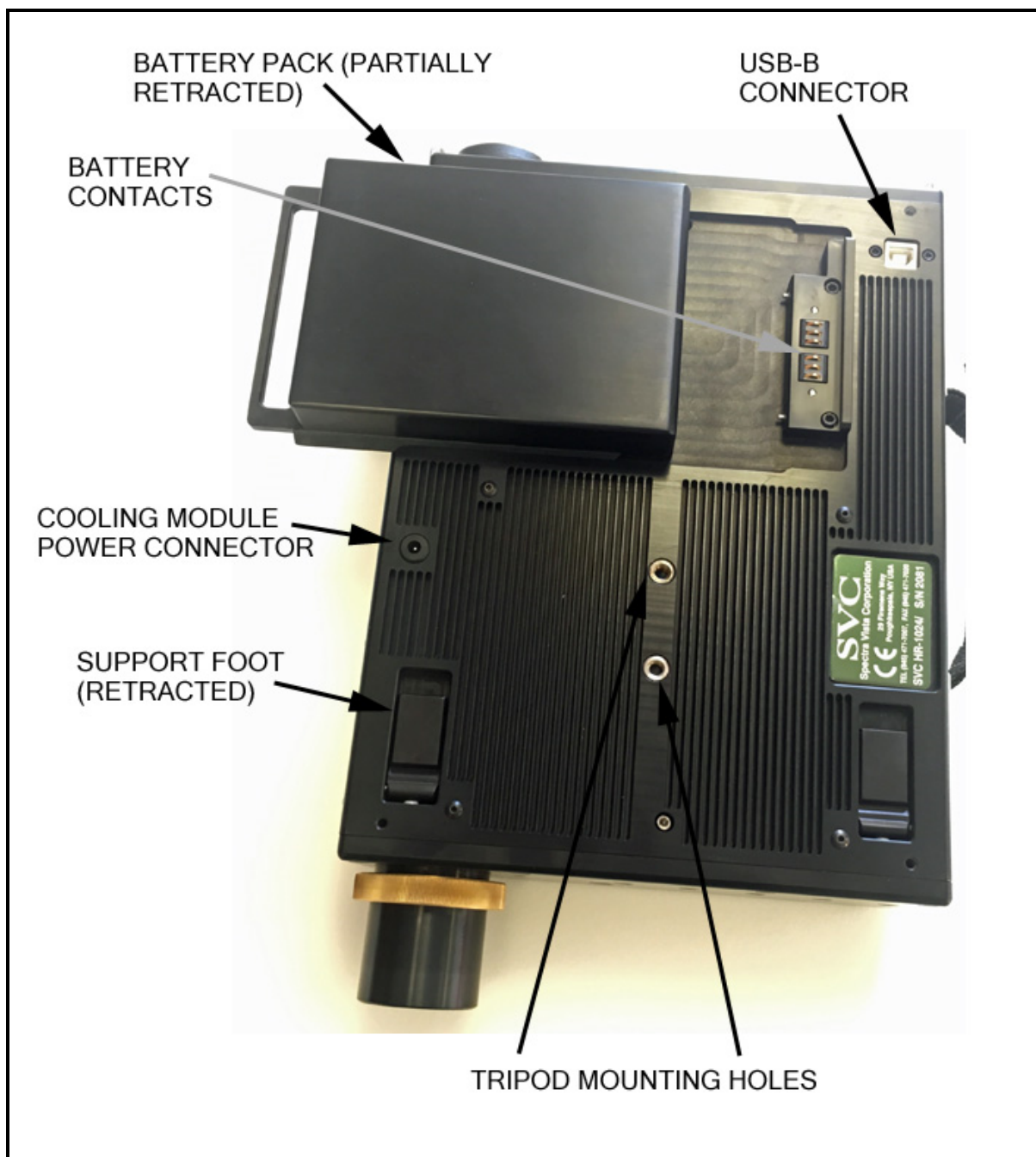


Figure 2 Instrument Bottom View

Foreoptic Options

Different foreoptics can be attached to provide different fields of view (FOV) to enable the instrument to look at different size targets. Normally the target is at a distance of 1m from the optical element. See section 3 item 1 for the field of view setup. Measurements can be taken at further distances but the target size will be larger. The foreoptics are mounted in lens barrels or adapters that are secured to the unit by

hand using a threaded collar and are keyed to go in at the same clock angle to maintain consistent radiometric readings.

The following foreoptics are available, which are listed along with the typical optic “slot” number that is associated with each foreoptic:

Typical Optic Slot Number	Foreoptic Description
1	4° nominal FOV lens (standard)
2	8° nominal FOV lens (optional)
3	14° nominal FOV lens (optional)
4	Uncalibrated (RAW DN) optic (standard)
5	25° FOV fiber optic light guide, various lengths (optional)
6	180° Irradiance Sphere (optional)

NOTES: When optics are installed or changed, the proper radiometric calibration must be applied to the software. The entry of this information is provided for in the Control -> Setup software section of the manual. Always screw the threaded brass collar until hand tight point. Do not over-tighten.

Battery Pack

The instrument’s Battery Pack provides power for portable operation. The battery pack installs via dovetail slides and automatically latches into position to engage the battery contacts on the bottom of the instrument. The Li-Ion battery is 7.4V nominal and connects directly to the instrument. This battery is designed to provide over 3 hours operation following a full charge. When the battery is inserted, power can be applied to the instrument. When the battery voltage is too low to power the instrument, it will turn off and go to 0 volts. The battery can then be charged via the charging connector with the supplied charger. Refer to Section 4 – Battery Charger for charging details.

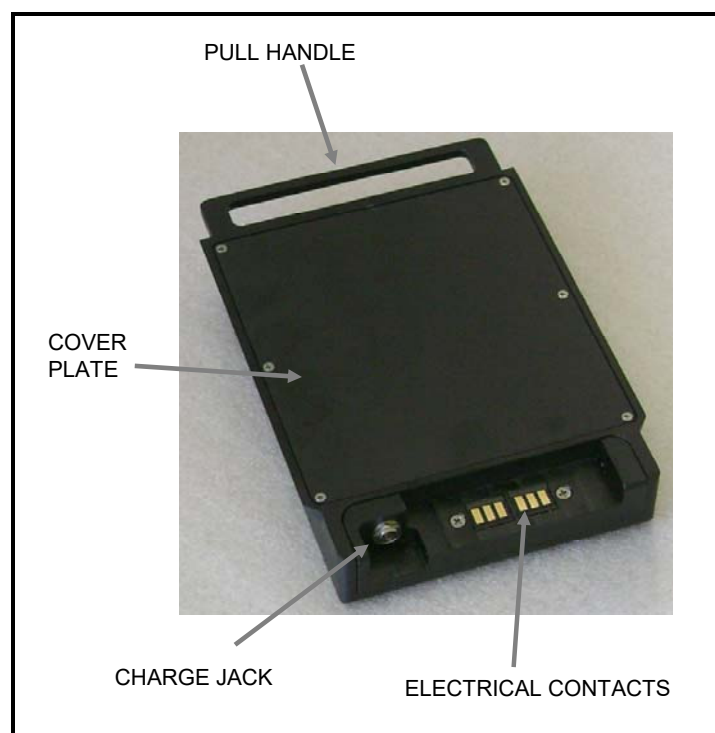


Figure 3 Battery Pack

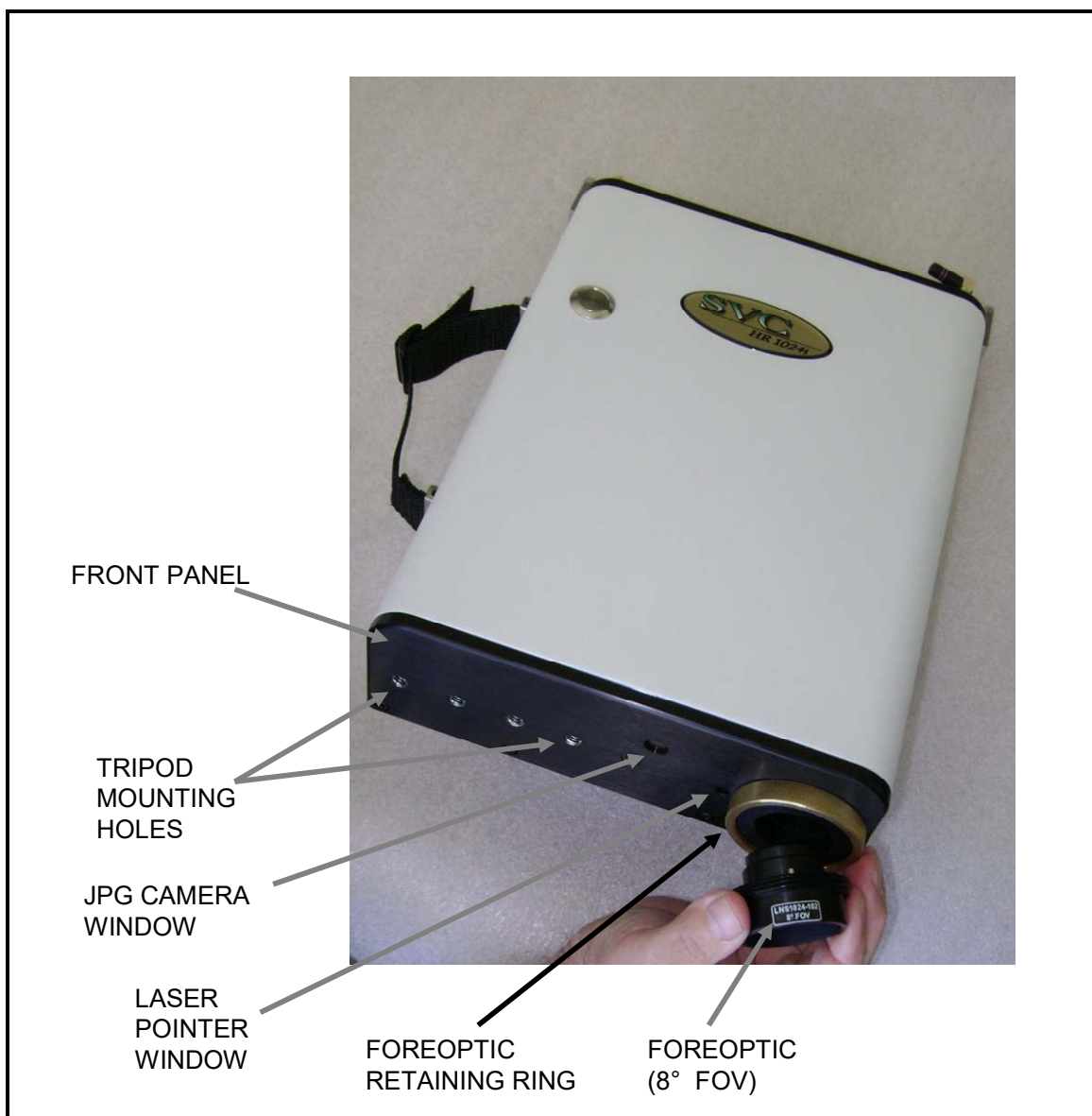


Figure 4 Front Panel

Support Feet

Support feet can be moved between the retracted and extended positions. When extended they allow the instrument to rest level on a flat surface when the battery pack is installed.

Hand Strap

The hand strap is adjustable and allows holding of the instrument with the right hand whilst actuating the laser scan switch with the right index finger.

Laser Scan Switch

The Laser Scan Switch is a pushbutton type that can both actuate the sighting laser and initiate the start of the spectral scan. (More detail is provided in Section 3).

Laser Pointer

The laser beam projects thru the laser pointer window close to the objective lenses and is aligned parallel to the axis of these lenses.

CAUTION: *Do NOT look into the sighting laser window or laser beam.*

JPEG Camera

The internal JPEG camera views thru a protective window as shown above. It is a miniature camera with a 40° field of view and is focused for maximum depth of field from 1m to infinity. It outputs a either a 640x480 or 320x240 pixel image of the spectral target sample and surrounding area.

3. CONTROL PANEL

The control panel of the instrument contains the controls and displays that are required for stand alone operation.

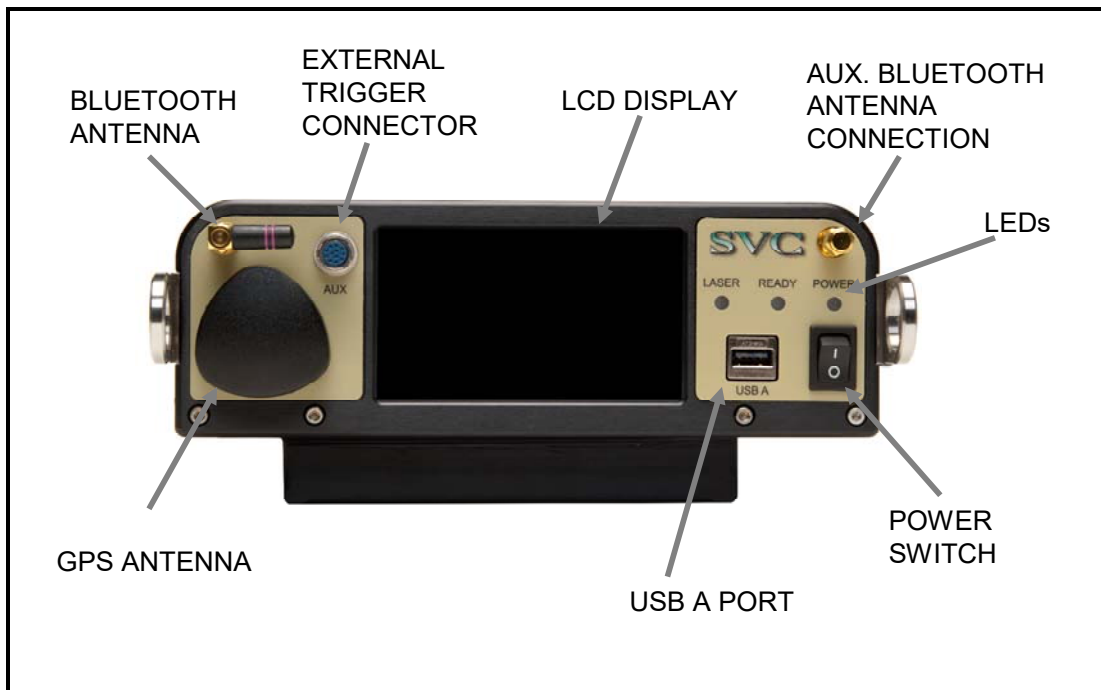


Figure 5 Control Panel

Data Connections

The instrument has 3 connectors that may be used to communicate with external devices; two of these connections are located on the Control Panel, one is located on the bottom of the unit:

AUX	The AUX port is a round 12-pin connector located on the left side of the control panel. It serves as an RS-232 serial port for computer-assisted operation, an external trigger connector, and as a source of power for optional equipment.
USB A	The USB A Port is for future use.
USB B	The USB client port on the bottom of the instrument is for computer assisted operation.
Bluetooth	The Bluetooth antenna is for computer assisted operation.

LED Indicators

The Control Panel has 3 LED status indicators.

Laser LED	The laser indicator is illuminated when the laser is on.
Ready LED	Is illuminated when the instrument is ready to take a scan.
Power LED	The power indicator is illuminated when the power is on.

Power Switch

The power switch provides on/off control of the unit.

GPS Antenna

The GPS Antenna dome is positioned on the control panel for optimal orientation for receiving signals from a GPS satellite when the instrument is in the typical orientation of pointing downwards to look at a spectral target.

LCD Display

The LCD display is useful during acquisition of spectral data in stand-alone operation (see section 3 item 3). It is a standard 4.3" diagonal module with LED backlighting and a 480 x 272 pixel resolution, incorporating a resistive touch screen designed to use finger inputs (no stylus needed.) Text and icons are high contrast, designed to be read in direct sunlight conditions. Spectral plots are viewable from the display.

Inclinometer

Internally, the instrument includes inclinometer hardware that measures, reports in real-time, and records the tilt of the instrument during scan acquisition. The inclinometer provides both visual and audible feedback to the operator to ensure that the instrument is held in the correct attitude for optimal data quality. See the section LCD SCREEN 7 – Inclinometer for more information.

Section 2 – PC Data Acquisition Software

1. INSTALLATION/REMOVAL

The instrument's PC Data Acquisition Software is supplied in single-executable installer format, and provides for step-by-step installation onto a Windows PC or notebook computer.

The installation media is used to install the PC Data Acquisition software into an installation directory (typically, C:\Program Files(x86)\Spectra Vista\[*Instrument Model*]\) on the computer's hard disk drive.

Recommended System Requirements

- Operating System: Windows 7 / 8 / 8.1 / 10
- CPU Speed: 500Mhz (or greater)
- Memory: 1G (or greater) of main memory, depending on Operating System version.
- Screen Resolution: 1024x768 (or greater)
- Serial / USB port: Provided by a built-in COM (9-pin D) port, a USB-to-Serial Adapter, or via a Bluetooth interface (Bluetooth Serial Port)

Installation Steps

Prior to installation, reboot the computer to ensure that Windows is in a stable state, and has as few applications running as possible.

Also, depending on the details of how security is setup on your Windows PC, you may have to be logged in to an account with Administrator privileges. **In addition, depending on your PC security settings, you may have to run the installation program using the special "Run As Administrator..." option.**

Run the installation program and follow the on-screen prompts in order to fully install the software. The initial installation screen is shown below:



Removing The Software

The Data Acquisition Software may be removed by using the standard Windows program removal tool found under *Control Panel -> Programs and Features*. Scroll down through the list of installed programs and select the entry “SVC HR-XXXXi Data Acquisition Software”. Press the “Remove” or “Uninstall” button to remove the software:

Note that only the software will be removed; any data files acquired will remain on the computer’s disk.

2. INTRODUCTION

The PC Data Acquisition Software provides the user with a flexible, easy to use software package. The PC Data Acquisition Software is used to acquire data from the instrument, store data to files on disk, and display data graphically.

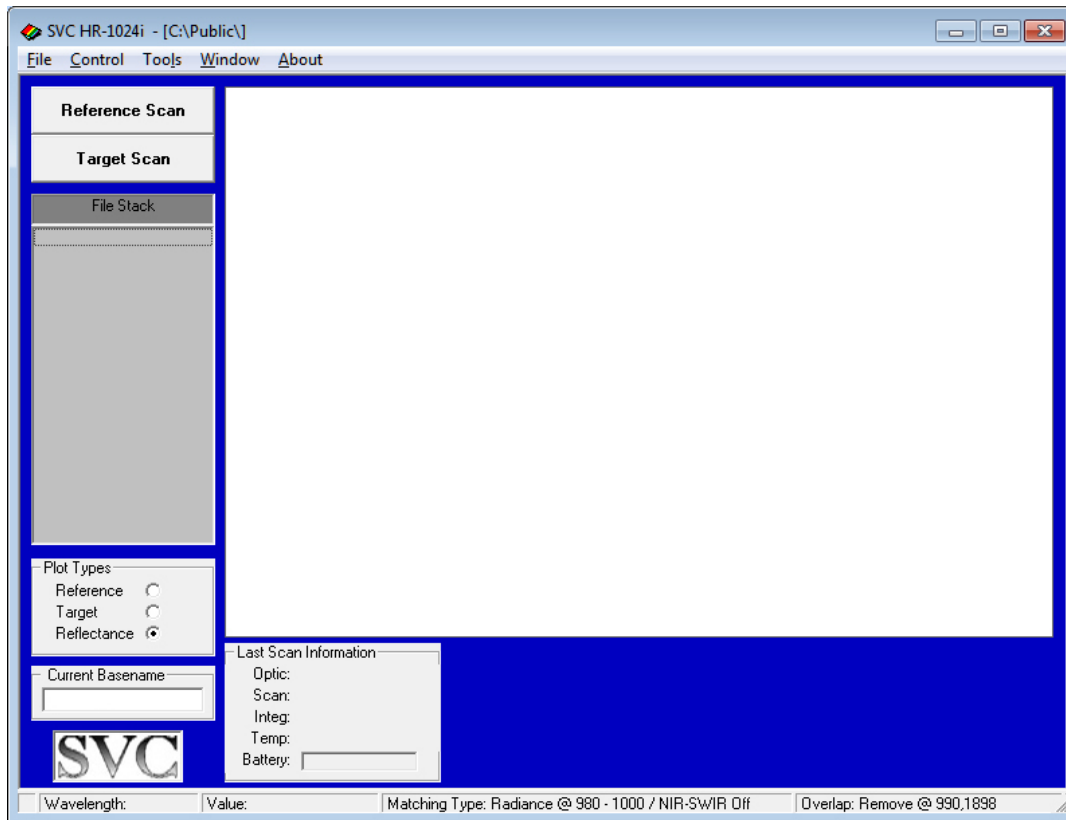
The PC Data Acquisition Software can modify spectroradiometer parameters, take spectral measurements, and download spectral measurements from the on-board memory of the instrument to files on disk. The PC Data Acquisition Software allows data to be configured and stored to a file on disk when taking a spectral measurement.

The software has a number of features that provide a great deal of flexibility for spectral data acquisition, display, and storage:

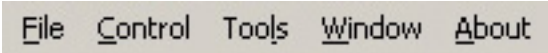
- Data files can be configured as SIG Files for use with PC Data Acquisition Software; SIG Files are saved to disk and may read back into the PC Data Acquisition Software at a later time for analysis.
- Data files can be configured as user defined ASCII Files for export to other software packages; **ASCII format files cannot be read back into the PC Data Acquisition Software** at a later time.
- Data files can be automatically created and given an incremental extension at the end of the acquisition.
- Data can be saved to a unique file name at the end of the acquisition.
- New data may be acquired manually by clicking the mouse on a button or pressing a “hot key”; or new data may be taken automatically using a configurable timer.
- Spectral data may be displayed in one of two ways:
 - “Single Graph” mode, which displays the Reference and Target Radiance on the left Y-axis and the Percent Reflectance on the right Y-axis for a single measurement.
 - “Multi Graph” mode, which displays the Reference Radiance, Target Radiance, or Percent Reflectance of up to 24 measurements in one graph.

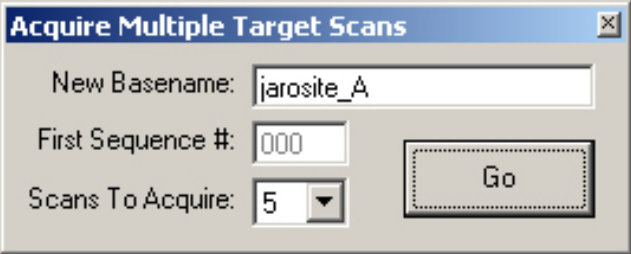
Several tools are built into the software that allow SIG format data files to be merged, re-sampled, or re-processed automatically (see the Tools Menu description below.)

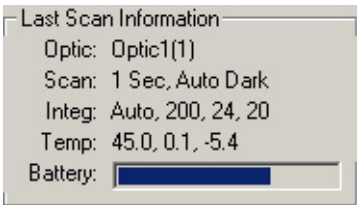
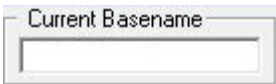
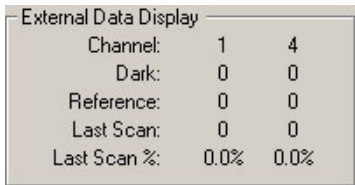
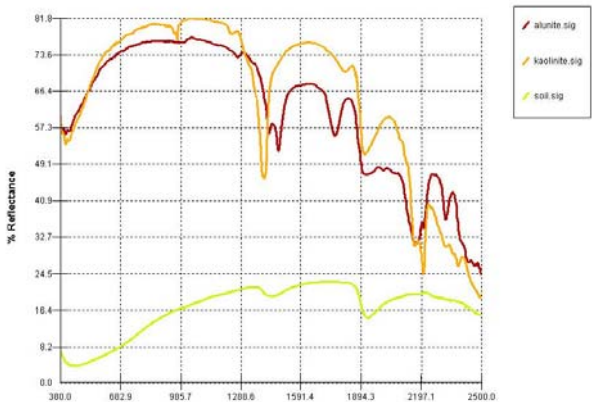
The Data Acquisition Software’s initial screen is shown below.



The screen above is divided into several areas, as described below:

	The application's Title Bar displays the Spectroradiometer Model Name and Current Working Folder – that is, the folder where new scans (real-time or downloaded) are saved, and where old scans may be opened and displayed. The location of the current working folder is retained in the Windows Registry. The location of the current working folder may be changed with the File->New... or File->Open... dialogs.
	The Main Menu allows the operator to open/save files and configure both the instrument settings and the display of spectral data. Each menu item is discussed in detail in subsequent sections.

	The Reference Scan Button causes the instrument to take and graphically display reference data. This button may also be activated by pressing either the “R” or “F1” key on the keyboard.
 	The Target Scan Button, when left-clicked, causes the instrument to take and graphically display a single target data set. This button may also be activated by pressing either the “T” or “F2” key on the keyboard. The Acquire Multiple Target Scans dialog allows the operator to take multiple scans of the same target sequentially. This pop-up dialog appears when the Target Scan Button is <u>right-clicked</u>, and may also be activated by pressing the “M” or “F4” key on the keyboard.
	The Ext Data Scan Button only appears in the GUI when the optional External Data Interface has been installed in the instrument, and has been configured for display. This button may also be activated by pressing the “E” key on the keyboard. See the section SETUP EXTERNAL DATA ... below for more details.
	The Ext Data Dark Button only appears in the GUI when the optional External Data Interface has been installed in the instrument, and has been configured for display. This button may also be activated by pressing the “D” key on the keyboard. See the section SETUP EXTERNAL DATA ... below for more details.
	The File Stack Area shows the list of data files that are currently open. It also allows the operator to select which files to graph in single-graph mode by double-clicking.
	The Plot Types Control allows the operator to select the type of data displayed in the graph area.

	The Status Bar Area displays the channel number, the wavelength, and the Radiance/ Reflectance value of a selected data point on a displayed graph. A data point in a graph is selected by moving the cursor to the point of interest, and clicking the left mouse button on the point.
	The Last Scan Information Box shows information about the last reference or target scan acquired by the Data Acquisition Software.
	The Current Basename Box displays the base file name that is being used to form file names in “Autosave” data acquisition mode. Clicking the mouse within the text entry box brings up a text entry dialog that allows the operator to quickly change the current basename.
	The External Data Display Frame only appears in the GUI when the optional External Data Interface has been installed in the instrument, and configured for display. See the section SETUP EXTERNAL DATA ... below for more details.
	The large white area displays single-graph or multi-graph data sets. The Graphing Area operates in two modes, Zoom Mode and Normal Mode. ZOOM MODE: A rectangular area of interest may be magnified by holding down the left mouse button while “dragging” a rectangular area across the graph. When the left mouse button is released, the graph is re-drawn to contain only the dragged rectangle. NORMAL MODE: Click the right mouse button anywhere in the graph area to restore the magnification to normal.

3. DESCRIPTION

Menu Selection Summary

The application's main menu bar is shown below:



<u>F</u> ile	Provides a menu of options pertaining to Data Files.
<u>C</u> ontrol	Provides a menu of options to control Spectroradiometer operations.
<u>T</u> ools	Provides a menu of available tools for enhancing spectral data files.
<u>W</u> indow	Provides a menu of available Window displays and settings.
<u>A</u> bout	Provides a dialog box of software and spectroradiometer data.

File Menu

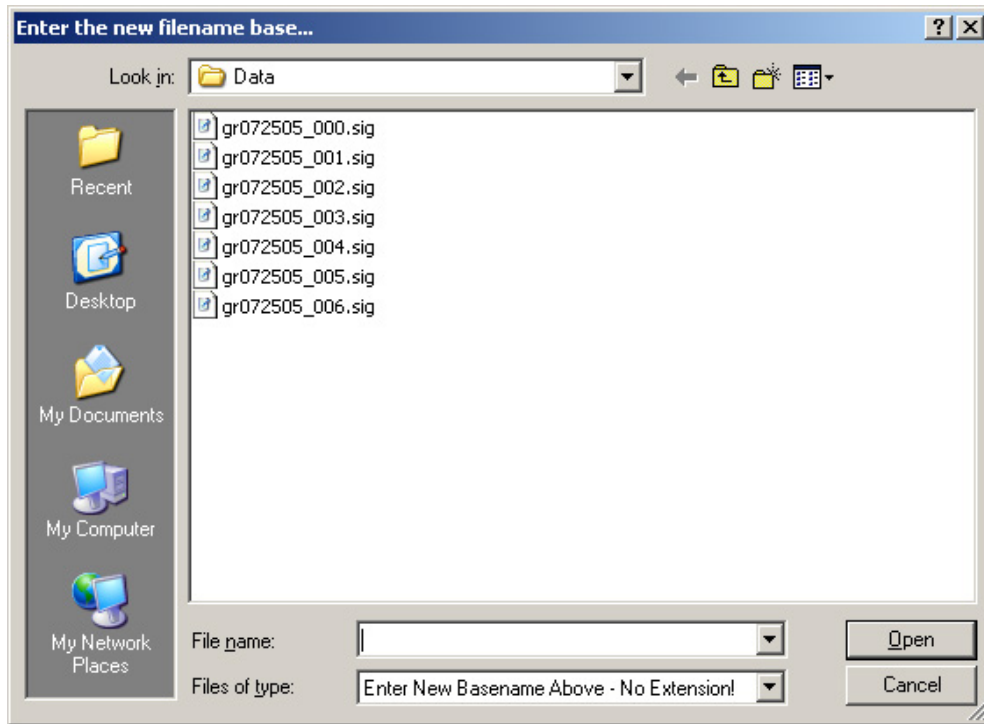
The File menu item provides the following drop-down menu of file controls.

NEW

Modifies the filename configuration used for storing spectral measurement when the 'Autosave' feature is being used.



The following standard Windows file dialog box appears when "New" is selected. Note that the dialog style may vary slightly with different versions of the Windows OS.



File name

Modifies the Base of the filename used with the Autosave feature. The filename of a spectral measurement acquired in Autosave mode is the File Base with an incremental extension automatically provided by the software.

Ex. If the file base is "Test", the first filename is "Test_0000.sig", the second is "Test_0001.sig", etc...

When entering the new base name into the File name box, **do not include any filename extensions**, as the software will automatically add the appropriate extension as each data set is acquired.

Files list

Displays the files that exist in the current drive, directory, and search pattern.

File of type

This box has no purpose other than to display instructions to the user.

Open

Accepts the new base file name entered.

Cancel

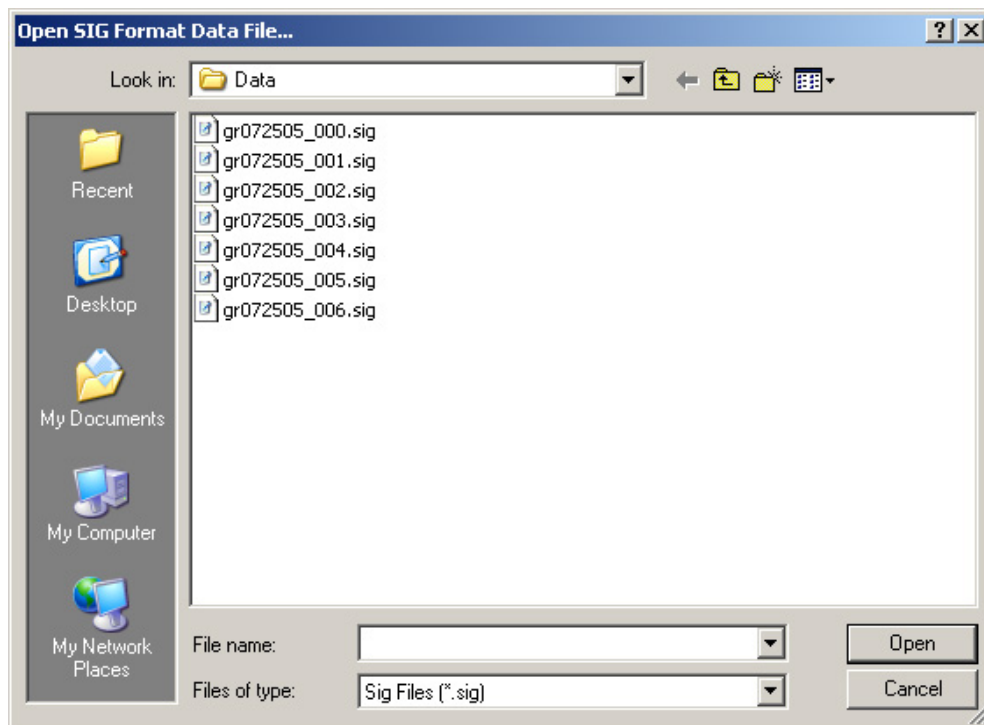
Cancels the operation.

OPEN

Opens the files to be displayed in multi-graph mode.



The following standard Windows file open dialog box appears when “Open” is selected. Note that the dialog style may vary slightly with different versions of the Windows OS.



File name

Selects the file(s) to be opened. A single left-mouse click selects a single fields or <Ctrl> + left-click or <Shift> + left-click can be used to select multiple files.

NOTE: The multi-graph is limited to twenty-four (24) plots.

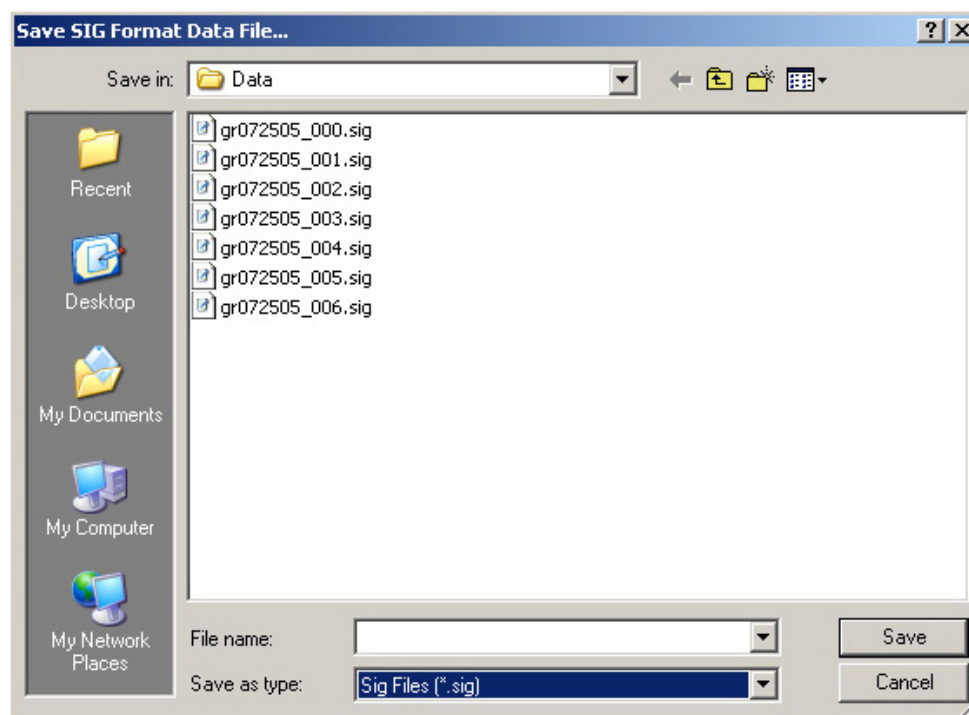
File Type	Modifies the current search pattern used for displaying the existing files in the Files Box. Only “SIG” format files may be opened by the Data Acquisition software.
Open	Opens the selected file(s) for display in the multi-graph.
Cancel	Exits the dialog box without opening a file.

SAVE AS ...

Saves the data that is currently contained in the single graph. **In order to use the Save As... function, the graphing software must be in Single Graph mode, displaying the data file that is to be saved.**



The following standard Windows file save dialog box appears when “Save As” is selected. Note that the dialog style may vary slightly with different versions of the Windows OS.



File name	Allows the desired filename for data storage to be entered. If no extension or an invalid extension is entered, then one is supplied, based on the current file format.
Files list	Displays the files that exist in the current drive, directory, and search pattern.
Save As Type	Modifies the current search pattern used for displaying the existing files in the Files Box. Only the currently selected format may be saved. The file format is selected by using the File->Data Options->Format menu to choose between “SIG” and “ascii” output file format.
Save	Saves the data contained in the single graph to the specified filename and exits the dialog box.
Cancel	Exits the dialog box without saving the file.

CLOSE

Closes the data files that are selected in the File Stack of the multi-graph. The multi-graph is redrawn after the files are closed.

CLOSE ALL

Closes all files in the File Stack.



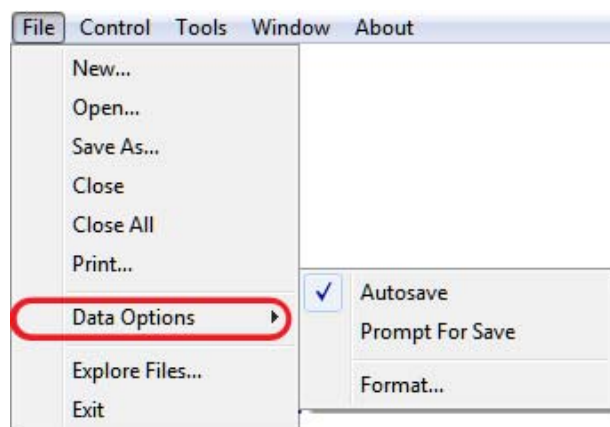
PRINT

Opens a standard Windows print dialog box. The current graph view may be printed using this dialog.



DATA OPTIONS

Configures how and when spectral measurements are stored to a files on a disk.



Autosave

When this item is checked, the Autosave feature is turned on. When Autosave is on, spectral measurements are automatically saved to files on disk using the current File Basename and an incremental extension. The default filename configuration uses the installation data directory (typically C:\Program Files\Spectra Vista\HR1024i\Data\) and the current File Basename determined by “gr” and the month, day, and year (Ex. “gr052105”).

When the Autosave is turned on, the Prompt For Save option is disabled.

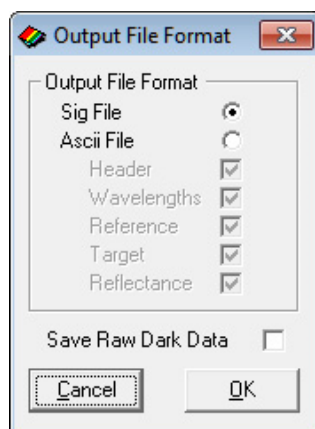
Prompt For Save

When this item is checked, the operator is prompted for a filename to store the data after the completion of each spectral measurement.

Format...

Configures the data output file.

Choosing “Format...” displays the following dialog box:



Output File Format

Provides for the selection of one of the following:

- **SIG File** This is the format compatible with the PC Data Acquisition software; spectral data saved in this format may be read back and displayed at some future time by the PC Data Acquisition Software

- **Ascii File** Check the column(s) of data that will appear in the ASCII data file. The options are:

Header information, including instrument settings, date/time, etc.

Wavelength in nanometers

Reference Radiance in $\text{W}/\text{cm}^2/\text{nm}/\text{sr} \cdot 10^{-10}$

Target Radiance in $\text{W}/\text{cm}^2/\text{nm}/\text{sr} \cdot 10^{-10}$

Percent **Reflectance**

Save Raw Dark Data

Checking this box will cause the software to save the “raw” dark detector data associated with each scan. With each target scan acquisition, this raw dark detector data is saved to a separate file with the file name extension “.raw”.

OK

Implements the selected file format and exits the dialog box.

Cancel

Ignores the modifications made to the file format and exits the dialog box.

EXPLORE FILES

Opens a independent Windows Explorer window, showing the list of files that are in the Current Working Folder.



EXIT

Exits the PC Data Acquisition Software program.



Control Menu

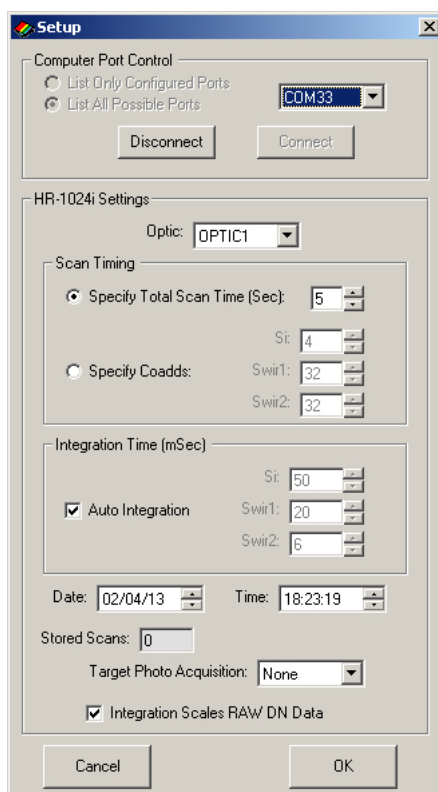
The control menu item provides the following drop-down menu of instrument controls:

SETUP INSTRUMENT...

Configures the Computer Port used for communications with the instrument. Once connected, sets the spectroradiometer operating parameters.

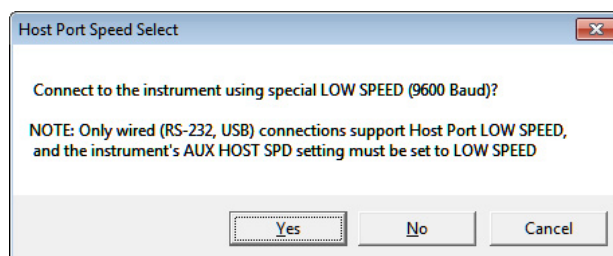


When “Setup Instrument...” is selected, the software presents the following setup dialog box:



Computer Port Control Frame:

List Only Configured Ports	When selected, only ports that are physically <u>configured</u> into the host computer are listed in the Port control. The list of configured ports is built only once each time the Data Acquisition Program is run.
List All Possible Ports	When selected, all 255 ports are listed in the Port control. Use this option if a virtual port (usually a Bluetooth port) that you want to use is not present in the “configured” list above.
Port	Selects/displays the serial communications COM port used to communicate with the instrument. The actual physical hardware associated with the COM port may be provided by a standard DB-9 hardware interface installed in the PC; It may also be realized by several other means that provide a “virtual” COM port (Bluetooth Serial Port, USB-to-Serial Adapter, or USB COM driver). See the Operations section, sub-section 2. Connecting The Instrument To A Computer, for more details.
Connect (<i>left-mouse-click</i>)	Establishes a connection to the instrument at the <u>standard baud rate</u> , and downloads the information necessary to acquire and process spectral data.
Connect (<i>right-mouse-click</i>)	Displays a confirming dialog box (see below), then establishes a connection to the instrument at the <u>low speed baud rate</u> , and downloads the information necessary to acquire and process spectral data. Note that using this option will significantly slow down spectrometer operation.



Disconnect	Closes the connection.
------------	------------------------

Settings Frame	Modifies the instrument settings. This frame is only enabled after the PC Acquisition Software has established a connection with an instrument.
Optic	Selects the entrance optic that is currently installed on the instrument. NOTE: The Special “RAW DN” entrance optic provides raw measurements without any radiometric calibration applied.
Scan Timing Frame:	Scan time may be control in two ways, either by specifying the amount of time spent on the light scan, or by specifying the number of sub-scans (Coadds) acquired from each spectrometer.
Specify Total Scan Time	Scan data will be acquired using a total light scan time, which controls how long (in seconds) a spectral scan will take. Higher settings can reduce noise, but scans will take longer to complete. Any integral scan time from 1 to 15 seconds is supported.
Specify Coadds	Scan data will be acquired by taking a specific number of Coadds from each spectrometer, regardless of the total amount of time required, up to a maximum of 15 seconds . If this time limit is reached before the number of specified Coadds has been acquired, then the scan is terminated and the actual number of Coadds is reported in the Sig file.
Integration Time Frame:	Using this control, the operator may select between auto- and fixed-integration modes.
Integration Time	The time in milliseconds allowed for the individual detectors to accumulate a single Coadd. Higher settings can reduce noise, but the scans will take longer to complete and may saturate.
Auto	The “Auto” check box allows the instrument to automatically select the optimum integration time for each spectrometer.

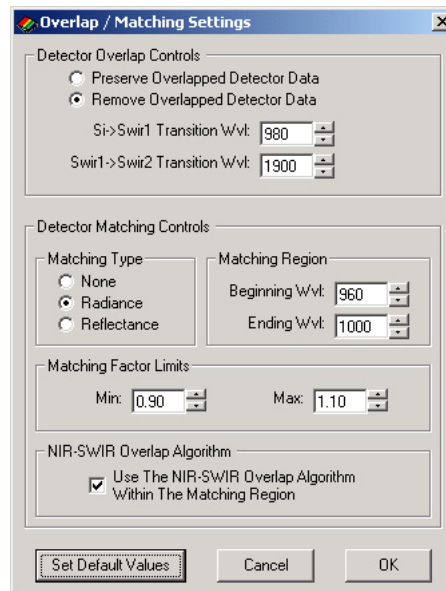
Date/Time	Displays the current date/time stored in the instrument's real time clock. If the displayed date/time is changed, the new settings are downloaded to the instrument if the "OK" button is selected.
Stored Scans	Displays the number of scans currently stored in the instrument's internal memory.
Integration Scales RAW DN Data	When checked, and when the instrument's "RAW DN" optic is selected, the RAW DN data will be scaled appropriately by the reported integration times. Use this checkbox when acquiring reflectance data in RAW DN mode. When unchecked, RAW DN data is not scaled by the integration times.
Target Photo Acquisition	Controls the type of photo that is acquired along with each target scan: OFF : No target photo. LO : Low resolution (320x240) photo. HI : High resolution (640x480) photo. Note that taking target photos will increase the amount of time to acquire and download scan data from the instrument to the PC.
OK	If connected to an instrument, the settings shown are downloaded to the instrument, and the dialog exits.
Cancel	Exits the dialog without saving or sending any instrument settings.

SETUP OVERLAP/MATCHING...

Configures how to handle overlapped detector data; note that this option is not available in the SVC HR-512i software:



When “Setup Overlap/Matching...” is selected, the software presents the following dialog box.



Overlap Frame:

Preserve Overlapped Detector Data

When this option is selected, the Transition Wvl setting is ignored and has no effect. All spectral data is preserved in the output file as each scan is acquired.

Remove Overlapped Detector Data

When this option is selected, the Transition Wvl fields control where the Si->Swir1 and Swir1->Swir2 transitions takes place. All overlapping spectral data is removed.

Si->Swir1 Transition Wvl

When in “Remove Overlap” mode, sets the point where the Si detector data ends, and the first InGaAs detector data begins.

Swir1->Swir2 Transition Wvl

When in “Remove Overlap” mode, sets the point where the first InGaAs detector data ends, and the second InGaAs detector data begins.

Matching Type Frame

Controls the method of matching Si and InGaAs detector data.

None

No matching take place. When this option is in effect, none of the other

	Matching parameters has any effect on spectral data.
Radiance	Uses the detectors' radiance curve data to perform matching on both the Reference and Target scans separately.
Reflectance	Combine the Reference and Target scans into a reflectance curve, and uses this curve's data to perform matching.
Matching Region Frame	Defines the region over which the Si and first InGaAs data is examined in order to perform detector matching.
Beginning Wvl	The beginning wavelength in the matching region. This settings has no effect if "Matching Type" is set to "None".
Ending Wvl	The ending wavelength in the matching region. This setting has no effect if "Matching Type" is set to "None".
Matching Factor Limits Frame	Defines the range of acceptable values for the factor used to match Silicon data to SWIR data.
Min	The minimum matching factor value.
Max	The maximum matching factor value.
Use NIR-SWIR Overlap Algorithm Frame	
Use The NIR-SWIR Overlap Algorithm...	When checked, the Silicon and SWIR1 data are combined (within the Overlap Region) using a combination of the two overlapping detector data sets. Using this option only affects the data within the Overlap Region. This algorithm is applied after the standard matching algorithm is applied (See Matching Type Frame instructions above).

SETUP EXTERNAL DATA ...

Configures the instrument's optional external data interface.



The instrument may optionally be configured with an external sensor data interface. This interface allows the PC software to digitize, display, and store the data from up to 16 external sensors that are attached to the external interface.

Currently, Spectra Vista supplies the WEDI-4 product to serve as an external data interface. Contact Spectra Vista for details on this product.

The Setup External Data dialog determines which, if any, of the external sensors' data is saved to SIG files and/or displayed in the main GUI display.

When selected, the Setup External Data command displays the following dialog:

External Data Setup

External Data Source: HR-1024i (via BT)

External Data Source Bluetooth Address

Bluetooth Address: 123412341234

Update Instrument

External Data Acquisition

	Enabled		Enabled
Ch. 1	☐	Ch. 9	☐
Ch. 2	☒	Ch. 10	☐
Ch. 3	☐	Ch. 11	☐
Ch. 4	☒	Ch. 12	☐
Ch. 5	☐	Ch. 13	☐
Ch. 6	☒	Ch. 14	☐
Ch. 7	☐	Ch. 15	☐
Ch. 8	☐	Ch. 16	☐

External Data Display

Display Ch. A: Ch. 2

Display Ch. B: Ch. 6

☐ Adjust Displayed Values For External Dark Current

Cancel OK

External Data Source:

Selects the source for the external data:

None – disables external data acquisition.

i-Series Model (Via BT) – use the optional external Bluetooth data interface within the connected instrument for external data acquisition.

External Data Source Bluetooth Address Frame

Bluetooth Address

Allows entry of the 12 digit hexadecimal address of the external data source. This address is used by the instrument to establish a connection between the instrument and the external data device.

The 12 digit Bluetooth address to use in this box may be found on the label attached to the bottom of the

customer's external data device (WEDI-16).

Update Instrument

Use this button to send the displayed Bluetooth Address up to the instrument. The instrument must be connected to the PC in order for this button to be enabled. Once the Bluetooth address is sent to the instrument, the instrument will attempt to establish a connection to the external data source each time it is turned on.

External Data Acquisition Frame

Ch 1-16 Enabled:

When checked, the data from that external channel will be saved in the header of the SIG file. When unchecked, that channel will be ignored, and a zero value will be stored in its place in the SIG header.

Use these check boxes to tell the software which of the 16 input channels actually have a sensor of interest attached to the external data device.

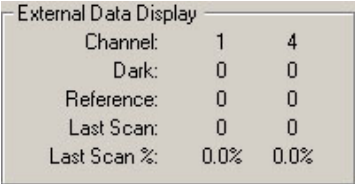
External Data Display Frame

Display Ch. A-B:

These two drop-down menus determine which (if any) of the 16 external channels are displayed in the main GUI screen.

When either 1 or 2 channels are selected for display, the main GUI screen shows two additional elements for control and display of the external data, as described below:

	Press the Ext Data Scan button when a new set of external data samples is desired. No spectral data is acquired when this button is pressed, and only the Last Scan values (see below) are updated.
	Press the Ext Data Dark button when a new set of external dark samples is desired. No spectral data is acquired when this button is pressed, and only the Last Scan values (see below) are updated.

	The External Data Display frame displays the data from the selected External Data channels: Channel: The Channel number for that column. Dark: The most recent Dark Scan values taken. When updated, the Reference and Last Scan fields are reset. Reference: The most recent Reference Scan values. When updated, the Last Scan fields are reset. Last Scan: The most recent values/percentages taken during either a Target Scan, or an External Data Scan. The percentage is computed by dividing Last Scan by Reference.
---	---

Adjust Display Values For External Dark Current:

When checked, dark current correction is applied to the data displayed in the main GUI. When unchecked, dark current correction is not applied. This control has no effect on the data saved to the SIG file header.

Cancel:

Dismiss the dialog without saving changes.

OK:

Apply the settings shown in the dialog.

READ MEMORY ...

Reads (downloads) the instrument's stand-alone scans from the internal memory into the PC Data Acquisition Software.



When selected, the Read Memory command displays the following dialog:

Read Scan Memory

Read Scans

Total Scans In Memory : 57 ☐ Edit Scan Types During Download

First Scan To Download: 1

Last Scan To Download: 57

Download Data Handling

☐ Do Not Apply Overlap / Matching Settings During Download

☒ Apply Current Overlap / Matching Settings During Download

Current Settings:

Overlap: Remove @ 1000,1867

Matching Type: Radiance @ 1002 - 1006 / NIR-S'WIR Off

Output File Information

Folder: C:\Data\ Change

File Base Name: Pine_Data

Output File Name Format: BaseName_XXXX

Download Control

Current Ref Scan Number:

Downloading Scan Number:

Next File Name: Pine_Data_0000.sig

Erase All Scans Download Now

Cancel

Read Scans Frame

This frame allows the user to define which internal scans are to be downloaded and controls the destination folders and file names.

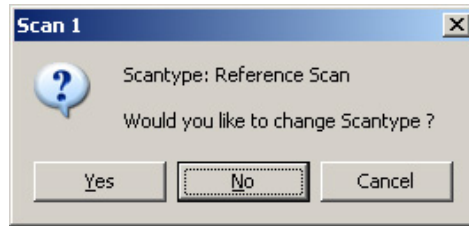
During download, a SIG data file is created for each target scan. A SIG data file contains the radiance of the target scan and the radiance of the most recent reference scan. The memory slots of the scans are recorded in the file header.

Note that each stored scan, either a reference or a target, occupies a sequentially numbered "slot" within the instrument. A slot number may vary from 0 to 999, and increment as each stand-alone is acquired in the field.

Total Scans In Memory:

Total number of scans currently stored in the instrument's internal memory.

First Scan To Download:	First scan to download. The first scan typically should be a reference scan.
Last Scan To Download:	The number of the last scan to download.
Edit Scan Type:	If checked, the user is prompted (see dialog below) to either accept or change the stored scan type (reference or target) as each stored scan is downloaded.



Download Data Handling Frame

This frame allows the user to optionally apply the current overlap and matching settings to the data as it is downloaded from the instrument.

Apply, Do Not Apply Overlap / Matching Settings During Download:	Select the desired radio button regarding this option. The current settings are displayed.
--	--

Note that it may be desirable to select the “Do Not Apply” option, and then later after the downloading is complete, use the “Tools->SIG File Matching / Overlap...” tool to post-process the SIG files. This process preserves the original downloaded SIG data, allowing it to be re-processed (if needed) with different overlap / matching parameters.

Output File Information Frame

This frame determines the location of the downloaded files and the names of the downloaded files.

Folder:	The current destination folder name
File Base Name:	The current SIG file base name. As each target scan is downloaded, a 4-digit incrementing number may be appended to this base name. This field <u>only</u> applies if the “BaseName_ xxxx” format is selected (see below).
Change:	Pressing this button will display a standard file open dialog box, which allows the user to select a new Folder and File Base Name.

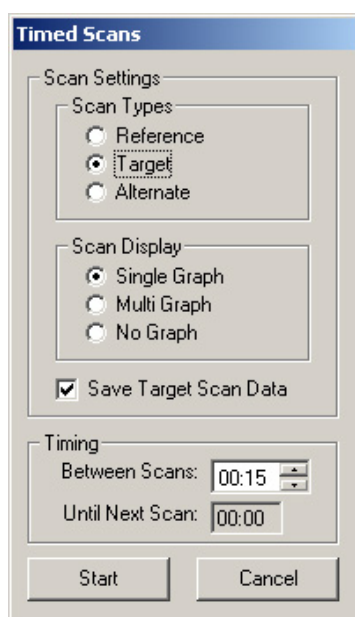
Output File Name Format:	This pull-down presents two choices regarding the format of the output SIG file name: BaseName xxxx: As each target is downloaded, the selected File Name Base is appended to an incrementing 4-digit number. Each SIG file written will have this combination of fields as a file name. yyMMdd HHmm RxxxTxxx: As each scan is downloaded, the target's date (yyMMdd), time (HHmm) and slot number (Txxx) will be used, along with the most recently downloaded reference scan's slot number (Rxxx), in order to form the SIG file name. This format provides a lot of information relating back to the original scans right in the SIG files' names.
Download Control Frame	This frame starts the download process and shows download progress.
Current Ref Scan Number:	This box displays the slot number of the most recently downloaded (the current) reference scan.
Downloading Scan Number:	This box displays the slot number of the most recently downloaded scan (reference or target).
Download Now:	Clicking this button begins the process of downloading of data from the instrument's internal memory. When all data are downloaded a summary message box indicates that the operation is complete.
Erase All Scans:	Pressing this button permanently erases all scans within the instrument's internal memory. Scans should only be erased when the user is sure that they are no longer needed.
Cancel	During download, this button stops the downloading process. During idle times, this button is used to dismiss the Read Scan Memory dialog.

TIMED SCANS ...

Acquire spectral data automatically using a scan timer. The user may select the type of scan (reference vs. target), how the acquired data is displayed, and whether the data is written to disk files.



When selected, the Timed Scans display the following dialog:



Scan Settings Frame

This frame determines the characteristics of the timed scan.

Scan Type: Reference

This option causes reference scans to be taken on a timed basis. Reference scans are always displayed in Single Graph format.

Scan Type: Target

This option causes target scans to be taken on a timed basis. Target scans may

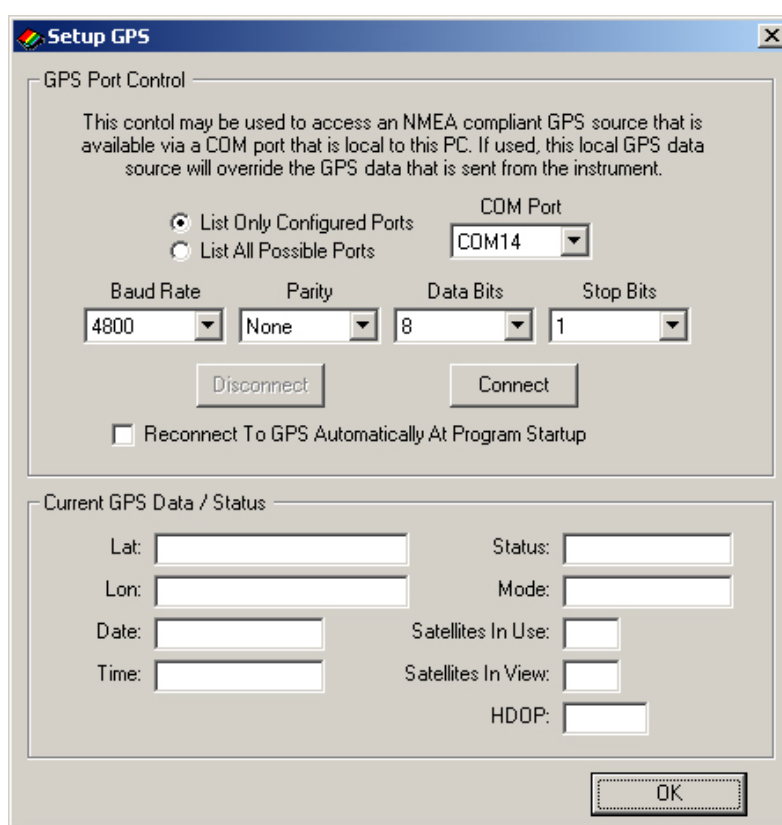
	be displayed in either Single Graph or Multi Graph format.
Scan Type: Alternate	This option causes scans to alternate between Reference and Target on a timed basis.
Scan Display: Single Graph	Target scans are displayed one at a time in a Single Graph display.
Scan Display: Multi Graph	Target scans are displayed together in a Multi Graph display format. Scans will stop being acquired, and a warning message displayed, if the maximum number of allowed Multi Graph data sets (24) is reached.
Scan Display: None	Acquired scan data is not displayed.
Save Target Scan Data	When checked, target scans cause a new data file to be written to disk. Otherwise the scan data is discarded.
Timing Frame	This frame controls scan timing.
Between Scans	Allows the user to set the time (minutes / seconds) between acquired scans. The time can range between 0 seconds and 1 hour.
Until Next Scan	When running, this box displays the time until the next scan is scheduled to be triggered.
Start	Pressing this button begins timed scan acquisition using the settings shown. This button changes to “Stop” when the timer is running and may be used to stop timed scan acquisition.
Cancel	Dismisses the Timed Scans dialog.

SETUP GPS ...

Set the parameters that allow the Data Acquisition Software to access a source of NMEA V2.1 GPS data directly from a local PC COM port.



When selected, the Setup GPS menu item displays the following dialog:



NOTE: This dialog is used only to setup a GPS source that is internal to the acquisition PC or laptop. If enabled by this setup dialog, the data from this PC/Laptop source will override the GPS that is provided by the instrument's internal GPS source.

GPS Port Control Frame

This frame determines the COM port settings of the GPS data source that is installed in the computer.

List Only Configured Ports

When selected, only ports that are physically configured into the host

	computer are listed in the COM Port control. The list of configured ports is built only once each time the Data Acquisition Program is run.
List All Possible Ports	When selected, all 255 ports are listed in the COM Port control. Use this option if a virtual port (usually a Bluetooth port) that you want to use is not present in the “configured” list above.
COM Port	Determines the COM port number for the NMEA V2.1 GPS data..
Baud Rate / Parity / Data Bits / Stop Bits	These controls determine the exact communication settings when receiving GPS data.
Reconnect To GPS Automatically At Startup	Checking this option will cause the Data Acquisition Software to reconnect at program start, using the current COM Port settings.
Connect Button / Disconnect Button	Enables/Disables the connection to the GPS source.
Current GPS Data / Status Frame	
	When connected, displays the current GPS location, date, time, and status information.
Lat/Lon/Date/Time/HDOP	These fields contains valid GPS data when a sufficient number of GPS satellites are available to produce a valid fix; otherwise, these fields are blank.
Satellites In View / Satellites In Use	Tracks the number of satellites in view of the receiver, and the number being used to produce a location fix. At least 3 satellites are required to produce a 2D fix.
Status	The current COM Port connection status.
Mode	The current mode of operation, from “No Fix” to “3D Fix”.

SETUP WATER QUALITY INTERFACE ...

Set the parameters that allow the Data Acquisition Software to access water quality data from a Eureka Manta 2 Water Probe.



When selected, the Setup Water Quality Interface menu item displays the following dialog:

The dialog box is titled 'Setup Water Quality Probe Interface'. It contains a text area explaining that the control is used to receive water quality data from a Eureka Manta 2 Water Probe. Below this, there are two radio buttons: 'List Only Configured Ports' (selected) and 'List All Possible Ports'. A 'COM Port' dropdown menu is set to 'COM31'. There are 'Disconnect' and 'Connect' buttons. To the right, the 'COM Port Communication Settings' section includes a 'Set Defaults' button, a 'Baud Rate' dropdown set to '19200', a 'Parity' dropdown set to 'None', a 'Data Bits' dropdown set to '8', and a 'Stop Bits' dropdown set to '1'. A checkbox for 'Reconnect To Probe Automatically At Program Startup' is unchecked. The 'Probe Communication Status' is shown as a green bar with the text 'Receiving Probe Data OK'. At the bottom, there is a table with 8 columns: Temp_deg_C, pH_units, SpCond_uS/cm, CHL_ug/l, HDO_mg/l, HDO_%Sat, pH_mV, and an empty column. The table contains 5 rows of data.

Temp_deg_C	pH_units	SpCond_uS/cm	CHL_ug/l	HDO_mg/l	HDO_%Sat	pH_mV	
23.30	-0.51	0.0	0.167	8.79	105.9	445.78	
23.30	-0.34	0.0	0.169	8.79	105.9	435.43	
23.30	0.00	0.0	0.180	8.79	105.9	415.52	
23.30	9.55	0.0	0.173	8.79	105.9	-139.48	
23.30	9.23	0.0	0.000	0.00	0.0	-120.55	

An 'OK' button is located at the bottom right of the dialog.

This control gathers water quality data in real-time from an attached Eureka model Manta 2 Water Quality Probe. Water quality data is then reported in the SIG file header information.

See the document “SVC iSeries Spectroradiometer, Water Quality Measurement System, User Manual” for complete details on use of this option.

LONG TERM TIMER ...

Setup a special timer that is capable of acquiring data periodically over multiple days.

Note that a maximum of 10,000 SIG files may be acquired through use of the long term timer. This limitation comes from the 4-digit (0000 through 9999) file number appended automatically to each SIG file base name during timed acquisition.



When selected, the Long Term Timer menu item displays the following dialog:

Long Term Timer

Use this tool to automatically acquire scan data. Scan data will be acquired from the specified Start Date to the specified End Date (inclusive). During each active day, up to 8 active time periods may be specified, during which scan data will be acquired, according to the selected Start Time, End Time, and Interval between scan acquisitions.

Acquisition Active Days

Start Date: 6/26/2019 End Date: 6/26/2019

Daily Acquisition Settings

Period	Enabled	Start Time	End Time	Interval
1	☐	00:00:00	00:00:00	00:00
2	☐	00:00:00	00:00:00	00:00
3	☐	00:00:00	00:00:00	00:00
4	☐	00:00:00	00:00:00	00:00
5	☐	00:00:00	00:00:00	00:00
6	☐	00:00:00	00:00:00	00:00
7	☐	00:00:00	00:00:00	00:00
8	☐	00:00:00	00:00:00	00:00

Save Settings To File Read Settings From File

Acquisition Control

Current Date / Time: 6/26/2019 14:04:23

START Timed Target Scan Acquisition

Take A Reference Scan Now

Take A Target Scan Now

☒ Ignore Scan Warnings ☒ Always Use Default SIG File Name

Hide Settings

Cancel

Use the following controls and fields to setup a multi-day timer.

Acquisition Active Days Frame	This frame determines the range of dates to acquire spectral data.
Start / End Date	These two dates define the first and last calendar days to include in the data acquisition.
Daily Acquisition Settings Frame	This frame defines periods within each active day for spectral data acquisition. Up to 8 unique periods may be defined.
Enabled	Each of the 8 periods may be enabled or disable by use of this check box.
Start / End Time	These two times define the beginning and ending times for each period. Format is hh:mm:ss (24-hour time format).
Interval	This is the time between target scan acquisitions for each period; format is mm:ss.
Save / Read Settings To / From File	These two buttons allow you to save / read daily acquisition settings to / from a file. When pressed, you will be prompted for the file name to use.
Acquisition Control Frame	Controls both timed and immediate spectral data acquisition.
START/END Timed Target Scan Acquisition	This button starts (or stops) timed spectral data acquisition as currently defined by the current start/end dates and time periods.
Take A Reference / Target Scan Now	Pressing these button will cause immediate acquisition of either a reference or target scan.
Hide / Show Settings	Pressing this button will shrink / expand the Long Term Timer dialog box; this is useful to see graphs in the main window during timed acquisition.
Ignore Scan Warnings	Checking this box will cause the software to ignore warning that would normally cause time acquisition to halt.
Always Use Default SIG File Name	Checking this box will cause the software to <u>ignore</u> the currently set SIG file base name; instead, the file base name format:

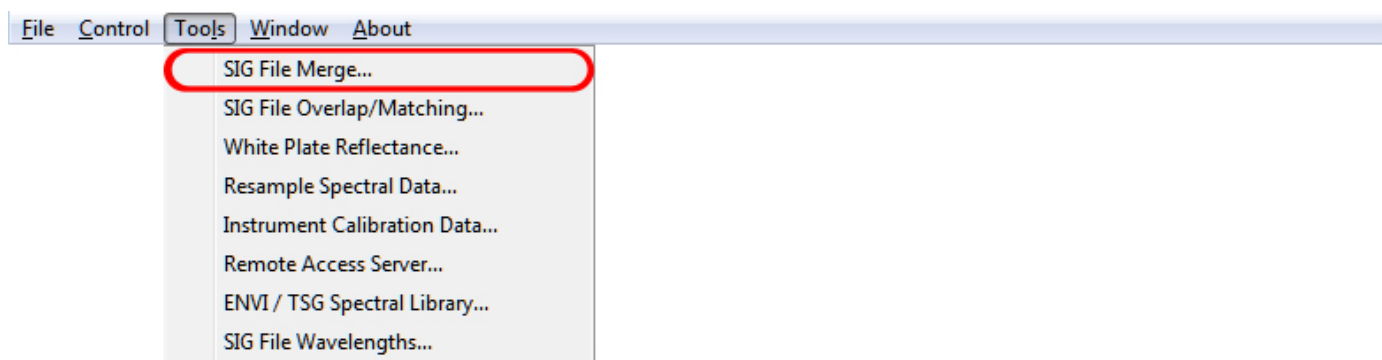
“grMMddYY” will be used instead. In this format, MM = current month, dd = current day of the month, YY = year.

Tools Menu

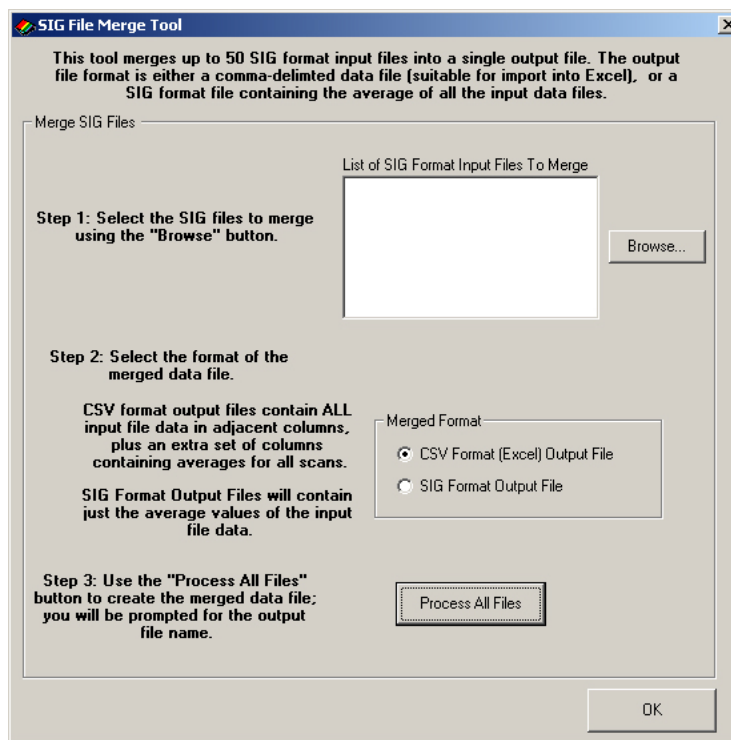
The Tools item provides a menu of useful tools for manipulating spectral data, or examining the instrument calibration tables.

SIG FILE MERGE...

Merges multiple SIG files into a single comma-delimited text data file, suitable for import into Excel or other analysis packages. Use this tool to save time when using a third-party analysis tool to examine multiple SIG files simultaneously.



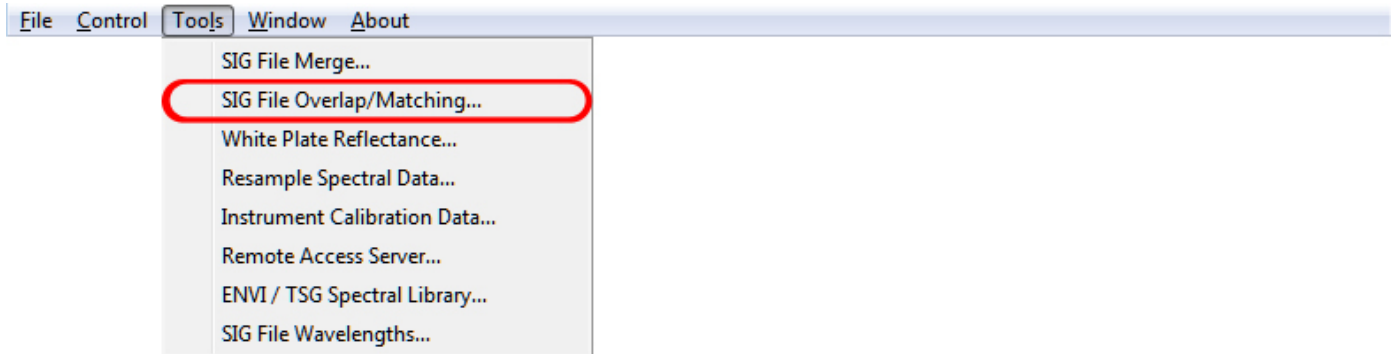
When “SIG File Merge...” is selected, the software presents the following dialog box.



List Of SIG Format Input Files:	Use the Browse button to select the SIG format files to merge.
Merge Format:	Determines whether the individual SIG files will be merged into a Comma Separated Values (CSV) suitable for import into MS Excel, or into a single SIG output file. When CSV Format is chosen, the individual SIG files appear side-by-side (across columns) in the output CSV file. In addition, another set of columns is added, in the rightmost columns, containing: • The wavelengths • The average of all Reference columns • The average of all Target columns • A calculated Reflectance, that is the average Target divided by the average Reference When SIG Format is chosen, the resulting SIG file contains: • The wavelengths • The average of all Reference columns • The average of all Target columns • A calculated Reflectance, that is the average Target divided by the average Reference
Process All Files	Begin processing the SIG files; the user is prompted for the single output file name for the merged data.

SIG FILE OVERLAP/MATCHING...

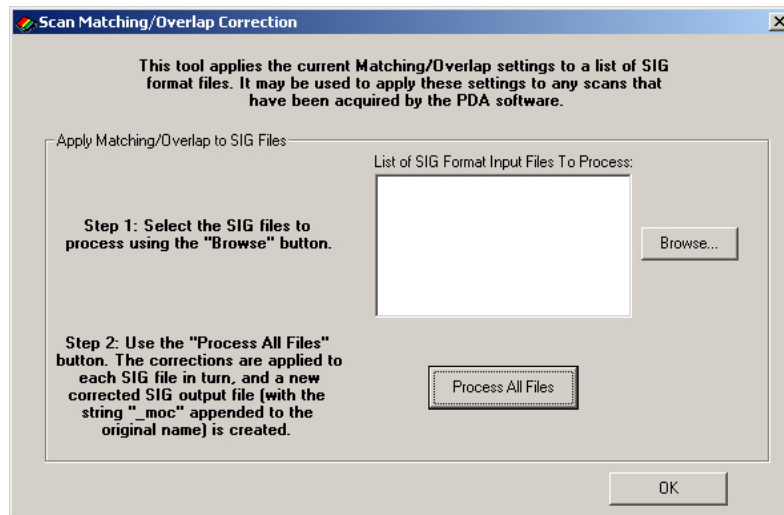
Applies the current overlap and matching settings to a set of input SIG format files; note that this option is not available in the SVC HR-512i software.



SIG data files that are acquired by the PDA Software do not have overlapping data removed, and have not had any matching algorithm applied. Use this tool to remove overlap and possibly apply detector matching algorithms.

Note that it is NOT necessary to use this tool on SIG file data acquired directly by the PC Data Acquisition Software; it is also NOT necessary to use this tool on SIG file data that was downloaded from the instrument's internal memory. In both cases, the current matching/overlap settings may have already been applied when the scans were acquired/downloaded.

When "SIG File Overlap/Matching..." is selected, the software presents the following dialog box:



List Of SIG Format Input Files: Use the Browse button to select the SIG format input files to process.

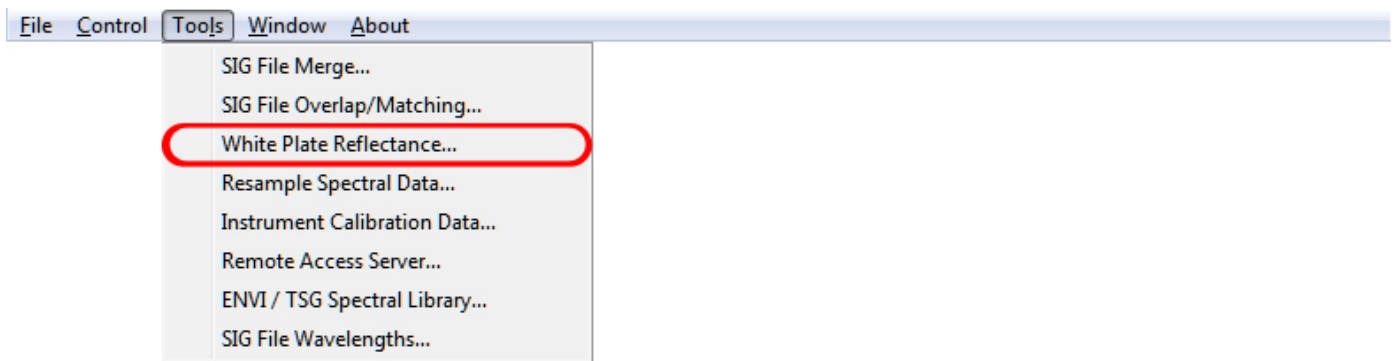
Process All Files Begin processing the SIG files; each SIG file in turn is read and processed, and the

modified data is written to an output file with the string “_moc” (Matching Overlay Correction) appended to the original name.

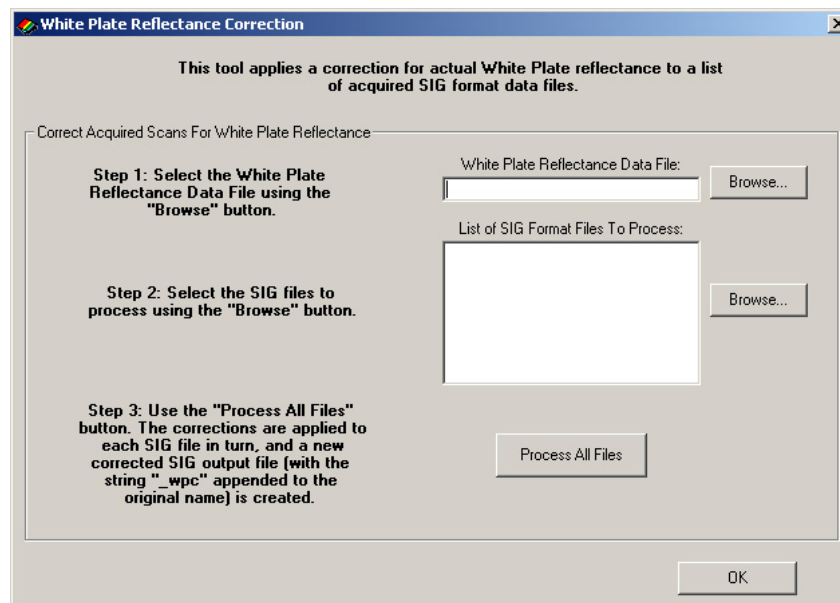
For example, if the input SIG file name is: “gr062606.sig”, then the output SIG file name is: “gr062606_moc.sig”

WHITE PLATE REFLECTANCE...

Corrects for non-ideal white (reference) plate reflectance.



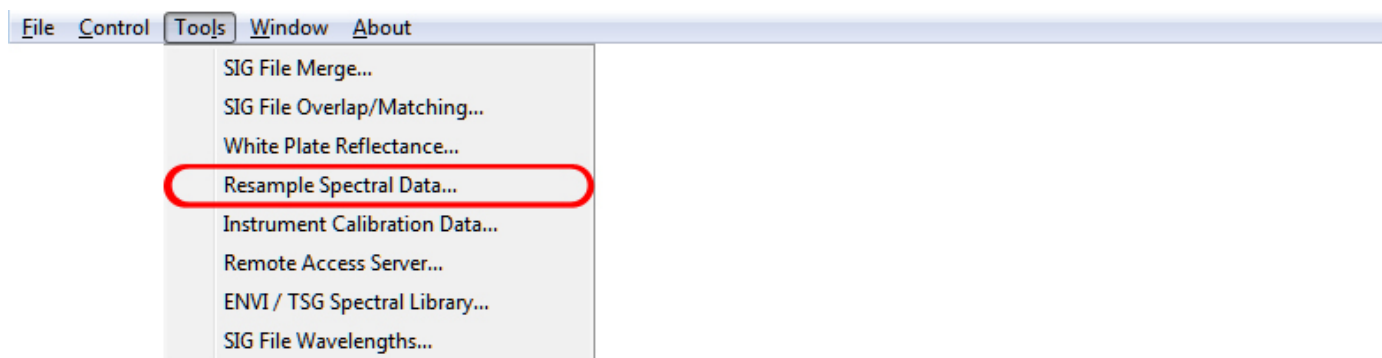
When “White Plate Reflectance...” is selected, the software presents the following dialog box:



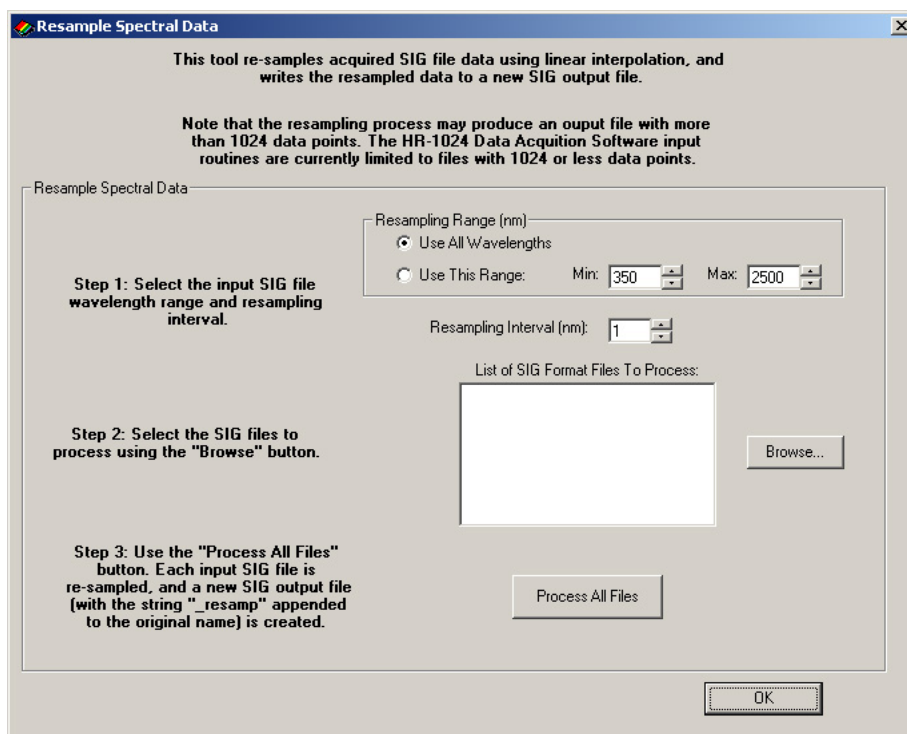
White Plate Reflectance Data File:	Use the Browse button to select the calibration file that corresponds to the white plate used to take the SIG file reference data. The White Plate Reflectance Data Files is a two-column text file. The first column is a list of wavelengths (in nm), and the second column is the actual white plate reflectance at that wavelength. See Appendix B – File Descriptions for a description of this file).
List Of SIG Format Input Files:	Use the Browse button to select the SIG format input files to process.
Process All Files	Begin processing the SIG files; each SIG file in turn is read and processed, and the modified data is written to an output file with the string “_wpc” (White Plate Correction) appended to the original name. For example, if the input SIG file name is: “gr062606.sig”, then the output SIG file name is: “gr062606_wpc.sig”

RESAMPLE SPECTRAL DATA...

Resample the input SIG format data files, and write the resampled data to new output SIG format data files.



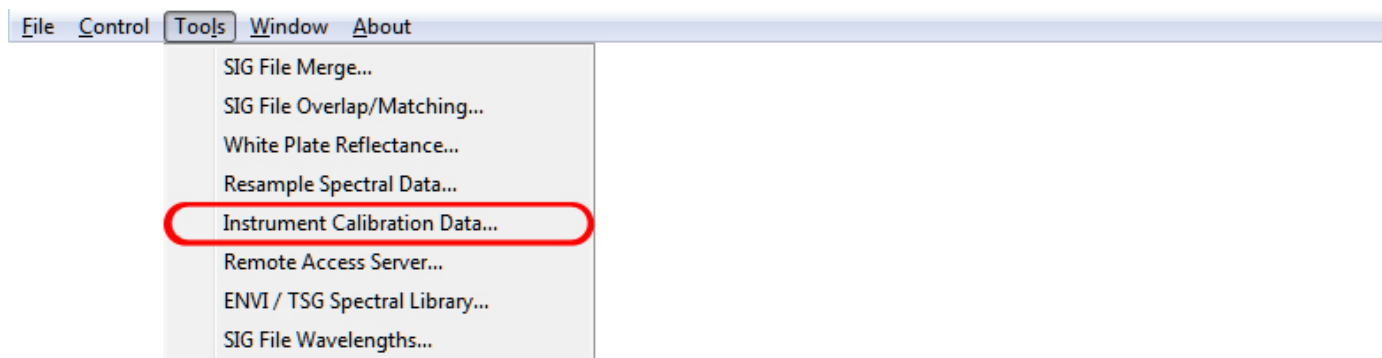
When “Resample Spectral Data...” is selected, the software presents the following dialog box:



Resampling Range	Select either all wavelengths present within the input SIG file, or a specific range of interest.
Resampling Interval	The interval used to perform the resample process.
List Of SIG Format Input Files:	Use the Browse button to select the SIG format input files to process.
Process All Files	Begin processing the SIG files; each SIG file in turn is read and is linearly re-sampled at the specified Resampling Interval, and the modified data is written to an output file with the string "_resamp" appended to the original name. For example, if the input SIG file name is: "gr062606.sig", then the output SIG file name is: "gr062606_resamp.sig"

INSTRUMENT CALIBRATION DATA...

Read, display, and save the calibration and other configuration data stored within the instrument.

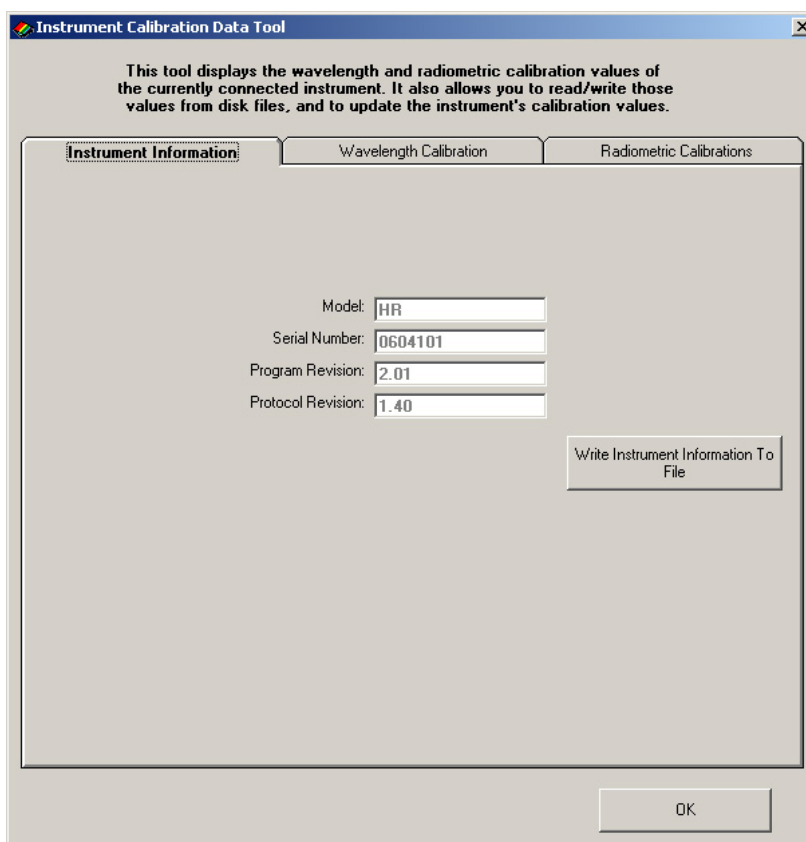


All instrument calibration information resides in non-volatile memory within the instrument. The Instrument Calibration Tool allows the operator to:

- View the connected instrument's calibration tables.
- Write the instrument's current calibration tables to files on the PC's hard disk.

When "Instrument Calibration Data..." is selected, the software presents the following 3-tabbed dialog box. Each tab on the dialog box allows the operator to view different information about the currently connected instrument.

Note that the information within these tabs will be blank if no instrument is currently connected (See the Control->Setup Instrument menu to connect to an instrument).



Instrument Information Tab: This tab simply displays the instrument's model and embedded firmware information.

Write Instrument Information To File:

The program will prompt the operator to save the displayed information to the PC's hard disk.

Instrument Calibration Data Tool

This tool displays the wavelength and radiometric calibration values of the currently connected instrument. It also allows you to read/write those values from disk files.

Instrument Information **Wavelength Calibration** Radiometric Calibrations

Calibration Date (mmddyy): 100412

Wavelength Calibration Table (xxx.x):

Band	Wvlen
1	338.3
2	339.8
3	341.2
4	342.7
5	344.1
6	345.6
7	347.1
8	348.6
9	350.0
10	351.5
11	353.0
12	354.5
13	355.9
14	357.4
15	358.9
16	360.3
17	361.8

Write Wavelength Calibration Table To File

Read Wavelength Calibration Table From File

Write Wavelength Calibration Table To Instrument

Set Instrument Calibration Password OK

Wavelength Calibration Tab: This tab displays the wavelength associated with each of the instrument's 1024-band wavelength calibration table. The date of the instrument's last calibration is also displayed in the "Calibration Date" text box.

Each button on this tab is described below:

Write Wavelength Calibration Table To File:

The program will prompt the operator to save the wavelength calibration table to the PC's hard disk. The actual calibration table within the instrument is **not** changed by this operation.

Read Wavelength Calibration Table From File:

The program will prompt the operator for a file name on the PC's hard disk that contains a wavelength calibration table, and display that information. The actual calibration table within the instrument is **not** changed by this operation.

Write Wavelength Calibration Table To Instrument:

The program will prompt the operator to upload the current wavelength calibration table to the connected instrument.

Note that this button will *only* appear if a) the wave cal file was just read in from disk, and b) the wave cal file has the correct corresponding “key” file associated with the “cal” file. Contact Spectra Vista for further information.

This tool displays the wavelength and radiometric calibration values of the currently connected instrument. It also allows you to read/write those values from disk files.

Instrument Information Wavelength Calibration **Radiometric Calibrations**

Select Optic Number (1-8): 1

Optic Name (6 chars max): LENS 4

Optic Type: RADIANCE

Radiometric Calibration Table:

Band	Wvlen	Factor
1	338.3	2.7460000E-04
2	339.8	2.9720000E-04
3	341.2	3.1640000E-04
4	342.7	3.2260000E-04
5	344.1	3.3390000E-04
6	345.6	3.5020000E-04
7	347.1	3.6000000E-04
8	348.6	3.6370000E-04
9	350.0	3.7240000E-04
10	351.5	3.7490000E-04
11	353.0	3.6720000E-04
12	354.5	3.5600000E-04
13	355.9	3.5030000E-04
14	357.4	3.4150000E-04
15	358.9	3.1760000E-04
16	360.3	2.9600000E-04
17	361.8	2.8850000E-04

Write One Radiometric Calibration Table To File

Read One Radiometric Calibration Table From File

Write One Radiometric Calibration Table To Instrument

Set Instrument Calibration Password OK

Radiometric Calibration Tab:

There are up to 8 radiometric calibration tables (corresponding to up to 8 optional instrument fore-optics) stored within the instrument.

This tab displays each of these eight 1024-entry tables, one table at a time. The table that is currently displayed may be selected via the “**Select Optic Number**” control at the top of the tab. **Note that buttons that are used to**

read or write radiometric data only
operate on a single table at a time –
 the table currently selected by the “Select
 Optic Number” control.

Each fore-optic’s radiance calibration
 information has an “**Optic Name**”,
 which is a 6-character familiar name that
 is displayed on the instrument’s LCD.

Each fore-optic calibration also has an
 “**Optic Type**” – the radiometric
 calibration data is either a “Radiance”
 mapping, a “RAW” (DN) mapping, or an
 “Irradiance” mapping.

Each button on this tab is described
 below:

Write One Radiometric Calibration Table To File:

The program will prompt the operator to
 save the currently selected (1-8)
 radiometric calibration table to the PC’s
 hard disk. The actual calibration table
 within the instrument is **not** changed by
 this operation.

Read One Radiometric Calibration Table From File:

The program will prompt the operator
 for a file name on the PC’s hard disk that
 contains a single radiometric calibration
 table, read that table into the currently
 selected table (1-8), and display that
 information. The actual calibration table
 within the instrument is **not** changed by
 this operation.

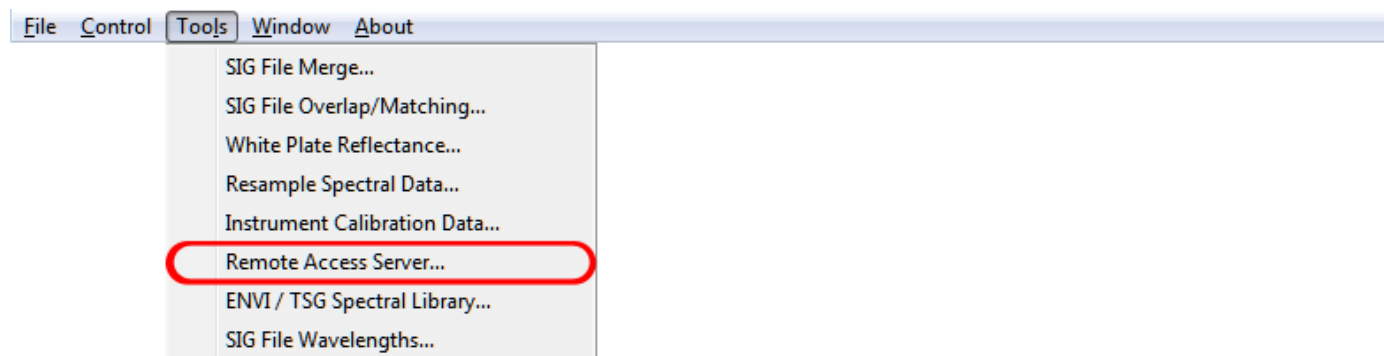
Write One Radiometric Calibration Table To Instrument:

The program will prompt the operator to
 upload the currently selected calibration
 table to the connected instrument.

Note that this button will **only** appear if
 a) the radiance cal file was just read in
 from disk, and b) the radiance cal file has
 the correct corresponding “key” file
 associated with the “cal” file. Contact
 Spectra Vista for further information.

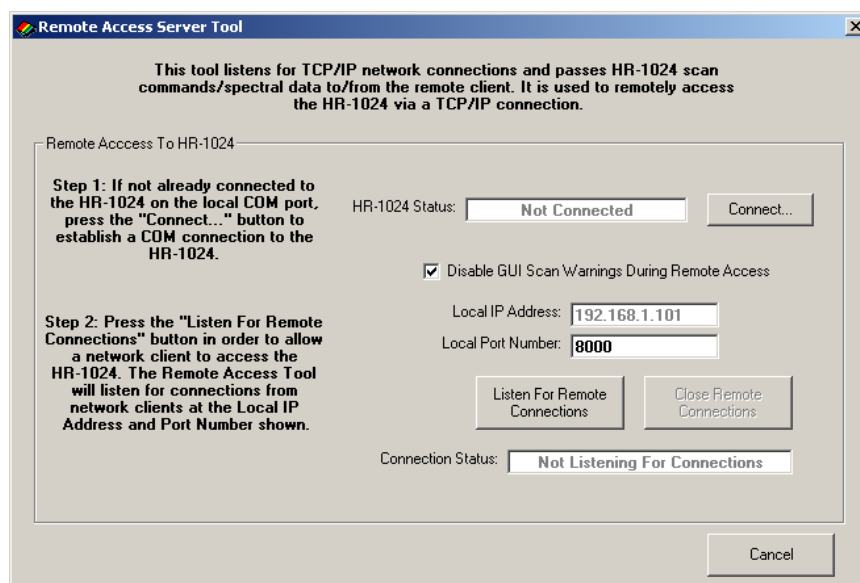
REMOTE ACCESS SERVER...

Allow remote network access to the instrument's control and data, using a TCP/IP client/server interface.



By using this tool, the PC Data Acquisition program becomes a network server that allows access from remote clients via a simple TCP/IP based protocol.

See Appendix H – Instrument Remote Access, for a description of the network protocol used to remotely access the instrument.



Connect...:

If the PC Data Acquisition program is not already connected to the instrument (via serial port, USB, or Bluetooth), use this button to bring up the standard Control/Setup menu to create that connection.

If already connected, this button will be disabled.

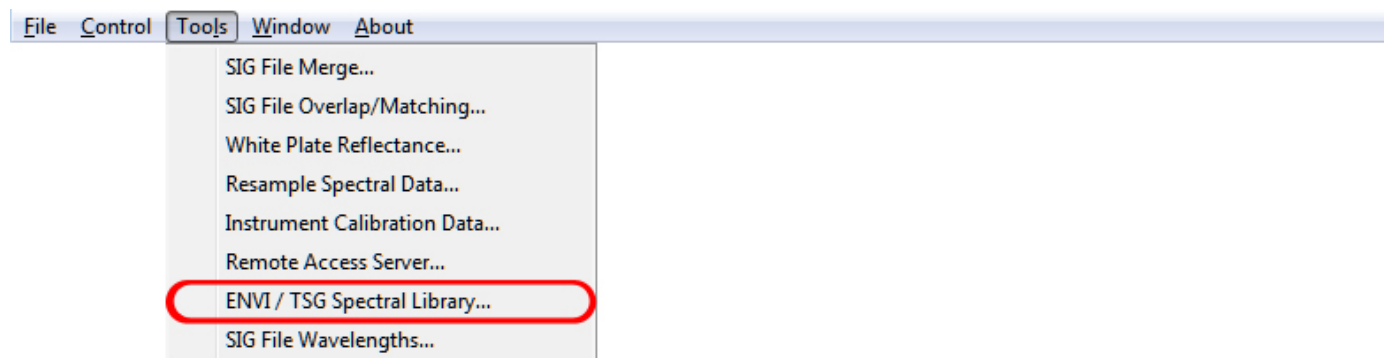
Listen For Remote Connections: Pressing this button causes the PC Data Acquisition program to act as a server, located on the network at the displayed IP Address and Port Number.

Close Remote Connections: Pressing this button causes the PC Data Acquisition program to close down its server function and stop listening for client connections.

SVC XHR-1024i / HR-1024i / HR-768i / HR-640i / HR-768si / HR-512i USER MANUAL REV 1.17

ENVI / TSG SPECTRAL LIBRARY...

Used to create ENVI® / TSG™ compatible Spectral Libraries from SIG format data files.



This tool reads SIG format files from disk and combines them into an ENVI format Spectral Library.

The ENVI library created by this tool may subsequently be read by either the Harris Geospatial's ENVI® image analysis software, or by CSIRO's The Spectral Geologist (TSG™) software tool.

The Spectral Library may be created from the SIG files' reflectance, target, or reference data. Each SIG file can optionally be associated with a familiar "ENVI Spectra Name", which is included within the ENVI Spectral Library.

Note that since ENVI Spectral Libraries only support a single wavelength table, **all SIG format files selected for inclusion into a library must have identical wavelength tables**. So, for example, SIG files with overlap preserved cannot be combined with SIG files with overlap removed within a single ENVI Spectral Library.

When "ENVI Spectral Library..." is selected, the software presents the following dialog box:

SIG File ENVI / TSG Spectral Library Tool

This tool creates an ENVI Spectral Library from SIG format files. A maximum of 256 SIG files may be included in the library. The SIG data file can contain a maximum of 1024 data points. In addition, all SIG files must have the same wavelength vector.

ENVI libraries created with this tool may also be imported into CSIRO's The Spectral Geologist (TSG) software for further analysis.

Create ENVI Spectral Library

Step 1: Select the SIG files to include in the spectral library using the "Browse" button.

SIG File Name	ENVI Spectra Name

Step 2: If needed, click / edit the "ENVI Spectra Name" field above to describe each spectra; otherwise, the "SIG File Name" field itself will be used for the Spectra Name.

Step 3: Select which data from the SIG file to include in the spectral library.

Step 4: Enter the Spectral Library description.

Step 5: Use the "Create Library" button to create the ENVI Spectral Library file; you will be prompted for the output file name.

Browse:

Pressing this button presents the normal "File Open" selection dialog, which is used to select the SIG format files that will be included in the Spectral Library.

Once selected, the SIG files are listed in the "SIG File Name" column of the text grid.

A descriptive name for the spectra found in each SIG file may optionally be entered by clicking on the corresponding box in the "ENVI Spectra Name" column and typing a descriptive name.

If no "ENVI Spectra Name" is entered, then the "SIG File Name" will be used to name the spectra within the ENVI Spectral Library.

SIG File Data:

Selects whether the reflectance, target, or reference data columns from the SIG file is used to create the ENVI Spectral Library.

Spectral Library Description:

Enter the ENVI Spectral Library description into this box. This name will become the name of the Spectral Library within the ENVI application.

Create Library:

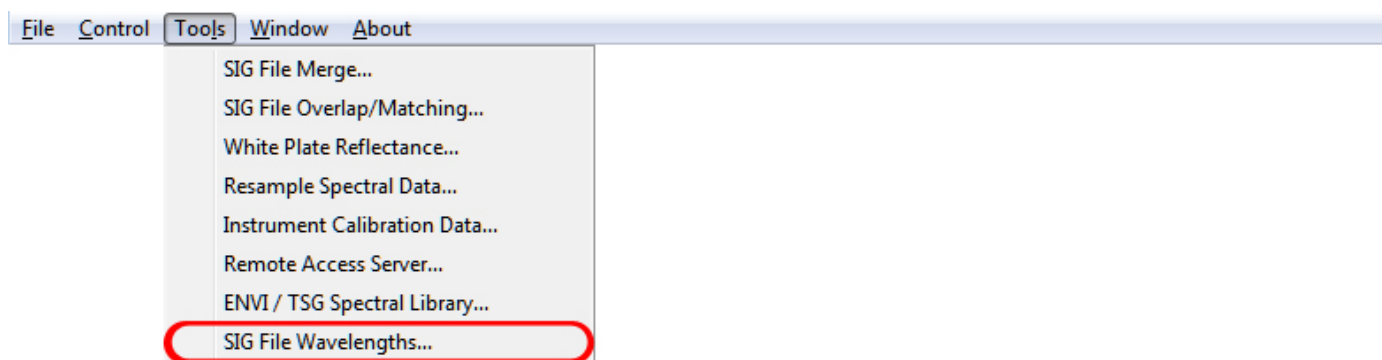
Pressing this button causes the software to create the ENVI Spectral Library header and data files.

OK:

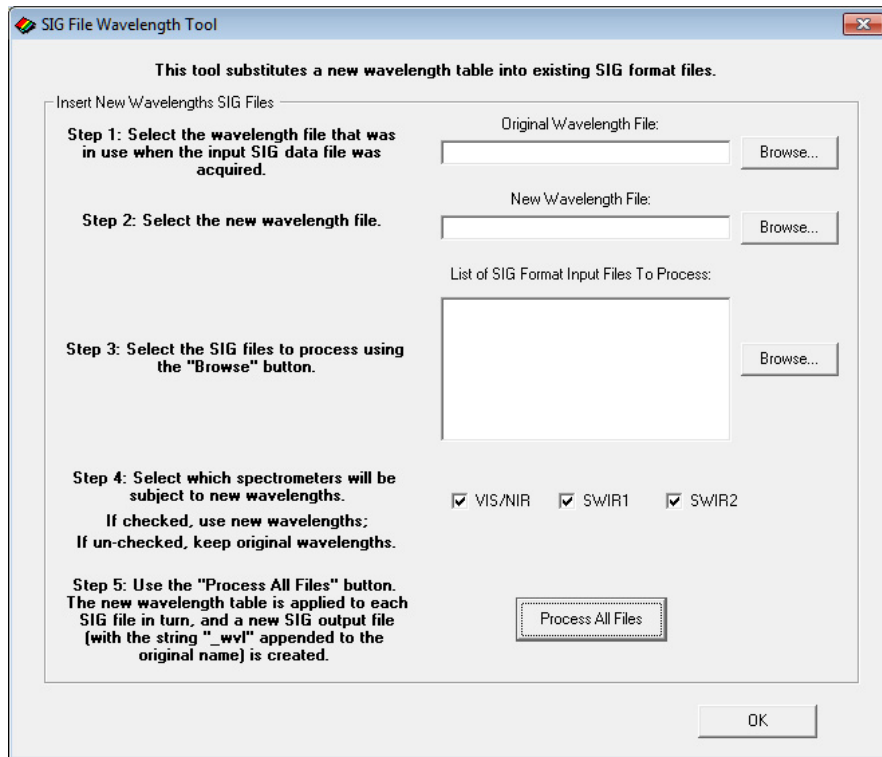
Pressing this button dismisses the dialog and returns control back to the main window.

SIG FILE WAVELENGTHS...

This tool may be used to insert a new wavelength vector into a set of existing SIG files.



When “SIG File Wavelengths...” is selected, the software presents the following dialog box:



Original Wavelength File

This is the wavelength calibration file that was installed in the instrument at the time that the SIG files were acquired.

New Wavelength File

This is the wavelength calibration file that should be substituted in place of the original wavelength file.

List Of SIG Format Input Files To Process

Use the Browse button to select the SIG format input files to process with this tool.

Spectrometers List

Use these check boxes to control which of the spectrometer's data within the SIG files should be subject to the new wavelengths.

Process All Files

Begin processing the SIG files; each SIG file in turn is read and processed. The new data containing the new wavelength vector is written to an output file with the string "_wvl" appended to the original name.

For example, if the input SIG file name is: "test.sig", then the output SIG file name is: "test_wvl.sig".

Window Menu

The Window item provides a menu of available window displays; it allows the operator to switch back and forth between the two types (Single- and Multi-Graph) displays, and allows the operator to control some characteristics of the graph (line width, data markers, etc.)

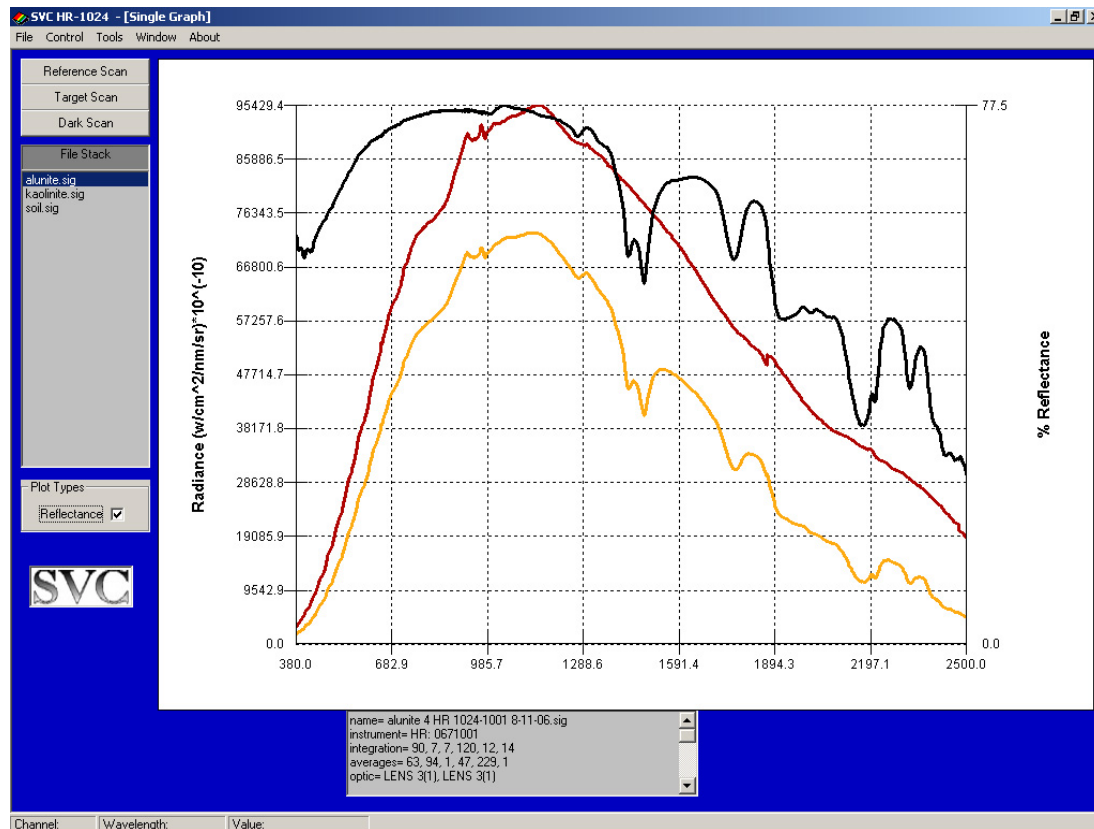
SINGLE GRAPH

Displays the currently selected file in a Single Graph.



A single measurement's Reference and the Target Radiance are scaled against the left Y-axis and the Percent Reflectance (Target/Reference *100) is scaled against the right Y-axis.

An example of this is shown below:



The graphs shown in the example above are:

Reference Radiance (Red)
 Target Radiance (Yellow)
 % Reflectance (Black)

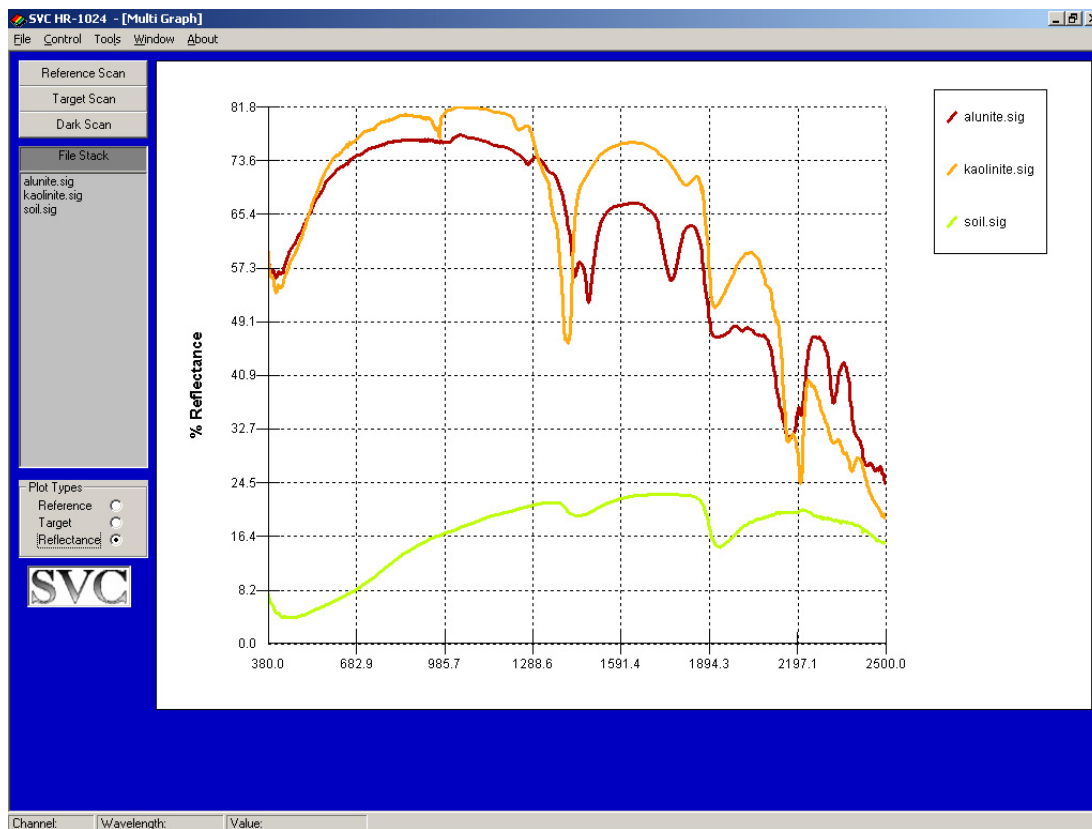
Each of these is displayed as a function of wavelength. The text box in the lower portion of the window displays the file header information.

MULTI GRAPH

Displays the current list of open files together in one graph. Files can be displayed as reference radiance, target radiance or percent reflectance.



The following is an example of a Multi Graph:



The Multi Graph displays the Target Radiance of the files in the File Stack versus wavelengths. The Multi Graph can display a maximum of 24 scans.

SHOW TARGET JPEGs



When the Show Target JPEGs option is checked, the target photo associated with a scan (if it exists) will be displayed along the target's spectral data.

In order to be displayed, the target photo must have the same file name as the target SIG file, with the extension “.jpg” appended. For example, if the target SIG file is named “jarosite.sig”, then the associated target photo must have the name “jarosite.sig.jpg” in order to be automatically displayed.

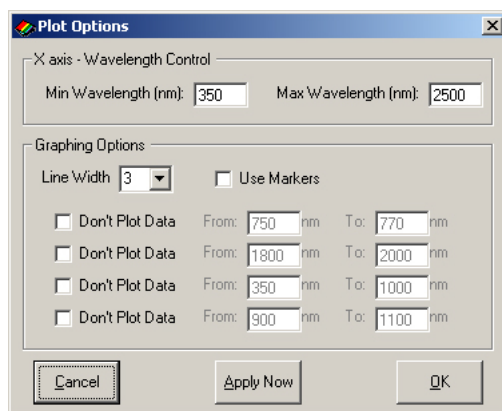
The target photo will be displayed any time the target SIG file is displayed in Single Graph Mode, i.e., when the target SIG file is double-clicked in the file stack.

PLOT SETTINGS ...

Configures the default X-axis, line width, and marker usage. You may also use this menu to exclude certain areas of the plot – for example, to exclude water absorption areas of the plot.



When Plot Settings is selected, the following dialog appears:



X-Axis – Wavelength Control Frame

Controls the default extent of the X-Axis of the plot.

Min Wavelength Sets the default minimum X-axis wavelength to use when displaying spectral data.

Max Wavelength Sets the default maximum X-axis wavelength to used when displaying spectral data.

Graphing Options Frame

Controls several feature of the plot itself.

Line Width Sets the relative width of the line used to graph spectral data.

Use Markers If checked, each data point plotted is marked by a square box.

Don't Plot Data / From / To If the Don't Plot Data checkbox is checked, the From / To text boxes define one area of the plot that will be excluded from display. The From / To boxes should contain valid wavelengths (in nanometers). Note that the check boxes affect only the displayed plot, they do not affect acquired data stored in SIG files.

Cancel Exits this dialog without changing settings.

Apply Now Applies changes and keeps the dialog open.

OK Applies changes and exits the dialog.

LARGER BUTTONS

Increases the size of the Reference, Target, etc. buttons.



When checked, the Larger Buttons option increases the size of the Reference, Target, Dark, and External Data Scan buttons. This option is useful on tablet PCs with a touch screen, in order to make it easier to press the on-screen buttons.

About Menu

The About menu item displays a dialog box that provides model, the current serial number, as well as the PC Data Acquisition Software version. Note that the Instrument Serial Number field can only be displayed after the PC Data Acquisition Software has successfully connected to an instrument.



When About is selected, the following dialog box appears:



Note that your software version may be different from this example. The “PROGRAM CHANGE LOG” contains a list of recent changes to the PC Data Acquisition Software in reverse chronological order.

Section 3 – Operation

1. Setup

Battery Pack

To use the instrument, first install a fully charged battery pack. This is done by sliding a charged battery pack into the battery slot located on the bottom of the instrument. Push in far enough to latch it in place. When in position, 6 copper contacts on the battery pack come in contact with 6 mating contacts on the bottom of the instrument, enabling the battery pack to power the instrument.

Instrument Power-on Time

When initially turned on the instrument displays its model number on the initial LCD screen. During this initial power-on period the instrument is waiting for the thermal-electric coolers to reach operating temperature before data collection can begin. This period lasts less than a minute and when complete the “Ready” light on the control panel comes on.

Instrument Warm-up Time

For optimal performance, the instrument should initially be turned on for a period of approximately 15 minutes prior to first use. During this period, the instrument should be placed in the environment where scanning will take place. This period allows the detector temperatures and other electronics to stabilize before taking spectral data.

Instrument Orientation

The instrument should be setup with its cooling slots on the bottom oriented as close as possible to vertical. Also the bottom of the instrument should be oriented away from the sun in order to maximize internal heat dissipation.

Laser (Tripod or Hand-held)

A sighting laser is used to aid in aligning the instrument to the target to be measured. The reference or sample (target) should be placed to completely fill the field of view of the instrument. The sighting laser can be actuated by pressing the laser scan switch on the top of the instrument.

CAUTION: Do NOT look into the sighting laser window or laser beam.

Field of View

A field of view diagram is provided with each instrument and shows the relative position of the sighting laser spot and the active field of view for the instrument at the distance specified on the diagram (normally 1m). If optional entrance optics are included, the corresponding FOV diagrams will also be supplied. (See Appendix Z for inserted hardcopies of FOV diagrams supplied with your instrument, when applicable).

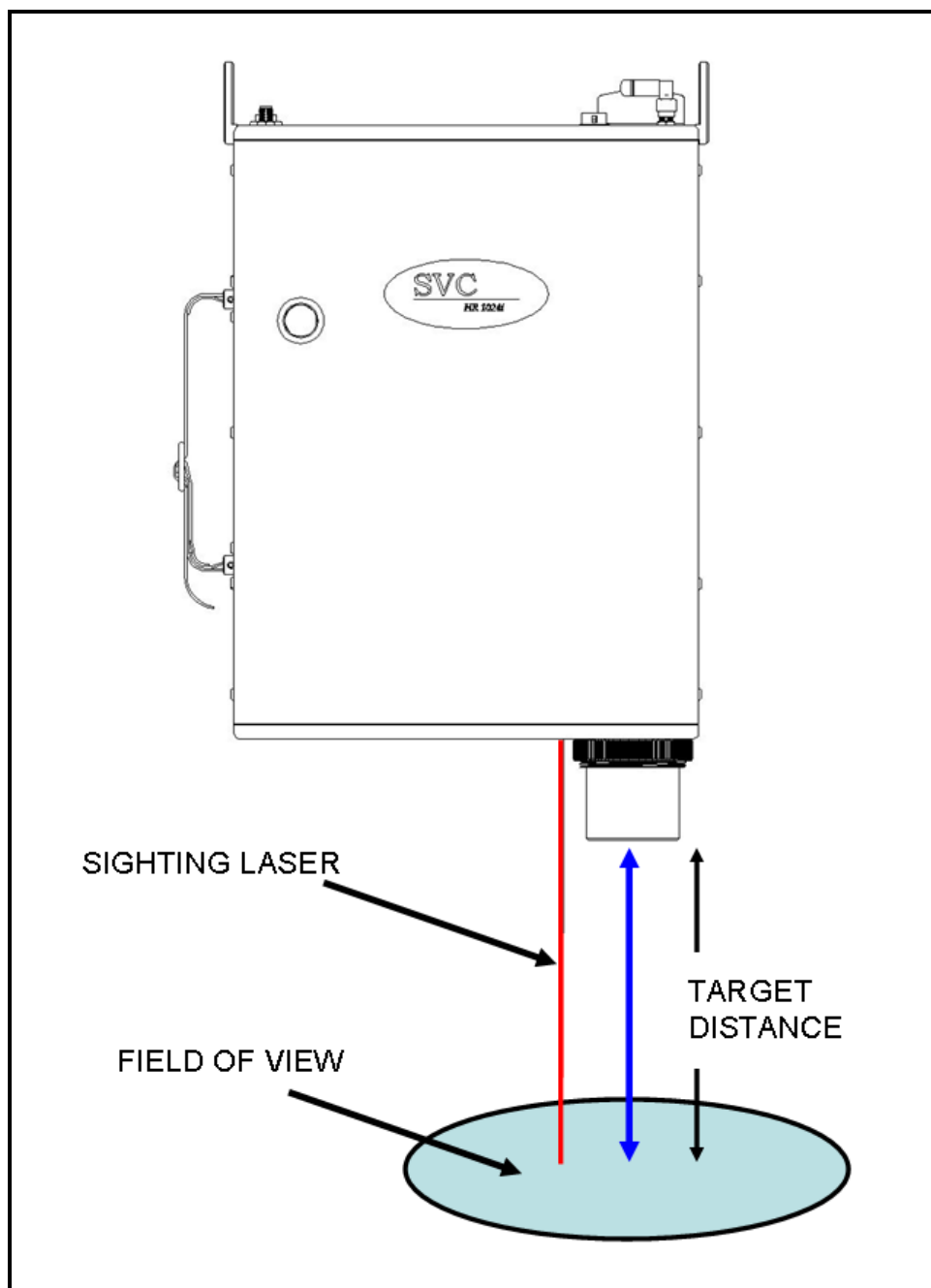


Figure 6 Setup for FOV map

CAUTION: Do NOT look into the sighting laser window or laser beam.

2. Connecting The Instrument To A Computer

Prior to downloading spectral data stored within the instrument, and prior to using the PC Data Acquisition Software to directly acquire spectral data, the instrument must first be connected to a (physical or virtual) port on a Windows PC. This connection can take place in one of two ways:

- By physically connecting the PC's USB connector to the instrument's USB-B connector (on the bottom plate) using a standard USB cable. Installation of special device drivers (supplied with the instrument) on the Windows PC is required. See Appendix E – PC USB Device Drivers for installation instructions.
- By creating a Bluetooth “virtual” COM port connection between the computer and the instrument. This option requires installation of third-party Bluetooth hardware and drivers (not supplied with the instrument) on the Windows PC.

3. Stand-Alone Data Acquisition

Operation Via The LCD Display

The LCD touch screen display on the control panel allows for stand alone operation of the instrument, allowing for the acquisition and viewing of spectral data.

The display consists of several screens with the following functions:

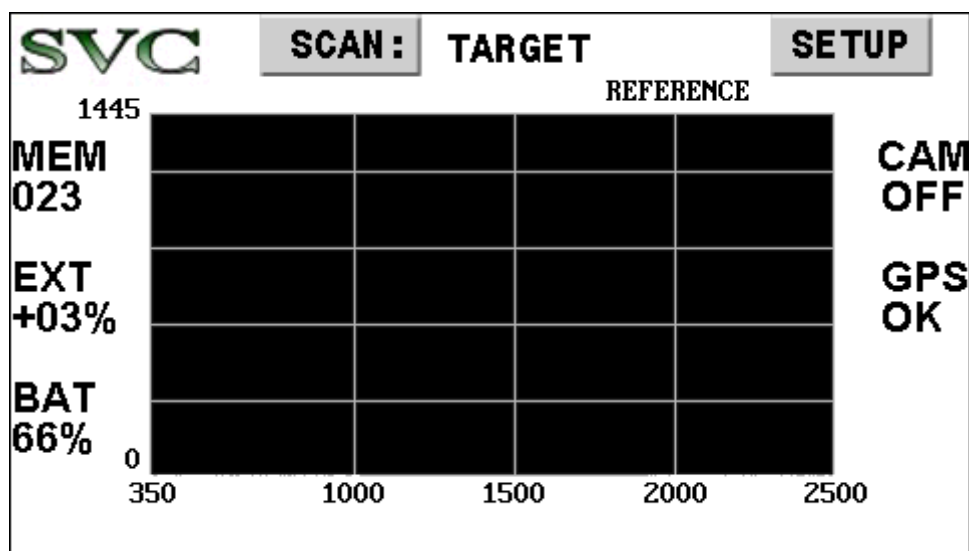
LCD SCREEN 0 - Power-On Screen

This screen appears for 10 seconds after power-on, and includes:

- o **INSTRUMENT MODEL/SERIAL NUMBER**
- o **FIRMWARE VERSION**
- o **LAST CALIBRATION DATE**

LCD SCREEN 1 - Main Screen

This is the main screen that is typically displayed during stand-alone operation:



- o **SETUP** BUTTON
Touch **SETUP** to go to the LCD Screen #2 to change various scan settings.
- o **SCAN:** BUTTON
Touch **SCAN** to choose the type of scan (Reference or Target) that will be acquired the next time the manual trigger button is pressed
- o **CURRENT SETTINGS / STATUS** AREAS – shows current instrument operating mode and status:

- **BAT** - Battery level – 0 to 100%
- **MEM** - Most recent internal scan memory slot used – 000 (empty), 001 through 999.
- **GPS** - The current GPS receiver mode/status:

OFF	Turned off.
N/L	Turned on, but no position information is available.
OK	Turned on, position valid.

- **CAM** - Current internal JPEG camera mode:

OFF	Turned off.
N/S	Turned on, but no sync with the camera.
LO	Target JPEGs will be acquired in low-resolution.
HI	Target JPEGs will be acquired in high-resolution.

- **EXT** – The current External Data interface mode/status:

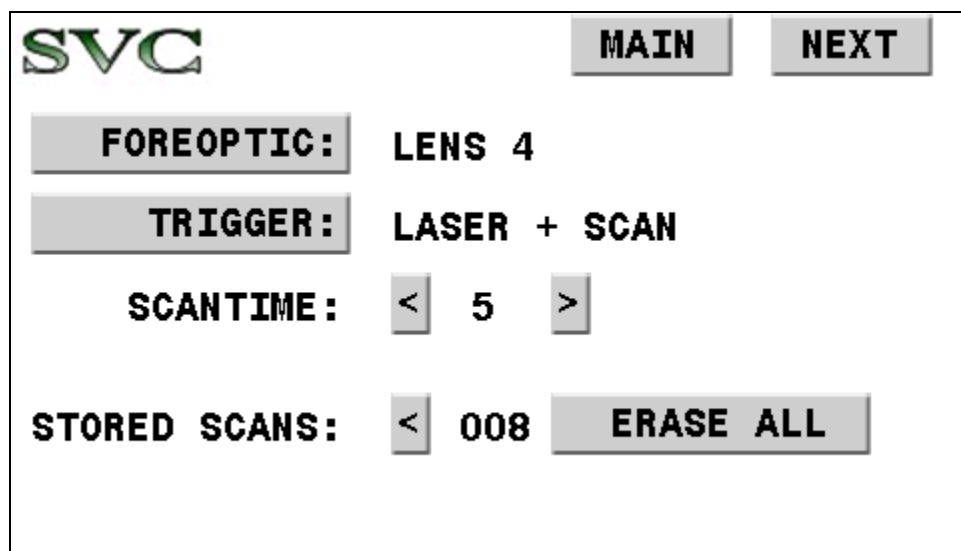
OFF	Turned off.
N/C	Turned on, but no Bluetooth connection established.
N/S	Turned on, Bluetooth connected, but no valid data.
REF	Valid data is being received, awaiting a reference scan; a reference scan will establish the baseline for calculating percent change in the currently selected external data channel.
XX%	Valid data is being received, and we've acquired Reference

	data. This field is now displaying the percent change in the currently selected external data channel when compared to the reference, in the range -99% to +99%.
--	--

- o **LAST ACQUIRED DATA GRAPH** AREA – a small graph showing last acquired scan:
 - For a Reference scan – always displays radiance.
 - For a Target scan – displays either radiance or reflectance (depending on settings)
 - **Touching any area of the graph** cycles you between displaying target radiance and target reflectance; the graph type that is currently displayed is called out just above the upper right corner of the graph area.
- o **GENERIC STATUS / WARNING / ERROR TEXT** LINE – a single line of text at the bottom of the main screen which displays status, warning and/or error messages.

LCD SCREEN 2 – Spectrometer Setup

This setup screen controls various spectrometer settings:

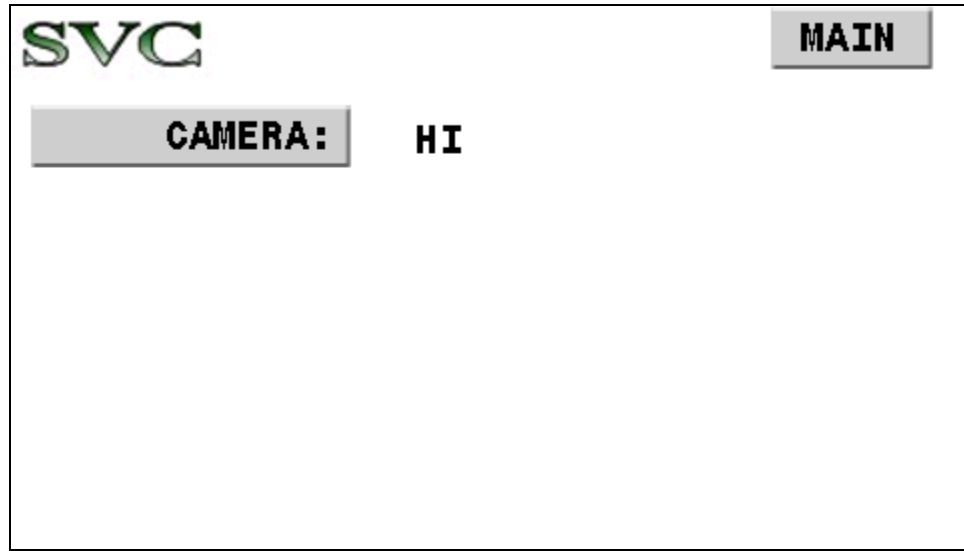


- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **NEXT** BUTTON:
Touch **NEXT** to continue on to LCD Screen #3.
- o **FOREOPTIC** BUTTON:
Touch **FOREOPTIC** until the foreoptic that is currently installed on the instrument appears – 4 degree lens, fiber optic, etc.
- o **TRIGGER** BUTTON:
Touch **TRIGGER** to control the action performed when the LASER/SCAN trigger button is pressed:

LASER ONLY	Laser is lit while the trigger is pressed.
LASER + SCAN	Laser is lit while the trigger is pressed; in addition, a stand-alone scan is acquired when the trigger is released.
SCAN ONLY	A stand-alone scan is acquired when the trigger is released
- o **SCANTIME** BUTTONS:
Touch the < or > button to decrease or increase the instrument scan time.
- o **STORED SCANS** BUTTONS:
 - Touch the < button to remove just the most recently acquired stand-alone scan.
 - Touch the **ERASE ALL** button to permanently remove all internal stored scans; once this button is pressed, a confirmation button will appear before the scans are actually removed from memory.

LCD SCREEN 3 – JPEG Camera Control

This setup screen controls the instrument's internal JPEG camera.

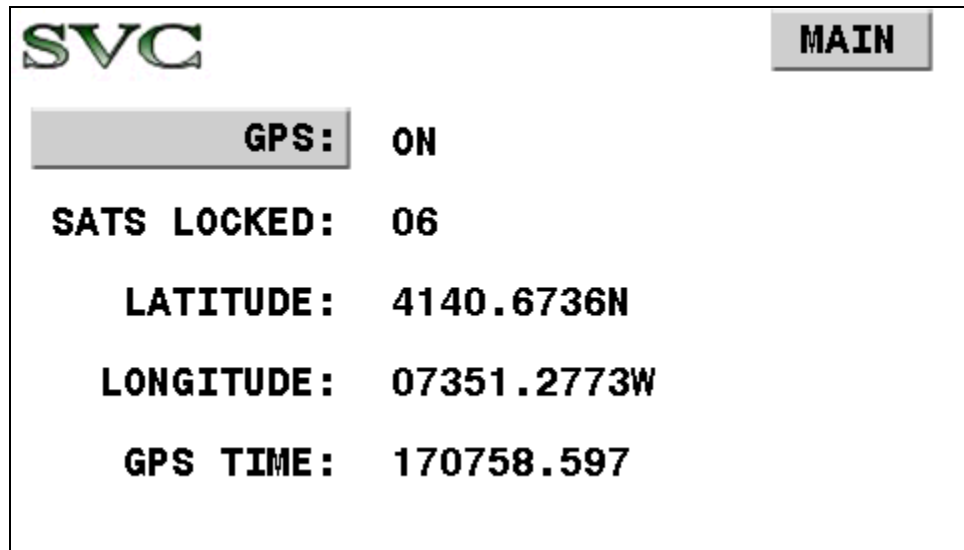


- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **NEXT** BUTTON:
Touch **NEXT** to continue on to Screen #4 (Setup Screen C).
- o **CAMERA** BUTTON:
Touch **CAMERA** to cycle among the internal JPEG camera target scan settings:

OFF	No photos are taken.
LO	Low resolution (320x240) photos are taken with each Target scan.
HI	High resolution (640x480) photos are taken with each Target scan.

LCD SCREEN 4 – GPS Control

Controls the GPS and displays the current GPS location and status:



- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **NEXT** BUTTON:
Touch **NEXT** to continue on to Screen #5 (Setup Screen D).
- o **GPS** BUTTON:
Touch **GPS** to turn the internal GPS function ON/OFF.
- o **GPS** STATUS FIELDS:
 - **SATS LOCKED**: Number of satellites currently being used to compute position.
 - **LATITUDE/LONGITUDE/GPS TIME**: Current position and GPS reported time stamp; these fields will remain blank until a position solution has been computed.

LCD SCREEN 5 – External Data Control

This screen controls the External Data Interface function:

SVC **MAIN**

EXTERN DATA: ON

BT ADDRESS: 646E6CC0B62F

STATUS: CONNECTED, DATA OK

DISPLAY CHAN: 1 [11791]

DARK CORR: ON

TAKE DARK **SAVE SETTINGS**

- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **NEXT** BUTTON:
Touch **NEXT** to continue on to Screen #6 (Setup Screen E).
- o **EXTERN DATA** BUTTON:
Touch **EXTERN DATA** to turn the External Data Interface function ON/OFF. When turned ON, the instrument will attempt to actively connect via Bluetooth to an External Data Interface.
- o **BT ADDRESS** FIELD:
Displays the 12-digit hexadecimal Bluetooth address of the External Data Interface that is currently configured into the instrument. This address determines the source of external data acquired by the instrument during spectral scans. This field may only be changed via the PC Software. See section **SETUP EXTERNAL DATA ...** for more information on how to change this field.
- o **STATUS** FIELD
Displays the current state of the Bluetooth connection between the instrument and the External Data Interface:

NOT CONNECTED	The interface is turned off, or the interface is turned on but no Bluetooth connection can be established to the external data source.
CONNECTED, NO SYNC	The interface is connected to the external data source via Bluetooth, but the data source is not sending correctly formatted data.
CONNECTED, DATA OK	The interface is connected and receiving correctly formatted data.
- o **DISPLAY CHAN** BUTTON

Touch **DISPLAY CHAN** to select the channel (1 to 16) whose real-time data values are displayed in **[BRACKETS]** ; this allows the operator to monitor external data values for each channel. This is also the channel selected to compute the percent change (Target to Reference) found in the EXT field on the LCD Main Screen.

o **DARK CORR** BUTTON

Touch **DARK CORR** to turn external data dark correction ON/OFF. When turned on, the most recent dark values are subtracted prior to computing the percent change found in the EXT field on the LCD Main Screen.

o **TAKE DARK** BUTTON

Touch **TAKE DARK** in order to store the most recently received set of external data samples as dark values. The dark values are stored in non-volatile memory. When this button is pressed, it is assumed that the external sensor has been configured to be in a “dark” condition. Note that when new dark values are taken, the current set of external data reference values are invalid; a new reference scan is required before the LCD Main Screen will begin to show Target/Reference percent change values.

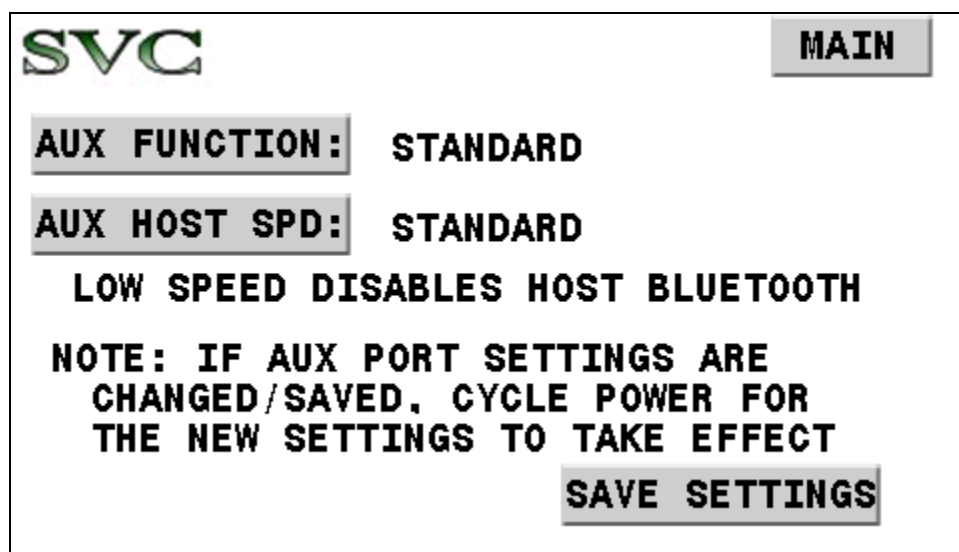
o **SAVE SETTINGS** BUTTON

Touch **SAVE SETTINGS** to save all external data settings to non-volatile memory. When the instrument is re-powered, the most recently saved settings are restored, including the most recently acquired dark values.

LCD SCREEN 6 – AUX Port Settings

This screen controls the function of some pins on the instrument's AUX Port functions; see Appendix K – Aux Port Specification - for more information about the standard functions provided by the round AUX port connector located on the instrument's control panel.

For normal Aux Port usage, the operator should leave both the AUX FUNCTION and AUX HOST SPEED values set to the default "STANDARD" setting. Contact Spectra Vista Corporation for further information regarding special non-standard AUX Port settings:



- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **AUX FUNCTION** BUTTON:
Touch **AUX FUNCTION** to cycle among the possible uses for the round AUX port located on the spectrometer's control panel:

STANDARD	The AUX port's pins 3 thru 6 are inputs to the spectrometer's Trigger function. This is the standard configuration for these pins – Do Not Change this setting, it should only be changed for special purpose applications.
STEPPER	The AUX port's pins 3 thru 6 are configured as outputs which may control an external device.

- o **AUX HOST SPD** BUTTON:
Touch **AUX HOST SPD** to cycle among the possible speeds of the RS-232 host communications interface which is located on the round AUX port connector.

STANDARD	The RS-232 speed is standard (115200 baud), which is compatible with the default baud rates used by the PC Data Acquisition Software and the PDA software.
LOW SPEED	The RS-232 speed is lowered (9600 baud) for applications that may use long communication cables back to the controlling PC. <u>Note that choosing this option will also disable the instrument's Bluetooth host interface.</u>

	The PC Data Acquisition Software, by default, uses the standard 115200 baud connection speed; however, this may be over-ridden by right-clicking (instead of left-clicking) the “Connect” button within the software’s <i>Control->Setup Instrument</i> dialog. See the section Control Menu, SETUP INSTRUMENT... for more information.
--	--

o **SAVE SETTINGS** BUTTON

Touch **SAVE SETTINGS** to save all AUX port settings to non-volatile memory.

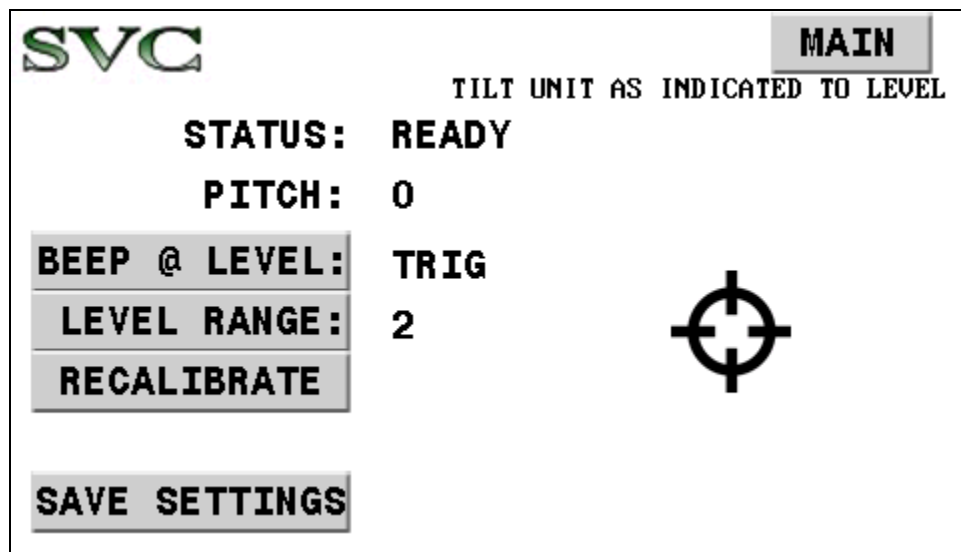
When the instrument is re-powered, the most recently saved settings are restored.

LCD SCREEN 7 – Inclinator

This screen contains controls and displays information related to the instrument's built-in inclinometer. The inclinometer informs the operator, visually and (optionally) audibly, when the instrument is being held level, with the foreoptic pointing straight downward at a calibrated nadir point.

Visually, the LCD screen uses graphic symbols to indicate levelness. The cross-hairs graphic  appears when the instrument is held level, while arrow graphics    will instead appear in order to indicate that the instrument is out-of-level, as well as to indicate the direction that the top of the instrument must be rotated in order to achieve level.

The inclinometer can optionally emit a tone when the instrument is held level. The tone may be emitted either only when the instrument's trigger button is held down, or continuously:



- o **MAIN** BUTTON:
Touch **MAIN** to return to Screen #1 (Main Screen).
- o **STATUS** FIELD:
Displays the current status of the instrument's inclinometer hardware:

READY	The inclinometer hardware is ready for use.
N/S	The inclinometer hardware option is either not installed in the instrument, or is not responding.

- o **PITCH** FIELD:
Continuously displays the instrument's current pitch in degrees from the calibrated nadir. Physically, the pitch direction is aligned along the LCD main screen graph's Y-Axis, and the roll direction is aligned along its X-Axis.

o **BEEP @ LEVEL** BUTTON:

Controls whether the inclinometer emits a tone when the instrument is held level:

OFF	No tone is emitted.
TRIG	A tone is emitted only while the instrument's trigger button is pressed when the instrument is held level.
CONT	A tone is emitted continuously whenever the instrument is held level.

o **LEVEL RANGE** BUTTON:

Controls the range, in +/- degrees from the calibrated nadir, where the inclinometer will indicate that the instrument is level.

For example, if the Level Range is set to 2 degrees, then any time the instrument is held within +/- 2 degrees of nadir in both the pitch and roll directions, the LCD will display the cross-hairs graphic and optionally emit a tone indicating level.

The Level Range may be set to any value from 1 to 8 degrees, depending on the accuracy and sensitivity desired in the inclinometer level indicators.

o **RECALIBRATE** BUTTON:

Re-sets the internally-stored calibrated nadir (roll = 0, pitch = 0) position in pitch and roll.

Before pressing this button, ensure that the instrument's front plate has been placed on a level surface.

When pressed, a confirmation button will then appear to ensure that the operator wants to actually perform the nadir recalibration.

o **SAVE SETTINGS** BUTTON:

Store the current "BEEP @ LEVEL" and "LEVEL RANGE" settings to non-volatile memory, so that they are preserved across power cycles.

Reflectance Measurement Overview

A reflectance measurement takes the scan of a reference (white plate) and calculates the ratio of the target scan to that reference.

If any of the below conditions exist, a **new reference measurement** should be taken before a series of target measurements:

- Beginning of a new set of scans.
- Any change in the collection optics.
- The instrument has been idle for an extended period, or has been turned off.
- Lighting conditions have changed or are variable.

When in doubt, take a reference measurement.

Refer to

Appendix L – Firmware V1.33 Changes for some additional information regarding instrument stand alone operation.

Taking a REFERENCE Measurement

1. Check that the LED Power Indicator is illuminated, indicating that the power is enabled.
2. On the LCD's "Setup Screen A", ensure that the trigger field is set to "Laser + Scan" by pressing the "TRIGGER:" button until the correct setting appears.
3. On the LCD's "Main Screen", ensure that the scan type is set to "Reference" by pressing the "SCAN:" button until the correct setting appears.
4. Place a reference at the point of measurement making sure to completely fill the field of view of the instrument. (The optional Spectralon® plate is recommended for this purpose.) This reference measurement is associated with all succeeding measurements until a new reference measurement is established.
5. Press and hold the LASER/SCAN button to activate the sighting laser and verify the placement of the reference plate relative to the laser spot as required to fill the field of view.
6. Release the LASER/SCAN button to record the measurement. The MEM field on the LCD's Main Screen increments to display the memory slot of the scan that was just acquired.

Taking a TARGET Measurement

1. Check that the LED Power Indicator is illuminated indicating that the power is enabled.
2. On the LCD's "Setup Screen A", ensure that the trigger field is set to "Laser + Scan" by pressing the "TRIGGER:" button until the correct setting appears.
3. On the LCD's "Main Screen", ensure that the scan type is set to "Target" by pressing the "SCAN:" button until the correct setting appear.
4. Place the target material at the point of measurement, making sure to completely fill the field-of-view of the instrument.
5. Press and hold the LASER/SCAN button to activate the sighting laser and verify the placement of the target to be measured relative to the laser spot as required to fill the field of view.
6. Release the LASER/SCAN button to record the measurement. The MEM field on the LCD's Main Screen increments to display the memory slot of the scan that was just acquired.

Additional target measurements may be made by pressing and releasing the LASER/SCAN switch.

Downloading Stand-alone Measurements

To view and analyze the data acquired in stand-alone mode, the data must be downloaded from the memory of the instrument to a computer running the Windows PC operating system. In order to download data, the PC Data Acquisition Software must be installed on the Windows PC (See Section 2 – PC Data Acquisition Software – INSTALLATION for instructions on how to install this software).

Downloading also requires a connection between the instrument and the PC; see the sub-section above “2. Connecting The Instrument To A Computer” in order to establish this connection.

Once the instrument/PC connection is established, refer to the following sub-sections in Section 2 – PC Data Acquisition Software in order to download the stand-alone measurements to the PC:

- See “Control->Setup Instrument...” for basic software connection instructions.
- See “Control->Setup Overlap/Matching...” for instructions on how to configure these settings, as they are applied to both downloaded and newly acquired spectral data.
- See “Control->Read Memory...” for downloading instructions.

4. PC Data Acquisition

The same general rules regarding Reference and Target measurement acquisition (described above in sub-section 3. Stand-Alone Data Acquisition) apply when acquiring data using the PC Data Acquisition software. Please read and understand those rules.

Connect the instrument and the Windows PC; see the sub-section above 2. Connecting The Instrument To A Computer in order to establish this connection.

Once the instrument/PC connection is established, refer to the following sub-sections in Section 2 – PC Data Acquisition Software in order to acquire new measurements on the PC:

- See “Control->Setup Instrument...” for basic software connection instructions.
- See “Control->Setup Overlap/Matching...” for instructions on how to configure these settings, as they are applied to both downloaded and newly acquired spectral data.
- Use the PC Data Acquisition’s “Reference” and “Target” buttons (or their equivalent keyboard hot-keys “R” and “T”) to acquire spectral data.

5. Windows Mobile PDA Data Acquisition

See Appendix F – Windows Mobile PDA Data Acquisition Software for instructions on how to use the optional Windows Mobile PDA device to acquire spectral data.

6. Android Phone/Tablet Data Acquisition

The Android SVCScan app, available via download from the Google Play store, supports i-Series instrument data acquisition under the Android operating system.

Operation of the spectrometer using the Android SVCScan app is covered in a separate document titled “SVCScan Android Data Acquisition Software User Manual”. This manual may be downloaded from the Spectra Vista web site.

Section 4 – Battery Charger

1. Introduction

The battery charger charges one (1) lithium-ion battery. The charger operates from 110/240 volt AC line voltage. The battery charger directions must be followed for safe, correct operation. Charging should only take place indoors, in a cool place.

The normal charge time for a completely discharged battery is approximately 5 hours. Two chargers are provided with the instrument so that both batteries can be charged overnight.

2. Charger Specifications

STANDARD CHARGER:

Input: 100 – 240VAC, 0.3A (Max), 50/60Hz, Output: 8.4 VDC, 1.3A (Max)

FAST CHARGER:

Input: 100 – 240VAC, 0.9A (Max), 50/60Hz, Output: 8.4 VDC, 2.7A (Max)

3. Battery Pack Information And Warnings

Cells: 2 Li-Ion cells connected in series: 9000mAh, 3.7V each.

Protection: 4A max polyswitch is installed in the battery pack for double over-discharging protection.

The battery will be automatically cut off to zero (0) volts when its voltage is too low for proper operation.

WARNING:

- Avoid heavy impact to the battery pack; this may cause the battery pack to malfunction.
- Do not expose the battery pack to fire or extreme heat; this may cause the battery pack to malfunction.
- Do not use any DC power supply or charger other than the charger supplied with the instrument, this may cause the battery pack to malfunction.



Figure 7 Battery Pack (Left), Fast-Charger (Middle), and Standard Charger (Right)

4. Charging

1. Plug the charger's round male plug into the battery pack's female jack.
2. Plug the charger into an AC outlet.

The charger's RED LED will be illuminated during charging; when charging is complete, the charger's GREEN LED will be illuminated.

Charging a battery from empty to full capacity will take approximately 5 hours.

Appendix A1 – HR-1024i Specifications

Spectral Range	350 nm to 2500 nm
Number of Channels	1024 (including overlapped channels)
Type of Sensors	512 Channel Silicon Array 256 Channel InGaAs array 256 Channel Extended InGaAs array
Resolution	≤ 3.3 nm FWHM @ 700 nm ≤ 9.5 nm FWHM @ 1500 nm ≤ 6.5 nm FWHM @ 2100 nm
Integration Speed	1 - 1000ms
Unit Size	8.75" wide, 3.0" high, 11.5" long 22 cm. wide, 8 cm. high, 29 cm. long
Unit Weight	8.5 lbs (3.8 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and Diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix A2 – HR-768i Specifications

Spectral Range	350 nm to 2500 nm
Number of Channels	768 (including overlapped channels)
Type of Sensors	512 Channel Silicon Array 128 Channel InGaAs array 128 Channel Extended InGaAs array
Resolution	≤ 3.3 nm FWHM @ 700 nm ≤ 16 nm FWHM @ 1500 nm ≤ 14 nm FWHM @ 2100 nm
Integration Speed	1 - 1000ms
Unit Size	8.75" wide, 3.0" high, 11.5" long 22 cm. wide, 8 cm. high, 29 cm. long
Unit Weight	8.4 lbs (3.8 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix A3 – HR-640i Specifications

Spectral Range	350 nm to 2500 nm
Number of Channels	640 (including overlapped channels)
Type of Sensors	512 Channel Silicon Array 64 Channel InGaAs array 64 Channel Extended InGaAs array
Resolution	≤ 3.3 nm FWHM @ 700 nm ≤ 30 nm FWHM @ 1500 nm ≤ 28 nm FWHM @ 2100 nm
Integration Speed	1 - 1000ms
Unit Size	8.75" wide, 3.0" high, 11.5" long 22 cm. wide, 8 cm. high, 29 cm. long
Unit Weight	8.4 lbs (3.8 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix A4 – HR-768si Specifications

Spectral Range	350 nm to 1900 nm
Number of Channels	768 (including overlapped channels)
Type of Sensors	512 Channel Silicon Array 256 Channel InGaAs Array
Resolution	≤ 3.3 nm FWHM @ 700 nm ≤ 9.5 nm FWHM @ 1500 nm
Integration Speed	1 - 1000ms
Unit Size	8.75" wide, 3.0" high, 11.5" long 22 cm. wide, 8 cm. high, 29 cm. long
Unit Weight	8.0 lbs (3.6 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and Diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix A5 – HR-512i Specifications

Spectral Range	350 nm to 1050 nm
Number of Channels	512
Type of Sensors	512 Channel Silicon Array
Resolution	≤ 3.2 nm FWHM @ 700 nm
Integration Speed	10 - 1000ms
Unit Size	8.75" wide, 3.0" high, 7.0" long 22 cm. wide, 8 cm. high, 18 cm. long
Unit Weight	5.2 lbs (2.4 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and Diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix A6 – XHR-1024i Specifications

Spectral Range	350 nm to 2500 nm
Number of Channels	1024 (including overlapped channels)
Type of Sensors	512 Channel Silicon Array 256 Channel InGaAs array 256 Channel Extended InGaAs array
Resolution	≤ 2.8 nm FWHM @ 700 nm ≤ 8.0 nm FWHM @ 1500 nm ≤ 6.0 nm FWHM @ 2100 nm
Integration Speed	1 - 1000ms
Unit Size	8.75" wide, 3.0" high, 11.5" long 22 cm. wide, 8 cm. high, 29 cm. long
Unit Weight	8.5 lbs (3.8 kg)
Battery Weight	1.5 lbs (0.7 kg)
Internal Memory	1000 scans maximum
Sighting	Diode laser at 635 nm
Display	4.3" LCD touchscreen with real time spectral and sample image display Laser, Ready & Power LEDs Real time spectral display with PDA/PC
I/Os	2 Bluetooth, USB A & B ports, & 12 pin external trigger
Standard Foreoptics	4° Nominal FOV
Optional Foreoptics	8° Nominal FOV 14° Nominal FOV 25° Fiber Optic Light guide, various lengths Irradiance Spheres and Diffusers
Power	Li-Ion Battery operated, 6.5 – 8.0 VDC @ 2.5A
Environmental	Temperature: -10C to +40C Humidity: <90%, non-condensing
Cleaning	Wipe down exterior with soft, dry cloth only.

Appendix B – File Descriptions

Signature File Format

The Signature (SIG) File Format is a text file that contains a header followed by spectral data information. Each line of the header begins with a keyword tag, as defined below. NOTE: Some header fields (scan method, scan time, optic, etc) should NOT change from reference to target, but are explicitly called out in the header for completeness).

Line	Keyword Tag	Fields	Description
1	/*** Spectra Vista SIG Data ***/	None	This line must appear <u>exactly</u> as shown in order for the file to be recognized as a current SIG format file.
2	name=	F1	F1 is the SIG file name.
3	instrument=	F1 :F2	F1 is the instrument's model number. F2 is the instrument's extended serial number.
4	integration=	R1,R2,R3,T1,T2,T3	R1, R2, R3 are the reference scan integration times (in milliseconds) for the Si, InGaAs1, and InGaAs2 detectors respectively. T1, T2, T3 are the target scan integration times (in milliseconds) for the Si, InGaAs1, and InGaAs2 detectors respectively.
5	scan method=	R1, T1	R1/T1 is the reference/target scanning method used to determine the length of the scan: "Time-based" – fixed scan time "Coadd-based" – fixed coadd number
6	scan coadds=	R1,R2,R3,T1,T2,T3	Rx/Tx is the reference/target coadds acquired as part of the scan for each spectrometer
7	scan time=	R1, T1	R1 is the reference scan time setting (seconds). T1 is the target scan time setting (seconds).
8	scan settings=	R1, T1	R1 is the reference scan integration type ("AI" = Auto integration, "FI" = Fixed integration, "UI" = unknown) T1 is the target scan integration type ("AI" = Auto integration, "FI" = Fixed integration, "UI" = unknown)
9	external data set1=	R01,R02,R03,R04,R05,R06,R07,R08,R09,R10,R11,R12,R13,R14,R15,R16,T01,T02,T03,T04,T05,T06,T7,T08,T09,T10,T11,T12,T13,T14,T15,T16	R01-R16 are set #1 of the reference scan external data samples, or [PC Software Only] 0 if disabled. T01-T16 are set #1 of the target scan external data samples, or [PC Software Only] 0 if disabled. Set #1 is sampled <u>just prior to the beginning</u> of spectral scan acquisition. Each sample is in the range –32K to +32K.
10	external data set2=	R01,R02,R03,R04,R05,R06,R07,R08,R09,R10,R11,R12,R13,R14,R15,R16,T01,T02,T03,T04,T05,T06,T7,T08,T09	R01-R16 are set #2 of the reference scan external data samples, or [PC Software Only] 0 if disabled. T01-T16 are set #2 of the target scan external data samples, or [PC Software Only] 0 if disabled. Set #2 is sampled <u>just after the end</u> of spectral

		,T10,T11,T12,T13,T14,T15,T16	scan acquisition. Each sample is in the range –32K to +32K.
11	external data dark=	D01,D02,D03,D04,D05,D06,D07,D08,D09,D10,D11,D12,D13,D14,D15,D16	D01-D16 are the most recent external dark data samples. These values are acquired when the “Dark Scan” button is pressed. Each sample is in the range –32K to +32K.
12	external data mask=	X1	[PC Software Only] X1 is the 8-bit decimal encoded value that represents which external channel has been enabled. A “1” -> channel enabled, or a “0” -> channel disabled. For example, a mask value of 5 would mean channels 1 and 3 are enabled. External Data Masks values range from 0 (no channels enabled) to 255 (all channels enabled).
13	optic=	R1, T1	R1 is the fore-optic name in use in the reference scan. T1 is the fore-optic name in use in the target scan. An example of a fore-optic name is “LENS 4” or “FIBER1”.
14	temp=	R1,R2,R3,T1,T2,T3	R1, R2, R3 are the reference scan reported temperature values (in °C) for the Si, InGaAs1, and InGaAs2 detector respectively. T1, T2, T3 are the target scan reported temperature values (in °C) for the Si, InGaAs1, and InGaAs2 detectors respectively.
15	battery=	R1,T1	R1/T1 is the reported battery voltage during the reference/target scan.
16	error=	R1,T1	R1/T1 is the reported error code (if any) during the reference/target scan. An error value of 0 (zero) means no error occurred. See the Error Codes section following this table.
17	units=	R1, T1	R1/T1 are the types of units associated with the scan data: “Radiance” – units of radiance “Irradiance” – units of irradiance “Counts” – raw counts uncorrected for calibration factors
18	time=	R1,T1	R1/T1 is the system time that the reference/target scan was acquired.
19	longitude=	R1,T1	R1/T1 is the GPS reported longitude when the reference/target was acquired. GPS fields are blank if the optional GPS unit is not installed, or was not producing position information at the time of the scan. (The exact format is described below in the SIG File GPS Data Format paragraph).
20	latitude=	R1,T1	R1/T1 is the GPS reported latitude for the reference/target scan.
21	gpstime=	R1,T1	R1/T1 is the GPS reported time for the reference/target scan.
22	comm=	F1	The F1 field provides space for any user-supplied comment.

23	memory slot=	R1,T1	If the spectral data was originally acquired by in stand-alone mode, R1/T1 are the internal scan numbers within the HR-1024i's scan memory. If the data was acquired by the PDA or PC Acquisition Software, these fields are both set to 0 (zero).
24	factors=	F1,F2,F3	F1/F2/F3 are the matching factors used to match Si and InGaAs1 detector data. F1 is the reference radiance matching factor. F2 is the target radiance matching factor. F3 is the reflectance matching factor.
25	inclinometer x offset=	R1,T1	R1/T1 is the instrument's offset in degrees from nadir in the X (roll) direction at the time of scan acquisition.
26	inclinometer y offset=	R1,T1	R1/T1 is the instrument's offset in degrees from nadir in the Y (pitch) direction at the time of scan acquisition.
27	sun zenith=	R1,T1	R1/T1 is the sun zenith angle, in degrees <i>[Android SVCScan Only]</i>
28	sun azimuth=	R1,T1	R1/T1 is the sun azimuth angle, in degrees <i>[Android SVCScan Only]</i>
29	weather=	{string}	Local weather information broken into several fields – temperature, humidity, etc. <i>[Android SVCScan Only]</i>
30	data=	None	This tag marks the beginning of the spectral data. The format of each line of spectral data is described below in the SIG File Spectral Data Format paragraph (below).

Error Codes

The “error=” SIG file field above has the following interpretations:

Prior To Instrument Firmware Version 1.33 / PC Software Ver. 1.21 / Android App Ver. 1.5.0

The error field should be ignored. In these versions of firmware / software, the error field was not yet implemented.

After Instrument Firmware Version 1.33 / PC Software Ver. 1.21 / Android App Ver. 1.5.0

The error field indicates the following:

- 0 = No Error
- 1 = VIS saturation occurred
- 2 = SWIR1 saturation occurred
- 4 = SWIR2 saturation occurred

Other values indicate that multiple saturation errors occurred; the error values for individual saturation are added together. For example:

- 3 = both VIS and SWIR1 saturation occurred
- 5 = both VIS and SWIR2 saturation occurred
- 6 = both SWIR1 and SWIR2 saturation occurred
- 7 = VIS, SWIR1, and SWIR2 saturation occurred

SIG File GPS Data Format

Note that the GPS information in the header is only available when an optional GPS data source is installed in the PDA. When present, the GPS components have the following format:

GPS Longitude	DDDmm.mmmmC
GPS Latitude	DDmm.mmmmC
GPS Time	HHmmSS.SSS

where:

D	is degrees
m	is decimal minutes
C	is quadrant (N, S, E, W)
H	is hours (GMT, 24-hour format)
S	is seconds

Note that depending on the GPS source, the exact format may vary; for example, there may be fewer significant decimal points present for the Lat/Lon “minutes” field, or the Time “seconds” field may be truncated to contain only integral values.

SIG File Spectral Data Format

Following the header are four columns of data:

Column 1: Wavelengths (nanometers)
Column 2: Reference Values (see below)
Column 3: Target Values (see below)
Column 4: Reflectance (percent)

Reference/Target Values’ units depend on which calibration table or fore-optic was in use:

Radiance: $10^{-10} * W / (cm^2 * nm * sr)$
Irradiance: $10^{-10} * W / (cm^2 * nm)$
RAW DN: counts

White Plate Reflectance Data File Format

The White Plate Reflectance Data File is a two-column text file. The first and second columns are separated by white space (either TAB or SPACE characters).

The first column contains wavelength values, in nanometers. The file should include all wavelengths in the instrument's spectral range.

The second column contains reflectance values that correspond to the white plate's true reflectance at each wavelength.

White Plate Reflectance Data File Example:

A typical (truncated) white plate reflectance data file appears below:

250	0.949
251	0.949
252	0.952
253	0.953
254	0.953
255	0.955
256	0.955
257	0.956
258	0.957
259	0.958
260	0.958
261	0.958
262	0.959
263	0.960
264	0.960
265	0.960
...	
2492	0.945
2493	0.945
2494	0.944
2495	0.943
2496	0.946
2497	0.946
2498	0.947
2499	0.948
2500	0.946

Windows Mobile PDA Software COM Port Configuration File

A software initialization file named:

“HR1024i_INI.txt” (**PDA software version 1.0**),

or

“SVCscan_INI.txt” (**PDA software version 1.1 and later**)

can be found in the PDA folder “My Device\Program Files\Spectra Vista\”. This file contains a list of COM ports that the PDA Software uses for communicating with the instrument.

The file has the following (example) 2-line format:

```
COM1  
COM8
```

Where:

Line 1 (COM1 in the example above) is the PDA’s 9-pin RS-232 serial port.

Line 2 (COM8 in the example above) is typically the PDA’s Bluetooth COM port.

The COM ports that are available to the PDA Software may vary, depending on the PDA’s configuration.

If the COM ports listed in the INI file do not match the COM ports configured into the PDA operating system, then the PDA will not be able to communicate correctly.

Appendix C – Laser Safety

The instrument uses a Class 3R laser beam that complies with IEC 60825-1, Ed. 1.2. The source of laser energy is a nominal 3 mw, continuous, 635 nm diode laser. No maintenance or adjustments are required in order to keep this product in compliance with the above specifications.

DANGER

Do not look into the laser beam at any time including instrument setup or operation.

The laser is operated by pressing the Laser Scan Switch located on the top surface of the instrument. See Section 2. HARDWARE DESCRIPTION, for the location of this control.

CAUTION - Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

There are no user serviceable components within this device. The unit should be returned to the manufacturer in case of malfunction.

LABELING

The following labels are required. Each label is described below. If the instrument does not have the appropriate labels, contact the manufacturer to ensure that your unit is in compliance.

(1) Explanatory Label



Location – Front panel near fore-optic mount.

(2) Warning Label



Location #1 – Front panel near fore-optic mount.

Location #2 – Laser Scan Switch push button.

Appendix D – PC Third Party Bluetooth Usage

The instrument supports Bluetooth connections from PCs for the purpose of transferring data and control information between the instrument and the connected PC. It is the customer's responsibility to purchase, install, and configure the appropriate PC Bluetooth hardware using this appendix as a guide.

Instrument-Specific Bluetooth Information

Bluetooth Versions Supported: V1.1

Bluetooth Class: Class I

Bluetooth Connection Type: Virtual COM Port, un-secure connection.

Bluetooth Connection Password: This password is exchanged between the PC or PDA Bluetooth driver and the instrument during the initial Bluetooth discovery/association process.

The password is dependent on the model number of the instrument, as defined below:

Model	Bluetooth Password
XHR-1024i	xhr1024i
HR-1024i	hr1024i
HR-768i	hr768i
HR-640i	hr640i
HR-768si	hr768si
HR-512i	hr512i

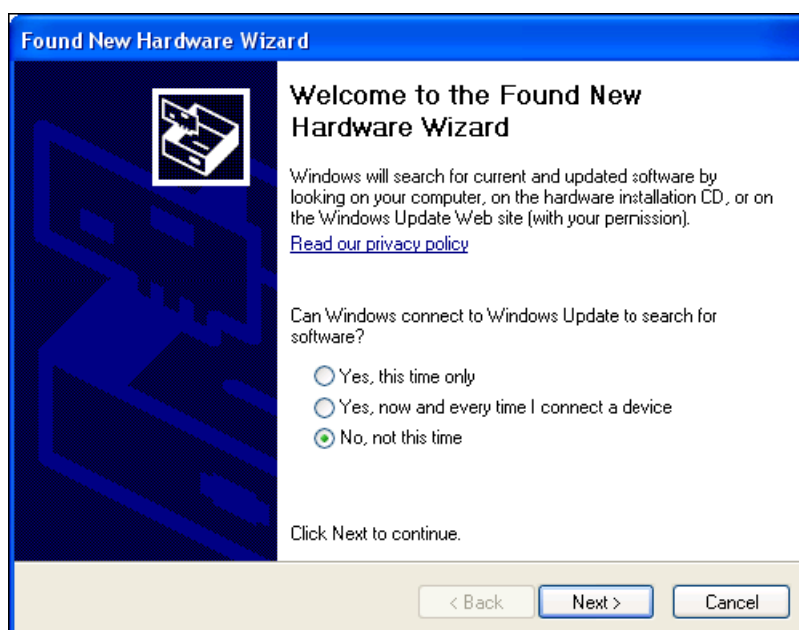
Appendix E – PC USB Device Drivers

Installing USB Device Drivers On The PC

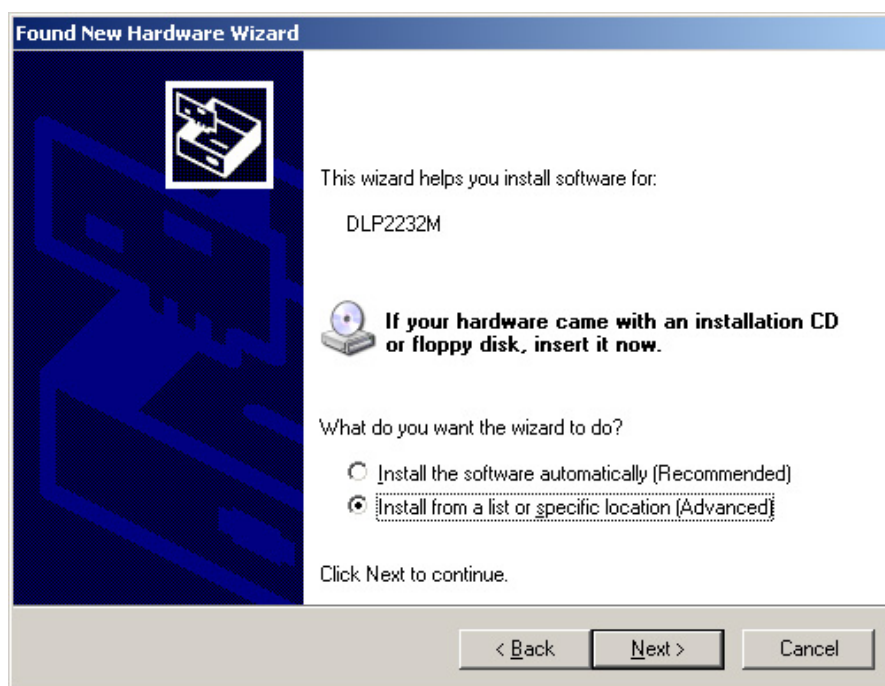
To install USB device drivers under Windows XP, follow the instructions below. Please read through all the instructions before beginning. **The USB Device Drivers are only necessary if you plan to connect your instrument to the PC using a USB cable.**

Do **not** plug the instrument into the Windows XP USB port until instructed below.

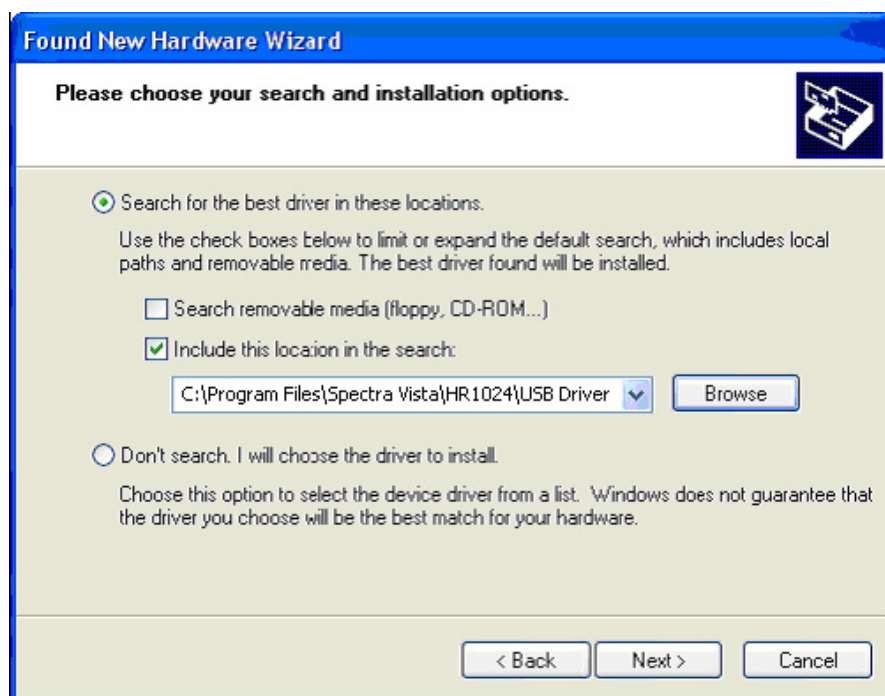
1. **You must install the PC Data Acquisition Software before following this procedure**, as the procedure uses files that are part of that installation. See Section 2 – PC Data Acquisition Software – Sub-section 1. INSTALLATION above installation instructions.
2. If you are re-installing the USB drivers, the original drivers need to be first uninstalled. Please refer to the Removing USB Device Drivers section below for details of this procedure.
3. If your Windows XP machine is connected to a network, temporarily disconnect from that network. This will prevent the Windows “Found New Hardware Wizard” from automatically connecting to the Windows Update website, and downloading and installing the wrong drivers for the instrument.
4. Connect the instrument to a spare USB port on your PC, and power-up the instrument. This will launch the Windows “Found New Hardware Wizard”. If there is no available Internet connection or if Windows XP SP 2 is configured to ask before connecting to Windows Update, the screen below is shown. Select "No, not this time" from the options available and then click "Next" to proceed with the installation.



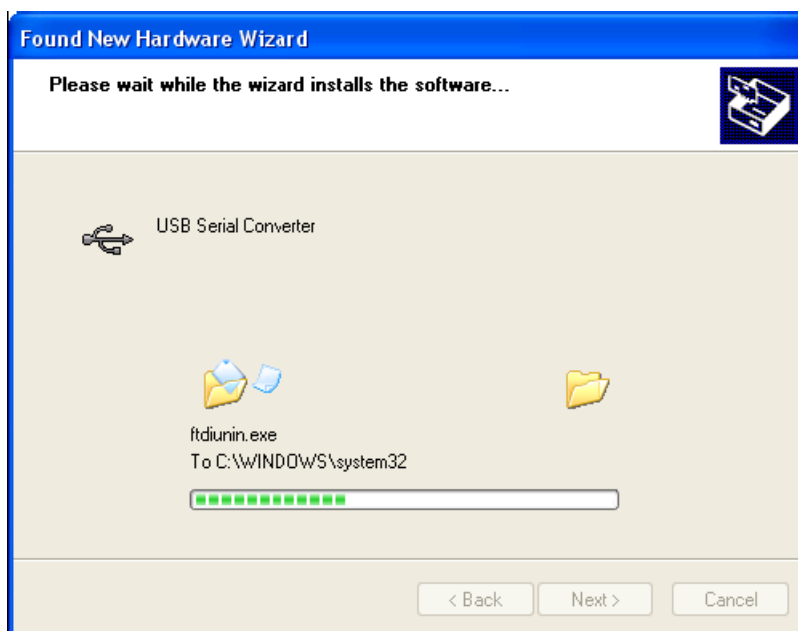
5. Select "Install from a list or specific location (Advanced)" as shown below and then click "Next".



6. Select "Search for the best driver in these locations" option, and use the "Browse" button to select the location of the USB driver files. The instrument's USB drivers are included as part of the PC Data Acquisition Software installation, in a folder called "USB Driver" that may be found in the PC Data Acquisition Software installation folder (typically, "C:\Program Files\Spectra Vista\HR1024i\"). Once the file path has been entered in the box, click "Next" to proceed.



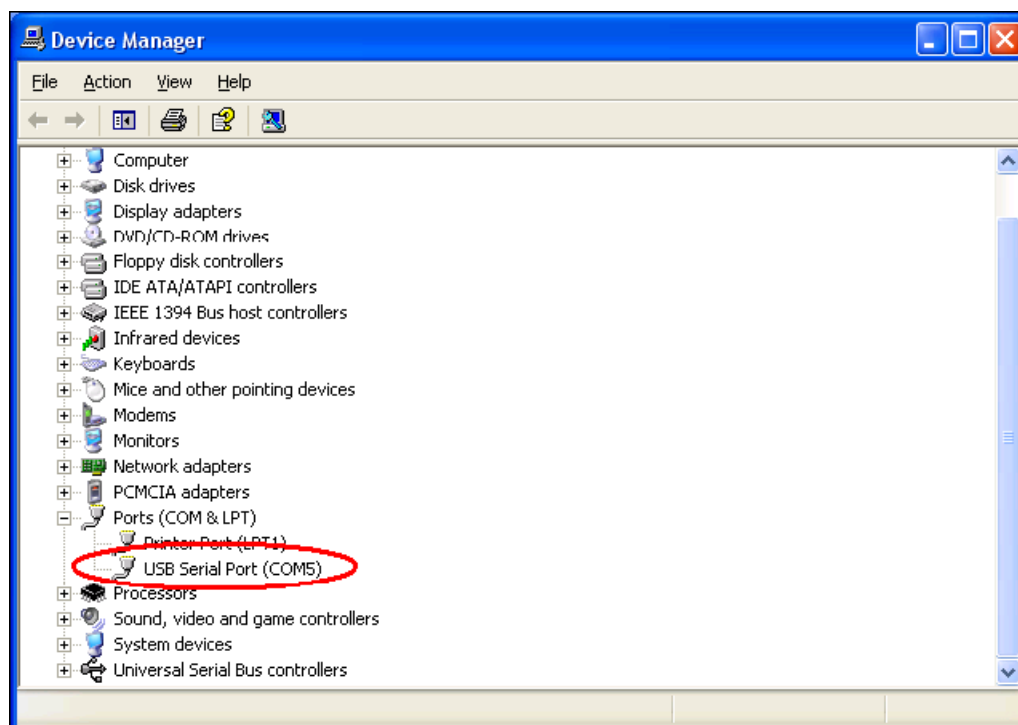
7. The following screen will be displayed as Windows XP copies the required driver files.



8. Windows should then display a message indicating that the installation was successful. Click "Finish" to complete the installation. This has installed the serial converter. The COM port emulation driver must be installed after this has completed.



9. After clicking "Finish", the **Found New Hardware Wizard** will **again appear** (See Step 4 above); Continue by installing the COM port emulation driver. The procedure is the same as that above for installing the Serial Converter driver above.
10. When both driver installations are complete, open the Device Manager (located in "Control Panel\System" then select the "Hardware" tab and click "Device Manger") and select "View > Devices by Type". The device appears as an additional COM port with the label "USB Serial Port".

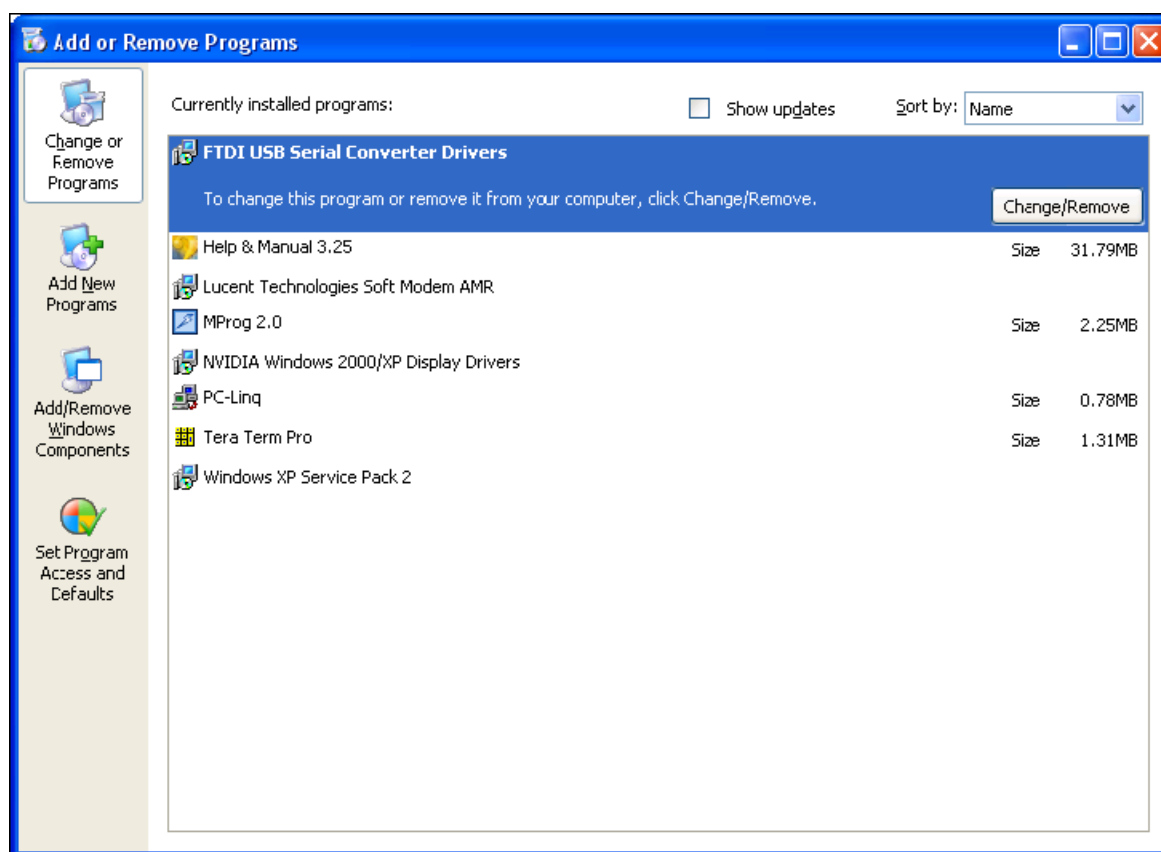


11. In the example above, the USB Driver was installed on "COM5". **You should note which COM port is assigned to your installation, and use this same COM port number in the PC Data Acquisition Software's "Control->Setup" dialog.**
12. Also, please note: If you later plug in the instrument to a different USB port on your PC, the USB Serial Port driver may assign a different COM number to the connection, and you will have to change that COM number in the PC Software's Control->Setup" menu. To avoid this, **try to plug the instrument into the same physical USB port on the PC each time.**

Removing USB Device Drivers From The PC

To uninstall USB Device Driver, follow the instructions below:

1. Disconnect the instrument from the PC (if attached).
2. Open the Add/Remove Programs utility located in "Control Panel\Add/Remove Programs". Select "FTDI USB Serial Converter Drivers" from the list of installed programs.



3. Click the "Change/Remove" button, and follow the instructions to uninstall the drivers.

Appendix F – Windows Mobile PDA Data Acquisition Software

Overview

This appendix documents the usage of PDA Data Acquisition Software application on a PDA (Personal Digital Assistant) using the Microsoft Windows Mobile operating system. The application will be referred to as just “the PDA application” throughout this appendix.

The main features and functionality of the PDA application are similar to that of its counterpart on the PC. It collects data from a hand held spectroradiometer, saves the data to the PDA’s internal memory, and displays the graph in different views.

For general knowledge about the spectroradiometer, please consult the earlier sections of this manual.

Information regarding the PDA, outside of its specific use with the spectroradiometer may be found in the PDA manufacturer’s User Guide.

PDA Requirements

PDA Type:	Windows Mobile
Operating System:	Windows Mobile 6.x
Interfaces:	RS-232, and/or Bluetooth virtual com port serial services.

PDA Software – Quick Reference

1. Power on the instrument.
2. Power on the PDA and start the application program by tapping

PDA Software 1.0: Start->HR1024ipda

PDA Software 1.1 or later: Start->SVCscan
3. Select “Connect” to display a list of COM ports.
4. Select appropriate COM port to establish communication to the instrument. COM1 is the serial cable connection and the other COM port is typically the Bluetooth connection.
5. The PDA will communicate with the instrument and upload current settings and calibration files.
6. The instrument’s current settings will be displayed in the Settings Dialog Box.
7. Select the settings to be used and press OK.
8. Begin Acquisitions by
 - a. Pressing “REFERENCE” with a white plate in the targeting area to acquire a reference scan.
 - b. Pressing “TARGET” with a target in the targeting area.
9. View Data: While in Graph->Radiance mode, the radiance graph will automatically appear upon reference or target acquisition completion. In Graph->Reflectance mode, the reflectance graph will automatically appear on target acquisition completion.
10. Save Data: Data will automatically be saved to the PDA's memory when the AutoSave Menu option is checked. When AutoSave is not checked, the acquired data may be manually saved to file using File->Save menu; a dialog box will appear, prompting for a filename. Input a filename and click OK. Data will be stored to the selected file name.
11. Upload files to PC for analysis. SIG format data files can be uploaded for viewing by the PC Data Acquisition Software or other packages for analysis. The files are transferred from the PDA to the Host PC using the Microsoft ActiveSync software via a USB connection. The default location of SIG files on the PDA is in the folder:

PDA Software 1.0: \My Documents\HR1024i Data\

PDA Software 1.1 or later: \My Documents\SVC Data\

PDA Software Application – Detailed Usage

Introduction

The PDA Data Acquisition Software application is based on the interface and functionality of the PC Data Acquisition Software for the i-Series family of spectroradiometer.

The name of the application, as well as the INI file and default data folder varies with the version of the PDA software:

PDA SW Version	PDA App Name	INI File Name	Default Data Folder Under /My Documents/
1.0	HR1024i PDA	HR1024i_INI.txt	HR1024i Data
1.1 and later	SVCscan	SVCscan_INI.txt	SVC Data

Data File Format

Data files are stored as plain text in the SIG format. See Appendix B – File Descriptions for information on SIG format data files).

DATA FILE NAMING CONVENTION (AUTOSAVE)

When in AutoSave mode, the software automatically creates and saves SIG format data in files that are named according to the template “**HR.mmDDyy.CCCC.sig**”, where:

- The field **HR** is the literal string “HR”
- The fields **mm**, **DD**, and **yy** are the current month, day, and year respectively as determined by the PDA’s clock setting.
- The field **CCCC** is the value of the AutoSave File Counter field as shown on the PDA application main screen (see below).
- The **.sig** filename extension identifies this as a SIG format spectral data file.

DATA FILE LOCATION

See the default data folder column in the table above.

Startup

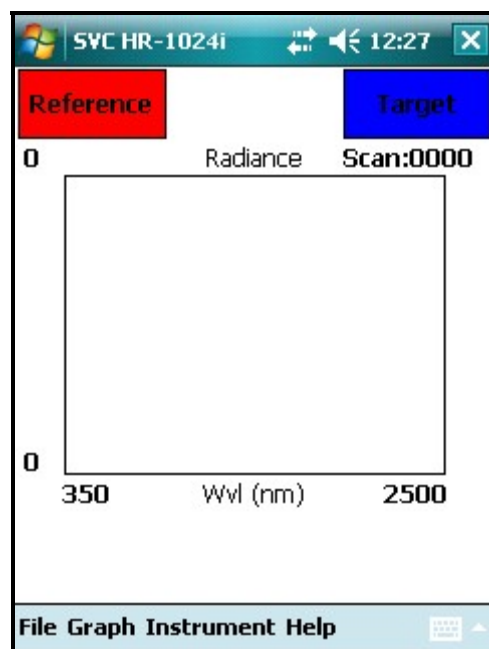
On the PDA press the On/Off button. The PDA application can be started by tapping the Start button in the corner of the PDA power-on screen, and selecting the PDA program.



Figure 8 PDA Start Menu Windows Mobile 6.0 (left) and Windows Mobile 6.5 (Right)

Operation

The **Menu Bar** for the PDA application is situated at the bottom of the window. There are 4 menu items on the Menu Bar, named from left to right as “File”, “Graph”, and “Instrument”, and “Help”; and two large **Command Buttons** on the top labeled “Reference” and “Target” (See below.)



The **Graph Type** text field just above the graph area displays which graph display (**Radiance** or **Reflectance**) is in effect within the rectangular **Graphing Area**.

The **AutoSave File Counter** (Scan:0000 in the example above) displays the value of the counter that is used to form the next AutoSave file name. The counter is incremented after each target acquisition while in AutoSave mode. This counter may be used to tie together AutoSaved file names with notes taken regarding targets during field acquisitions. By convention, the AutoSave file name has the format “HR.mmDDyy.XXXX.sig”, where XXXX is the value of the AutoSave File Counter. **Note that this counter displays the count of the next target acquisition, not the currently displayed target acquisition.**

The following sections describe these features in more detail.

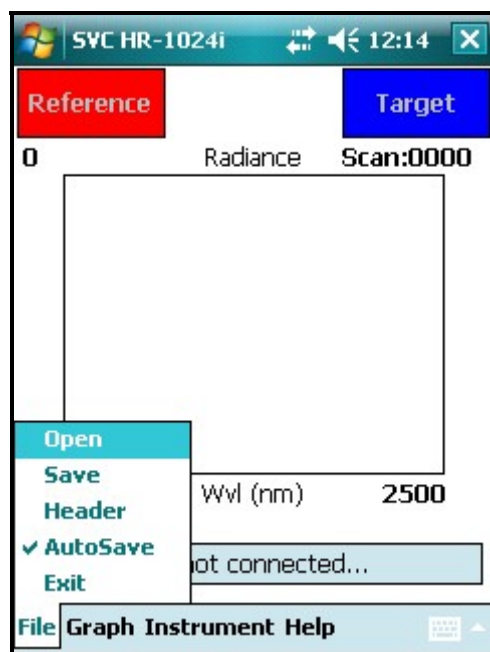
COMMAND BUTTON “REFERENCE”

The “Reference” button is used to initiate a Reference scan acquisition by the instrument. The instrument must be “connected” to the PDA for a successful scan to be acquired. Once the scan is acquired the Radiance result will be displayed graphically if the graph is configured to display Radiance.

COMMAND BUTTON “TARGET”

The “TARGET” button is used to initiate a Reference scan acquisition by the instrument. The instrument must be “connected” to the PDA for a successful scan to be acquired. Once the scan is acquired, the Radiance result will be displayed graphically (if the Graph mode is set to display Radiance) or the Reflectance result will be displayed (if the Graph mode is set to Reflectance). The acquired scan can then be saved to file manually, using the FILE>SAVE menu, or it will be automatically be saved if the File->AutoSave option is checked.

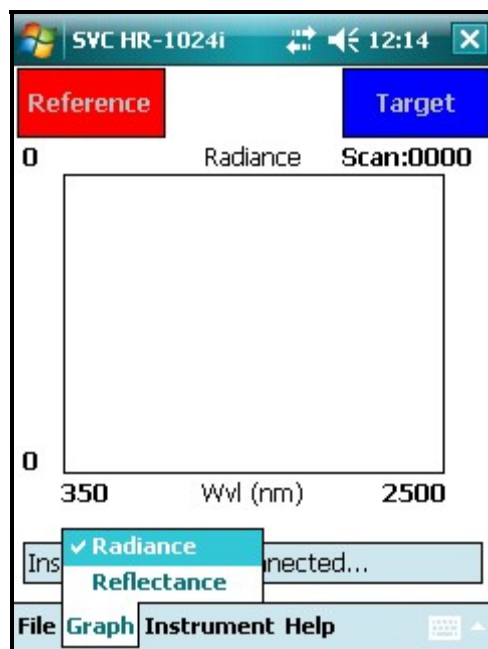
MENU BUTTON “FILE”



The drop down menu list for “File” consists of the following menu items:

Open	Displays a dialog for a user to select and open an existing PDA SIG file for display.
New	Displays a dialog for a user to specify a new base name for future scan acquisitions; a 4-digit value, starting at 0000, is appended to each future scan and incremented with each target acquisition.
Save	Displays a dialog for a user to select a filename and save the current scan to a SIG data file generated by the PDA application.
Header	This is a check box option which causes header information to be displayed in the application. When checked, a text box appears containing the header data. The header data is described in the SIG File Format. When unchecked, the header text box is not visible.
AutoSave	This is a check box option that causes measurements to automatically be saved to file after each target acquisition. Files will be generated in SIG format, and will be named according to the convention: "HR.mmddyy.nnnn.sig", where mmddyy is the date and nnnn is the current value of the AutoSave File Counter (ranges from 0000 to 9999.)
Exit	Exits the application.

MENU BUTTON “GRAPH”

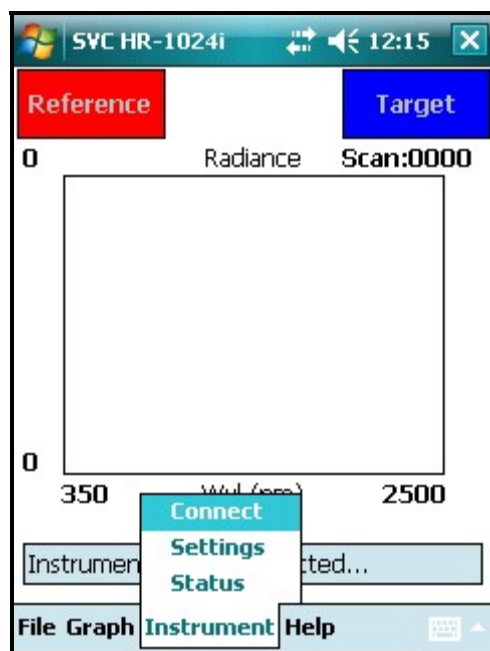


The dropdown menu list for “Graph” is composed of the following menu items:

- | | |
|-------------|--|
| Radiance: | Sets the graphic display mode to Radiance to display data in Radiometric Units. The Y-axis Radiance units are $\mu\text{w}/\text{cm}^2/\text{nm}/\text{sr} \times 10^{(-10)}$ and the X-axis is Wavelength in nanometers(nm). |
| Reflectance | Sets the graphic display mode to Reflectance to display data in Percent Reflectance units. The Y-axis Reflectance units are percent (%) and the X-axis is Wavelength in nanometers(nm). Reflectance is the ratio of the target scan vector to the reference scan vector. |

MENU BUTTON “INSTRUMENT”

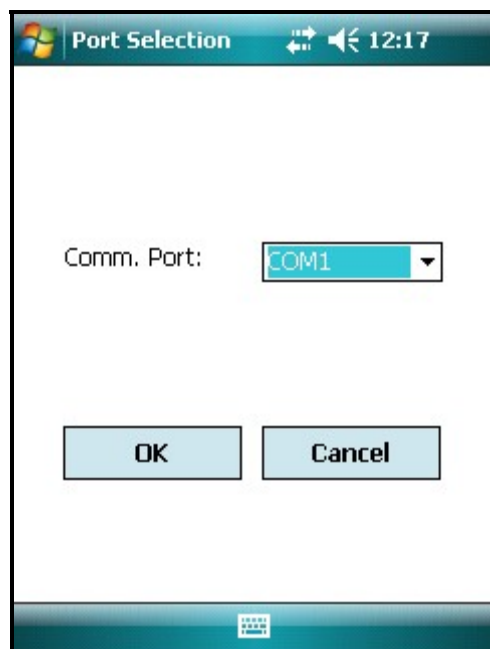
The dropdown menu list for “Instrument” has the following menu items:



Connect

Connect is used to connect the PDA to an instrument. When Connect is selected, the PDA displays a list of COM Ports in a new screen. The COM port that is to be used to connect to the instrument needs to be selected.

The COM Ports listed are controlled by a PDA Software configuration file; see Appendix B – File Descriptions, Windows Mobile PDA Software COM Port Configuration File for more information.



Selecting a COM port option causes the PDA to make a connection with the instrument. If successful, the Status menu will display a message indicating an instrument is connected and the Settings screen will automatically appear (See SETTINGS FORM below.)

Settings

The Settings menu is used to modify instrument data collection parameters. When selected a dialog box will appear, presenting various options (See SETTINGS FORM below.)

Status

This indicates the instrument connection status.

SETTINGS FORM

The image displays two versions of the 'Settings' dialog box. The left version has 'Scan Time' selected, showing a 'Sec.' (seconds) spinner set to 5. The right version has 'Coadds' selected, showing 'Vis' (Visible) set to 4, 'Swir1' (Swir1) set to 32, and 'Swir2' (Swir2) set to 32. Both versions have 'Photo Mode' set to 'None' and 'Integration Setting' checked for 'Auto'. The 'OK' and 'Cancel' buttons are at the bottom of each dialog.

Photo Mode: Control the JPEG target camera that is internal to the instrument. The setting are:

- ☐ OFF (no photos are acquired during target data acquisition),
- ☐ LOW-RES (320x240 photos acquired)
- ☐ HI-RES (640x480 photos acquired)

Scan Time / Coadds: These radio button control the way scans are acquired, either by setting a fixed scan time limit per scan (with varying number of co-adds), or by setting a fixed number of coadds per scan (and so varying the total scan time).

Depending on whether Scan Time or Coadds mode is selected, the appropriate control in the upper half of the setup dialog is enabled (see above):

- ☐ A single "Sec" setting – controls the total Scan Time per scan.
- ☐ Triple VIS/Swir1/Swir2 settings – controls the number of coadds acquired for each spectrometer, per scan.

Optic Sets the optic that is in use for data collection. This setting **must** match the actual foreoptic installed on the instrument, in order to obtain correct radiometric data. The menu options are determined by factory setting.

Integration Settings Auto When Auto is checked, the instrument automatically determines the optimal integration settings for all detector arrays (Si, Sw1, and Sw2). When Auto is not checked, the operator may set individual

integration times for each detector array, using the Si/Sw1/Sw2 controls in the lower half of the setup dialog.

OK

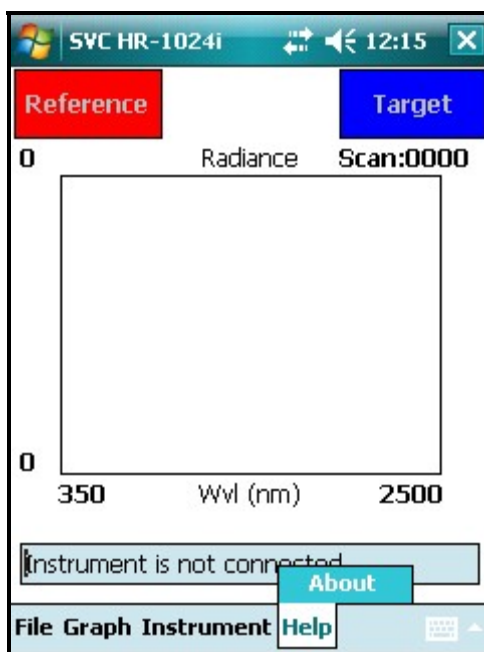
When OK is pressed the settings are sent to the instrument and used for operation.

Cancel

When Cancel is pressed, no changes are made to the settings.

MENU BUTTON “HELP”

The dropdown menu list for “Help” has the following menu items:



About:

Displays PDA software version information

Appendix G – Windows Mobile PDA Usage Tips

PDA Reset Procedures – Trimble Juno T41



Figure 9 Trimble Juno T41 PDA

Juno T41 Soft Reset / Turn-off Procedure

The Juno T41 PDA may be Soft Reset by briefly (1-2 seconds) pressing then releasing the green reset button located on the right-hand side of the PDA:



Figure 10 Juno T41 PDA Side View Showing Green Reset Button

This will bring up the Power Menu on the PDA screen (below). If this does not occur, a Hard Reset may be necessary.

To perform a Soft Reset, tap the “Reset” button to Soft Reset the PDA. The PDA screen will go blank and the PDA will reboot within several seconds.

To turn off the PDA, tap the “Shutdown” button. This is important to save battery life.

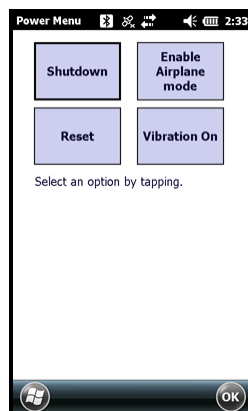


Figure 11 Juno T41 PDA Power Menu Screen

Note: Performing a Soft Reset simply restarts the PDA operating system. It is similar to re-booting a PC.

Juno T41 Hard Reset Procedure

If a Soft Reset does not work, then a Hard Reset may be necessary.

The Juno T41 may be Hard Reset by pressing and holding the green reset button located on the right-hand side of the PDA until the PDA screen goes blank and the reboot process begins. Note that it may be necessary to hold the button for as long as 20 seconds in order for the reset to take effect.

Performing a Hard Reset does not affect the contents of the software on the PDA and may be performed without causing any software issues.

PDA Reset Procedures – Trimble Recon 400X



Figure 12 Trimble Recon 400X PDA

Recon 400X Soft Reset Procedure

The Recon 400X PDA may be Soft Reset by pressing and holding the Power Key (shown below) until the PDA screen indicates the soft reset is in progress. Note that if the PDA is hung, **the key may have to be held down as long as 60 seconds before the soft reset takes place.**



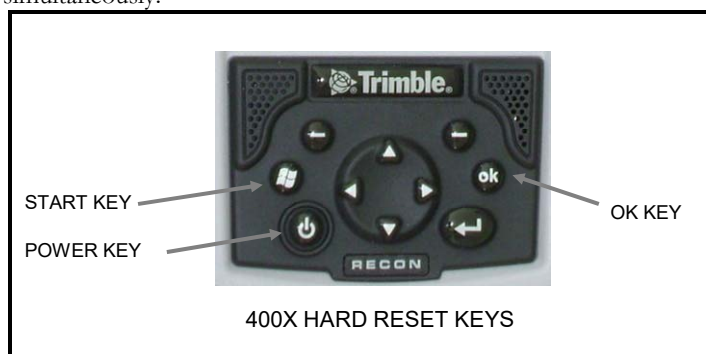
Note: Performing a Soft Reset simply restarts the PDA operating system. It is similar to re-booting a PC.

Recon 400X Hard Reset Procedure

Warning: Hard Resetting the PDA will require re-installation of the PDA application, either from internal PDA Backup, or from CD.

The Recon 400X PDA may be Hard Reset by:

1. Pressing and holding the Power Key (shown below) until the PDA displays a countdown.
2. Continue to hold the Power Key until it counts to zero and beeps twice.
3. Release the Power Key and immediately press and hold the Start and OK keys simultaneously.



PDA Power Up

If the PDA is turned off, or automatically powers down while the PDA application is in use, it may be necessary to perform a Soft Reset on the PDA, followed by a restart the PDA application.

In general, after turning on the PDA and before beginning a series of new scans with the PDA application, it is recommended that the operator always first perform a Soft Reset on the PDA.

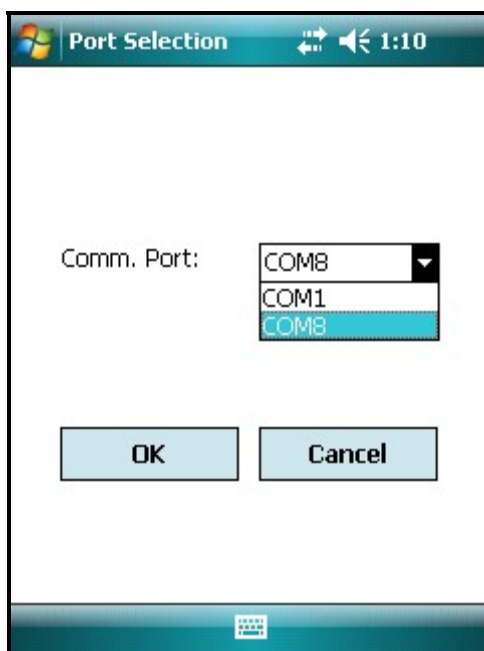
PDA Charging

Even when not in use, the PDA should be periodically recharged every few weeks. This process allows the PDA's internal backup battery to maintain the PDA application installation and configuration.

Failure to maintain the charge on the PDA's internal backup battery will cause the PDA application to be lost. In this case the PDA application must be re-installed.

PDA COM Ports And PDA Application Software

The PDA Application Software's "Instrument->Connect" menu displays a list of COM ports that are available to establish the connection to the instrument. A typical list, as displayed, is shown below:



The list of COM ports displayed in this dialog (COM1 and COM8 in the example above) may vary from one PDA installation to another.

The COM ports shown in the dialog are read from the application's initialization files at program startup. See the section Windows Mobile PDA Software COM Port Configuration File for the format of this initialization file.

The COM1 port is typically fixed and corresponds to the 9-pin RS-232 connector found on the PDA.

The other COM port listed is the Bluetooth COM port. The COM port listed **must** correspond to the COM port configured into the PDA's Bluetooth driver, or it will not be possible to communicate with the instrument via Bluetooth.

Appendix H – Instrument Remote Access

GENERAL PROTOCOL DESCRIPTION

The PC Data Acquisition Program can act as the access point that will allow a customer to remotely control the instrument from a custom software application. The access allows the customer's software to:

- trigger a reference or target scan
- receive the results of a target scan
- control spectrometer settings (fore-optic selection, scan timing, scan integration)

In order to control the instrument, the PC Data Acquisition Program acts as a network server. This network server function listens for TCP/IP socket connections from the customer's (client) software. The customer's client software sends trigger commands to the server, and receives responses and spectral data from the server.

Functionally, the operator first uses the PC Data Acquisition Program in the standard way - connect to the instrument and set the desired scan parameters (Scan Time, Integration, Auto Dark, Optic, Overlap/Matching, etc).

Once the desired scan parameters are set, the operator causes the PC Data Acquisition Program to enter into its "Remote Server" listening mode. At this point, the customer's client software is able to connect via a TCP/IP network socket and remotely control the instrument.

See the description of the Remote Access Tool in **Section 2 – PC Data Acquisition Software** for specific controls and descriptions on how to setup the PC Data Acquisition Software to act as a remote server.

DETAILED PROTOCOL DESCRIPTION

All the client/server communication uses simple human-readable ASCII commands and responses.

The client and server software may reside on the same (Windows) computer, or may reside on different computers that are connected via a TCP/IP network.

The PC program accepts and responds to the following commands from remote clients that want to remotely control the connected instrument:

Reference Scan Command

Command: "R *a*<cr>" - This one-parameter command triggers a reference scan acquisition in the instrument.

Parameter "*a*" (the scan sub-type) – determines the characteristics of the Reference scan, per these values:

0 – acquire a normal scan – the spectroradiometer acquires both dark and light data; the dark is subtracted from the light, and the resulting values are used as the reference.

1 – acquire a special light-only scan – the spectroradiometer acquires light data only; the dark data from a prior scan is subtracted from the light data, and the resulting values are used as the reference. Note that a prior normal scan must have previously been acquired by the spectrometer. Note also that this scan sub-type may not produce as accurate results as the normal scan, where dark/light data are both acquired.

Example: "R 0<cr>" – acquire a normal reference scan.

Example: "R 1<cr>" – acquire a special reference scan.

Normal Response: "r<cr><lf>" - This three character response is sent back to the client after the reference scan is complete. The reference scan may take up to 30+ seconds to complete, depending on the instrument's current "Scan Time" or "Coadds" setting.

Error Response: "E: Reference Scan Error<cr><lf>" - An error occurred while taking a reference scan. This could be caused by a timeout or other communication error between the PC and the instrument.

Error Response: "E: Wrong Number Of Parameters<cr><lf>" – The client sent the wrong number of parameters for this command.

Error Response: "E: Parameter Value Error" – The client attempted to set a parameter outside of its range of valid values for this command.

Target Scan Command

Command: "T a<cr>" - This one-parameter command triggers a target scan acquisition in the instrument.

Parameter "a" (the scan sub-type) – determines the characteristics of the Target scan, per these values:

0 – acquire a normal scan – the spectroradiometer acquires both dark and light data; the dark is subtracted from the light, and the resulting values are used as the target.

1 – acquire a special light-only scan – the spectroradiometer acquires light data only; the dark data from a prior scan is subtracted from the light data, and the resulting values are used as the target. Note that a prior normal scan must have previously been acquired by the spectrometer. Note also that this scan sub-type may not produce as accurate results as the normal scan, where dark/light data are both acquired.

Example: "T 0<cr>" – acquire a normal target scan.

Example: "T 1<cr>" – acquire a special target scan.

Normal Response: "t<cr><lf><SIG><Ctrl-Z>" - This response is sent back to the client after the target scan is complete. The target scan may take up to 30+ seconds to complete, depending on the instrument's current "Scan Time" or "Coadds" setting. Following the normal three character response "t<cr><lf>", the entire <SIG> format text file (containing the last reference, target, and reflectance data) will be transmitted back to the client, followed by the single <Ctrl-Z> (end-of-file, ascii code 26) character. The <SIG> file will typically be somewhere between 32K and 64K characters in length.

Note that the <SIG> format file itself has embedded <cr> and <lf> characters.

See Appendix B – File Descriptions for a complete description of <SIG> file formats.

Error Response: "E: Target Scan Error<cr><lf>" - An error occurred while taking a target scan. This could be caused by a timeout or other communication error between the PC and the instrument.

Error Response: "E: Target SIG File Open Error" - An error occurred while attempting to open the SIG file in preparation to sending the target data back to the client.

Error Response: "E: Wrong Number Of Parameters<cr><lf>" – The client sent the wrong number of parameters for this command.

Error Response: "E: Parameter Value Error" – The client attempted to set a parameter outside of its range of valid values for this command.

Optic Command

Command: "O *a*<cr>" - This one-parameter command changes the instrument's fore-optic.

Parameter "*a*" (the fore-optic) – min: 1 max: 8 increment: 1 - one of the eight available fore-optic values.

Example: "O 2<cr>" – sets the current fore-optic to 2.

Normal Response: "o<cr><lf>" - This three-character response is sent back to the client after the optic has successfully been selected.

Error Response: "E: Wrong Number Of Parameters<cr><lf>" – The client sent the wrong number of parameters for this command.

Error Response: "E: Parameter Value Error" – The client attempted to set a parameter outside of its range of valid values for this command.

Integration Time Command

Command: "I *a b c*<cr>" - This multi-character, three-parameter command changes the instrument's integration time setting, and selects between auto- and fixed-integration:

Auto-Integration: Parameter "*a*", "*b*" and "*c*" are all set to the special value of 0 (zero), and so the instrument determines the actual optimal integration times for each spectrometer.

Fixed-Integration: Parameters are set as follows:

Parameter "*a*" – (Si Spectrometer integration time, mS) – min: 10 max: 1000 increment: 10; values between 10 mS and 1000 mS that do not lie on boundaries of 10 mS will be silently rounded to the nearest 10.

Parameter "*b*" – (Swir1 Spectrometer integration time, mS) – min: 1 max: 40 increment: 1.

Parameter "*c*" – (Swir2 Spectrometer integration time, mS) – min: 1 max: 10 increment: 1.

Example (Auto-Integration): "I 0 0 0<cr>" – sets auto-integration mode.

Example (Fixed-Integration): "I 400 24 8<cr>" – fixes integration times (mS): Si: 400, Swir1: 24, Swir2: 8.

Normal Response: "i<cr><lf>" - This three-character response is sent back to the client after the integration mode/times have successfully been set.

Error Response: "E: Wrong Number Of Parameters<cr><lf>" – The client sent the wrong number of parameters for this command.

Error Response: "E: Parameter Value Error" – The client attempted to set a parameter outside of its range of valid values for this command.

Scantime Command

Command: "S *a*<cr>" - This one-parameter command sets the instrument's scan timing to a fixed number of seconds; note that this command **overrides** any prior Coadds Command that may have set the instrument's scan timing to a fixed number of coadds.

Parameter “*a*” (the scan time, seconds) – min: 1 max: 15 increment: 1 – the fixed scan time.

Example: “S 4<cr>” – sets the scan time on any subsequent reference/target scans to 4 seconds.

Normal Response: “s<cr><lf>” - This three-character response is sent back to the client after the scan timing has successfully been set.

Error Response: “E: Wrong Number Of Parameters<cr><lf>” – The client sent the wrong number of parameters for this command.

Error Response: “E: Parameter Value Error” – The client attempted to set a parameter outside of its range of valid values for this command.

Coadds Command

Command: “C *a b c*<cr>” - This three-parameter command sets the instrument’s scan timing to require a fixed number of coadds (subject to a maximum total scan time) from each spectrometer; note that this command **overrides** any prior Scantime Command that may have set the instrument’s scan timing to a fixed number of seconds.

Parameter “*a*” – (Si Spectrometer coadds) – min: 2 max: 500 increment: 2; values that are not even are silently rounded to the next even value.

Parameter “*b*” – (Swir1 Spectrometer coadds) – min: 2 max: 500 increment: 2; values that are not even are silently rounded to the next even value.

Parameter “*c*” – (Swir2 Spectrometer coadds) – min: 2 max: 500 increment: 2; values that are not even are silently rounded to the next even value.

Example: “C 10 200 300<cr>” – sets the scan timing to require 10 Si spectrometer coadds, 200 Swir1 spectrometer coadds, and 300 swir2 spectrometer coadds on any subsequent reference/target scans.

Normal Response: “c<cr><lf>” - This three-character response is sent back to the client after the scan timing has successfully been set.

Error Response: “E: Wrong Number Of Parameters<cr><lf>” – The client sent the wrong number of parameters for this command.

Error Response: “E: Parameter Value Error” – The client attempted to set a parameter outside of its range of valid values for this command.

Unknown Command

Command: <unknown> - an unrecognized command of some kind has been received by the server. The command is not one of those listed above.

Error Response: “E: Unknown Command - <unknown><cr><lf>” - This response flags any unrecognized command received by the server. The client should close down, and re-establish the socket connection in order to recover from this error.

Busy Error

If a new command is sent from the client before the previous command has completed, the server responds with an “E: Busy<cr><lf>” message.

Appendix I – Fiber Optic Installation

The instrument's fiber optic adapter installs the same way as the various foreoptic lenses. Similar to these lenses it has a pin to set a constant clock angle with the instrument which is especially important when using the fiber optic to maintain a stable radiometric calibration for correct data acquisition.

1. Remove any foreoptic lens that is currently installed by turning the knurled retaining ring **CLOCKWISE** while facing the instrument noting that the threads are right handed and the retaining ring has to move towards the instrument.
2. Orient the fiber optic adapter so that the clocking pin is face-up, as shown below: **CAUTION – DO NOT ALLOW ANY FOREIGN MATERIAL TO ENTER THE INSTRUMENT CAVITY AS IT MAY INTERFERE WITH THE FIBER ADAPTER INSTALLATION, DAMAGE THE INTERNAL OPTICS OR EFFECT THE INSTRUMENT'S RADIOMETRIC CALIBRATION.**

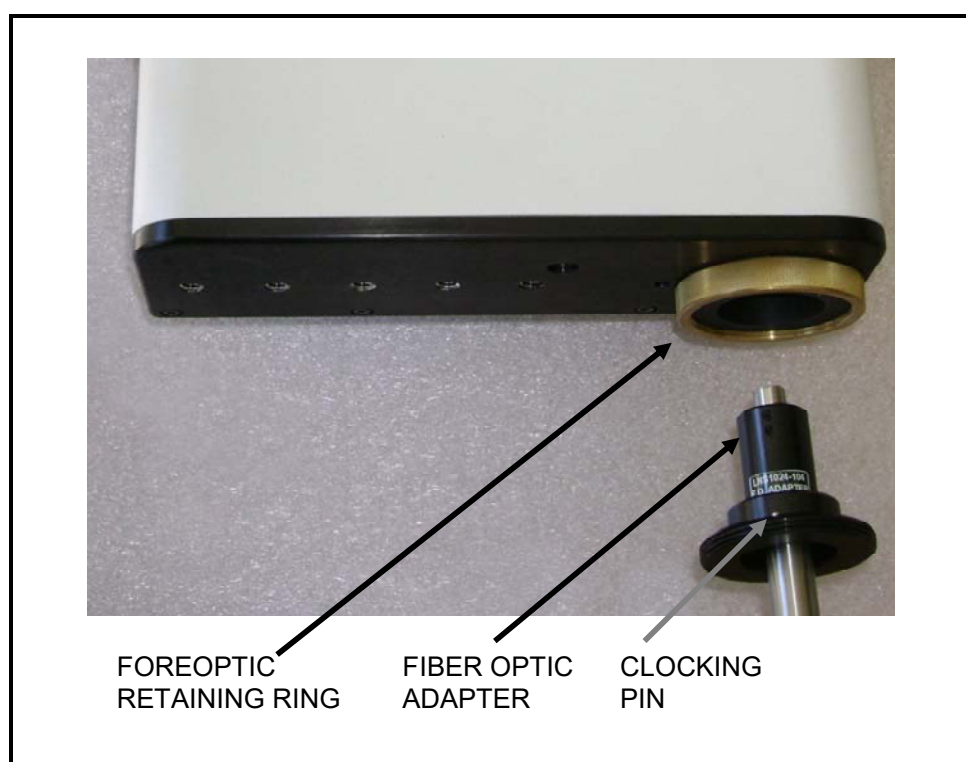


Figure 13, Fiber optic installation

3. Rotate the retaining ring counterclockwise while holding the fiber optic adapter clock angle until the pin feeds into its locating hole. Hand tighten the retaining ring.

Appendix J – Optional Fan Installation

WARNING: If the optional fan assembly is installed on the instrument's heat sink, do not operate the instrument unless the fans are actually turning and providing cooling air to the heat sink. Installing the fan assembly onto the heat sink without running the fans can cause undesirable thermal effects on the instrument.

The optional fan module (shown below) is attached to the bottom of the instrument to provide additional cooling.

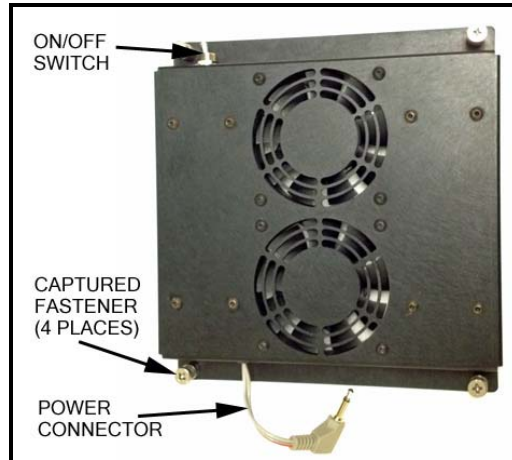


Figure 14 Fan Module

1. Carefully place the instrument on a sturdy work surface with the optical plate (bottom of the instrument) facing up, as shown below. If present, remove the four “Filler Fasteners” – these occupy threaded holes that will be used later by the installed fan module.

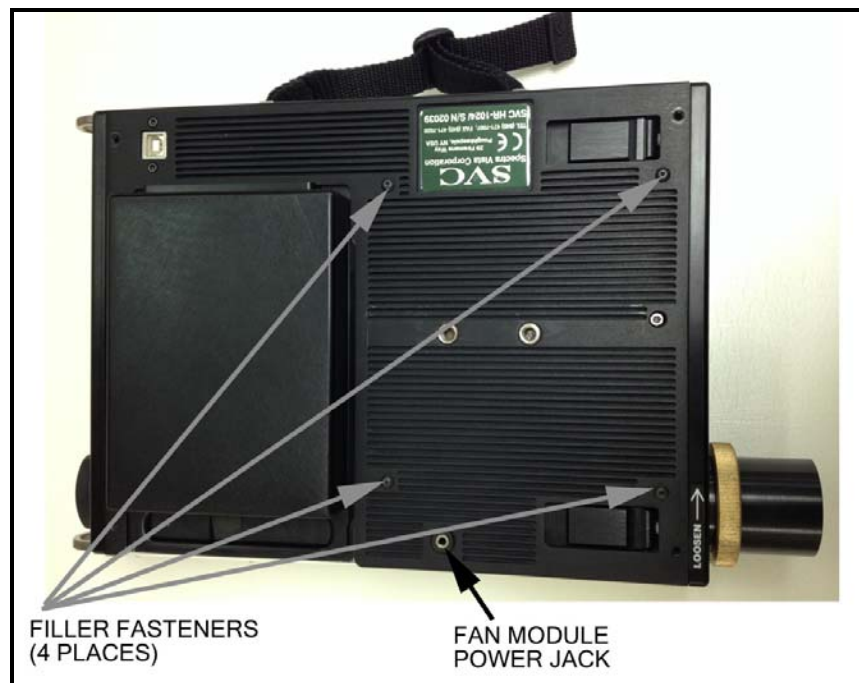


Figure 15 Instrument Optic Plate Prior To Fan Module Installation

2. Orient the fan module on the bottom of the instrument as shown below. Tighten the four captured fasteners mounted on the fan module, and plug the fan module power connector into the power jack on the bottom of the instrument.



Figure 16 Instrument Optical Plate After Fan Installation

Appendix K – Aux Port Specification

The round connector labeled “AUX” on the instrument’s back panel has multiple uses:

- An RS-232 serial communications interface for PC/PDA host communication.
- Multiple external trigger inputs – Reference, Target, and Current Setting.
- Scan-in-progress output – provides a signal to external devices when a scan is in progress.
- Battery power for powering external devices – V_{BAT} (6.5-8.5 VDC) @ 500 mA maximum current.

Connector (Female): Hirose P/N HR10-10R-12S(73)

Mating Connector (Male): Hirose P/N HR10-10P-12P(73)

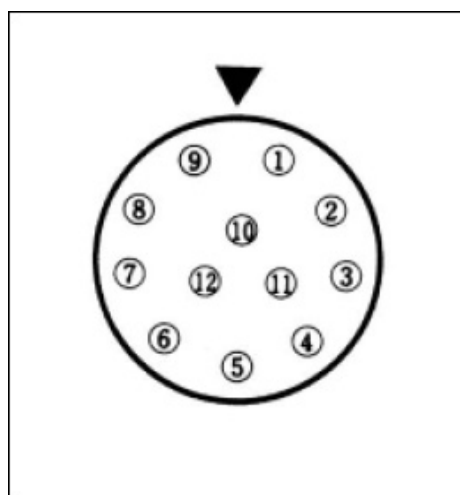


Figure 17 AUX Port Pinout

The following table specified the function associated with each AUX Port connector pin:

Pin	Direction	Signal Name	Function
1	Output	V_{BAT}	Supplies unregulated battery voltage to an external device; battery voltage is 6.5 – 8.5 VDC @ 500mA maximum current draw.
2	n/a	Reserved	Do Not Connect , reserved for future use.
3	Input	Reference Trigger	Normally open; shorting this pin to Ground will initiate a Reference scan.
4	Input	Target Trigger	Normally open; shorting this pin to Ground will initiate a Target scan.
5	Input	Current Scan Trigger	Normally open, shorting this pin to Ground will initiate a scan of the type currently selected in the LCD user interface.
6	n/a	Reserved	Do Not Connect , reserved for future use.
7	Output	Scan In Progress	Normally open; this pin will be shorted to Ground by the spectrometer electronics while a scan is in progress.
8	Output	Ground	Power Return / Signal Ground.
9	n/a	Reserved	Do Not Connect , reserved for future use.
10	Output	Host Comm TX	RS-232 PC/PDA host connection.
11	Input	Host Comm RX	RS-232 PC/PDA host connection.
12	Output	Host Comm GND	RS-232 PC/PDA host connection.

Appendix L – Firmware V1.33 Changes

The sequence of events that takes place during an i-Series instrument scan has changed with release of instrument Firmware V1.33. These changes are detailed in this section, offering the operator insight into how to make most efficient use of the instrument.

Background information:

Each spectral scan acquisition consists of two separate types of scans:

- A “light scan” taken with the instrument’s shutter in the open position, scanning the reflected light from the material within the instrument’s field of regard.
- A “dark scan” taken with the instrument’s shutter closed, scanning the amount of dark detector current present.

The results of the dark scan are then subtracted from the light scan, producing a value for each spectral channel that is proportional to the amount of light energy within each channel.

Prior to instrument firmware version 1.33, all instruments acquired the dark scan first, followed by the light scan:

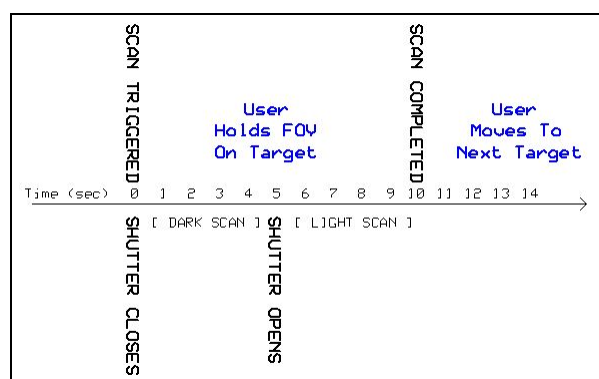


Figure 18 Typical 5-Second Scan Timing (Firmware < V1.33)

Starting with instrument firmware Version 1.33, the order has been reversed, so that the light scan is acquired first:

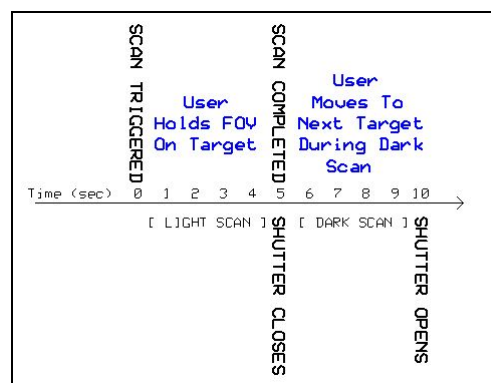


Figure 19 Typical 5-Second Scan Timing (Firmware ≥ V1.33)

This new light/dark arrangement has the advantage of allowing the instrument operator to begin moving the spectrometer viewing optic to another scan location earlier, at time = 5s instead of time = 10s.

When the instrument is used with an LC-RP Pro, this change allows the operator to begin removing the current sample from the LC-RP Pro earlier in preparation for inserting the next sample. This reduces the amount of lamp energy that the sample is exposed to and allows for samples to be processed more rapidly.

The V1.33 Firmware displays the following new LCD screen message at the start of scan acquisition:

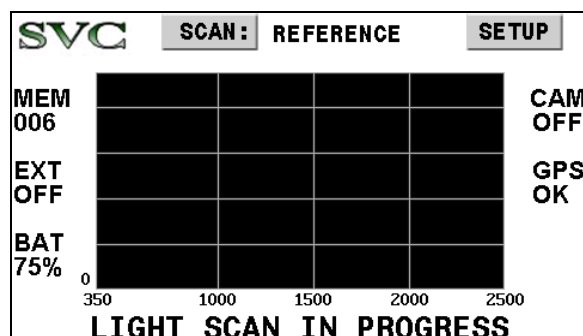


Figure 20 Firmware ≥ V1.33 LCD Message At Start Of Scan

When the light scan has completed, and it is safe to begin moving to a new target, the instrument produces a single “BEEP” and displays the following message on the instrument LCD:

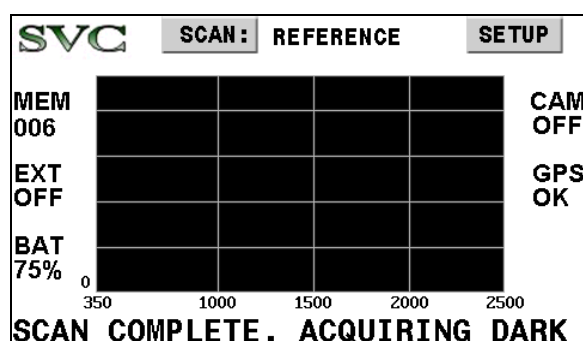


Figure 21 Firmware ≥ V1.33 LCD Message When The Shutter Has Closed

When the dark scan has also completed normally, the instrument signals this event with a single “BEEP”, clears the “SCAN COMPLETE...” message from the LCD, and displays the spectral data on the LCD graph.

One additional feature of the V1.33 Firmware: If detector saturation occurs during a scan, the instrument signals this by generating a three “BEEP” sequence and displaying the following message on the instrument LCD:

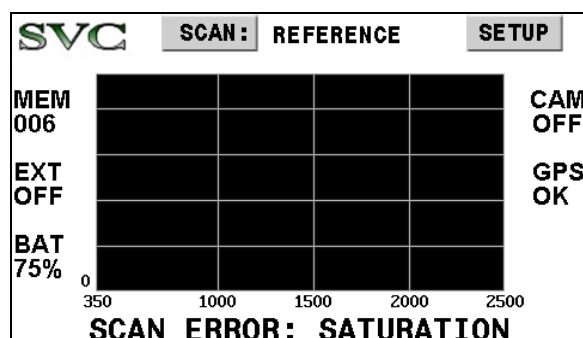


Figure 22 Firmware ≥ V1.33 LCD Message Signaling Saturation

If this error occurs, then the scan spectral data is suspect and the scan should be re-acquired.

AC-powered Spectral Lamps and Saturation Warnings

When AC (alternating current) powered noble gas spectral lamps are used to illuminate white panel or other target materials, the system may sometimes generate detector scan saturation warnings when none are apparent.

This is because the auto-integration algorithms that are programmed into the instrument firmware depend on stable illumination conditions in order to be effective and accurate. An AC-powered spectral lamp changes intensity many times per second, which causes these algorithms to produce inaccurate results.

When using AC-powered spectral lamps for illumination, SVC therefore recommends using fixed-integration mode in order to acquire spectral scans.

In order to achieve the best fixed-integration mode results:

- Setup the lamp, target and spectroradiometer in the desired configuration.
- After powering the lamps and allowing a sufficient warm-up time, acquire an initial auto-integration scan to establish a starting point for each spectrometer's integration timing.
- Now switch to fixed-integration mode, using the reported auto-integration values above as a starting point.
- If the system generates saturation warnings for a particular spectrometer in fixed-integration mode, reduce the integration time for that spectrometer until the warnings stop. Note that the VIS spectrometer must be reduced in 10 mS increments, while the SWIR1 and SWIR2 spectrometers may be reduced in 1 mS increments.

Appendix M – CE Declaration

**Spectra Vista Corporation
29 Firemens Way
Poughkeepsie, NY 12603 USA**

declares under our sole responsibility that the products listed below:

XHR-1024i, HR-1024i, HR-768i, HR-768si, HR-640i, HR-512i

are fully in conformity with the requirements of Council Directive:

2014/53/EU (RED - Radio Equipment Directive)

**This declaration is based on the full compliance of these products with the following
European standards:**

EMC: ETSI EN 301 489-1:2017
ETSI EN 301 489-17:2020
EN 61326:2013
EN 55032:2015 + A11:2020
EN 55035:2017

Electrical Safety: EN 61010-1:2016 + C1:2019

On the basis of this declaration, these products bear the following mark:



Appendix V – Mylar Sample

The following page may contain a Mylar sample which can be used in the field to verify SWIR wavelength calibration, when this manual is supplied in hardcopy form along with a new or re-calibrated instrument.

Please see the instructions included with the Mylar sheet for its usage with your instrument.

Appendix W – Spectralon® Care / Handling

The following pages may contain Spectralon reference panel calibration documentation, when this manual is supplied in hardcopy form along with a new or re-calibrated instrument that was delivered with a reference panel as part of the order.

General Care

Care should be taken to prevent contaminants such as finger oils from contacting the material's surface. Wear clean gloves when handling Spectralon.

Cleaning Instructions

If the material is lightly soiled, it may be air brushed with a jet of clean dry air or nitrogen. DO NOT use Freon. For heavier soil, the material can be cleaned by sanding under running distilled water with 220-240 grit wet or dry emery cloth until surface is totally hydrophobic (water beads and runs off immediately). Blow dry with clean air or nitrogen or allow to air dry.

In order to preserve the integrity of the frame it is suggested that the PTFE panel be removed prior to wet sanding.

Better results are produced using a flat surface sanding block during sanding operations.

The following pages may contain Spectralon® reference panel related information when this manual is supplied in hardcopy form along with an instrument order that includes a reference panel.

Appendix X – Instrument Quality Checklist

The following pages may contain an instrument-specific quality assurance checklist for a particular serial number instrument, when this manual is supplied in hardcopy form along with a new or re-calibrated instrument.

Appendix Y – Instrument Default Settings

The following pages may contain one or more instrument-specific default values (recommended overlap removal, matching parameters, etc.) for a particular serial number instrument, when this manual is supplied in hardcopy form along with a new or re-calibrated instrument

Appendix Z – Instrument FOV Diagram(s)

The following pages may contain one or more instrument-specific field-of-view diagrams for a particular serial number instrument, when this manual is supplied in hardcopy form along with a new or re-calibrated instrument.

SVC i-Series User Manual
Document Revision Sheet

Revision Level	Section	Revised By	Date	Revision Details
1.5	All	TC, LS	10/24/12	Initial Draft Issue
1.5a	All	TC	11/8/12	Updated Initial Draft
1.6	2, 3, Appendix P	LS	2/1/13	Updated PC Software Screenshots; Updated LCD Screenshots, Added Appendix P
1.7	Appendices	LS	5/29/13	Fixed appendix lettering; clarified SIG file header fields; added appendices X and Y.
1.8	Several	LS	8/29/13	Added new PDA SVCScan / WM 6.5 information. Updated new calibration table handling.
1.9	Appendix X/Y/Z	LS	10/12/13	Added breaks for insertion of external information (Spectralon, FOV diagrams, etc.)
	Section 3			Added description of the External Data Setup Screen.
1.10	Section 3	LS	11/21/13	Added description of the ENVI Spectral Library Tool
1.11	Appendix H	LS	12/9/13	Added description of new Remote Access commands.
1.12	Several	LS	7/31/13	Added descriptions for new instrument models – HR-640i, HR- 768si, HR-512i
1.13	Section 3, Appendix A6, Appendix J	LS	2/22/17	Updated Section 3 with new PC Data Acquisition Software controls and tools; added new XHR-1024i model information and fan module information.
1.14	Section 3, Appendix B	LS	6/1/19	Added new internal inclinometer information – new LCD screen and SIG file header additions.
1.15	Section 2, Appendices B, L	LS	11/7/19	Described new scan timing changes; added new SIG file header entries; described new main screen Current Baseline control.
1.16	Several	LS	8/10/20	Updated the ENVI library tool with TSG compatibility information. Updated CE declaration. Updated laser operation / warnings.
1.17	Appendix X	LS	3/17/2021	Added the special calibrated Mylar sample documentation to the manual.